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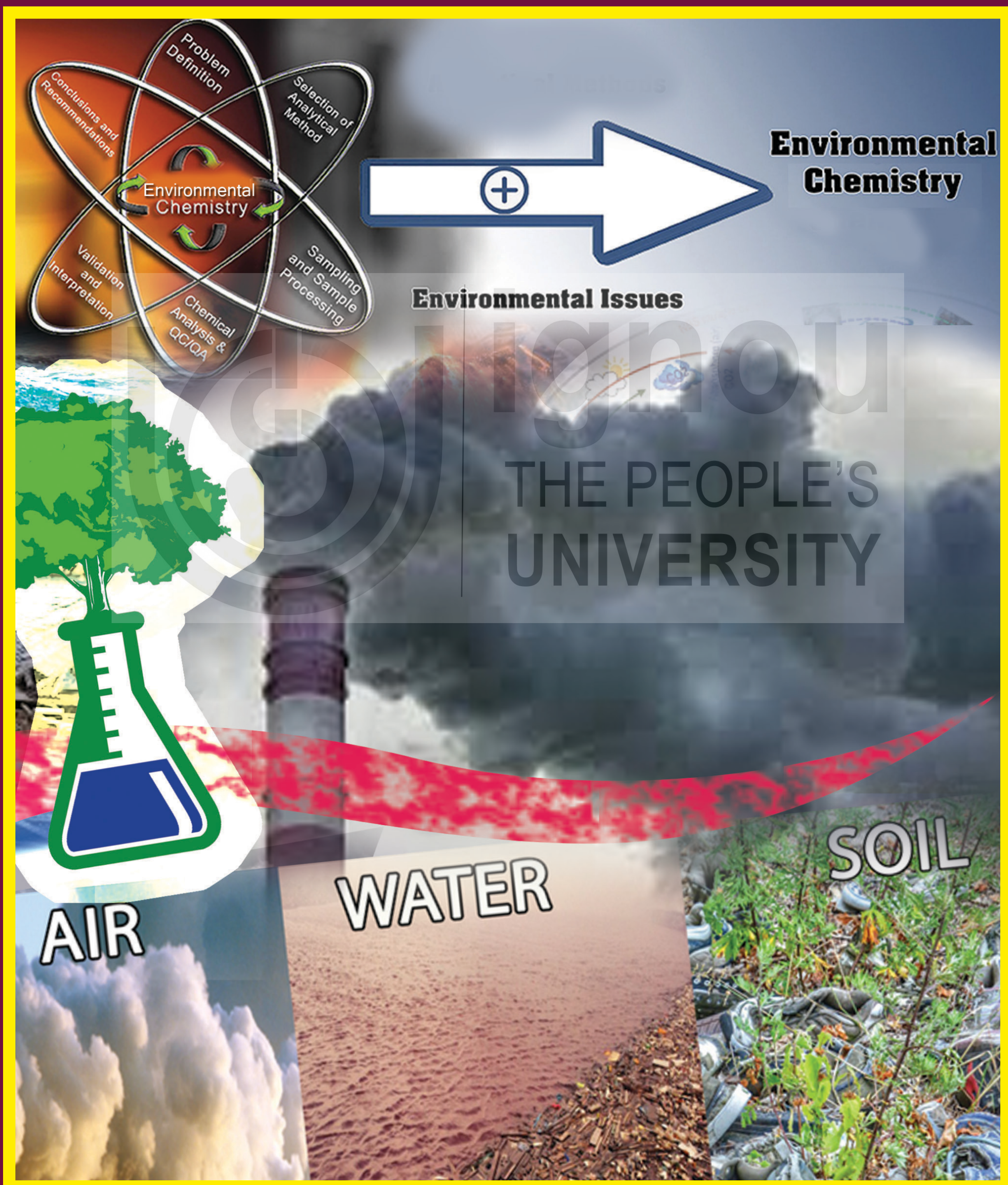
Indira Gandhi National Open University
School of Interdisciplinary and
Trans-disciplinary Studies

MEV-013
Environmental Chemistry



"शिक्षा मानव को बन्धनों से मुक्त करती है और आज के युग में तो यह लोकतंत्र की भावना का आधार भी है। जन्म तथा अन्य कारणों से उत्पन्न जाति एवं वर्गगत विषमताओं को दूर करते हुए मनुष्य को इन सबसे ऊपर उठाती है।"

– इन्दिरा गांधी

"Education is a liberating force, and in our age it is also a democratising force, cutting across the barriers of caste and class, smoothing out inequalities imposed by birth and other circumstances."

– Indira Gandhi

MEV-13

ENVIRONMENTAL CHEMISTRY

SCHOOL OF INTER-DISCIPLINARY AND TRANS-DISCIPLINARY STUDIES (SOITS)
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COURSE INTRODUCTION

Welcome to the MSc (Environmental Science) programme of IGNOU. The curriculum prepared for this programme is relevant and significant in the present-day scenario.

This core course is entitled "Environmental Chemistry (MEV-013)". This course consists of 4 blocks comprising of 15 Units.

Block 1 deals with **Fundamentals of Environmental Chemistry.** In this block we tried to explain the fundamentals of environmental chemistry. The block also deals with the state of atmosphere in the Earth's geological reservoirs. It provides a modern and concise introduction to environmental chemistry principles and the dynamic nature of environmental systems. It is this limited size that makes the atmosphere potentially so vulnerable to contamination. It explains how even the addition of a small amount of material can lead to significant changes in the way the atmosphere, and how the chemical species exist in equilibrium.

Block 2 deals with **Chemistry of Air, Water and Soil**. This block explains chemistry of air, water and soil, their composition. Distribution of chemical species in air and water, their properties, – Cation exchange capacity, pH, NPK in soils, soil organic matter, water soluble organic carbon and its significance, soil fertility have also been explained.

Block 3 deals with **Pollution Chemistry.** The block explains air, water, soil pollution and pollution parameters. The emphasis has been made on various organic and inorganic air pollutants, photochemical smog and their reactions, consequent health effects are addressed. It also elaborated parameters of water pollution, chemistry of hazardous substance and wastes.

Block 4 deals with **Analytical Techniques in Environmental Chemistry**. The block details on the classical methods of analysis required in environmental chemistry such as basic analytical techniques like titrimetry, gravimetry, spectrometry, chromatographic techniques and radio techniques. Finally, the block describes the application of these techniques in the environmental monitoring.

All these Blocks will provide you with sufficient knowledge about the pollutants, management, hazards, risk assessment, standards and laws pertaining to the industrial sector.

INTRODUCTION TO BLOCK 1

This block focuses on the fundamental principles of environmental chemistry. The block mainly deals with the state of atmosphere in the earth's geological reservoirs. It provides a modern and concise introduction to environmental chemistry principles and the dynamic nature of environmental systems.

Unit 1 deals with the **Environmental Chemistry Part 1.** In this we discussed the concept and scope of environmental chemistry, Fundamentals of Elemental Stoichiometry, Chemical equilibrium, Chemical potential, Chemical Kinetics

- Simple reaction mechanisms, Order and molecularity of chemical reactions; Chemical reactions, Catalysis, Adsorption. The study of the chemical balance in natural systems and how the disturbance in this chemical balance happens by the anthropogenic activities by releasing of chemicals into the environment which change the natural ecosystem levels.

Unit 2 deals with the **Environmental Chemistry Part 2.** This unit deals with the chemical reactions that occur between an acid and base and result into formation of salt and water are generally called acid- base reactions. These reactions play a critical role in maintaining balance in our body as well as our environment. It explained how highly purified water also behaves like a weak electrolyte and shows conductivity and how it ionizes in hydrogen and hydroxyl ions but hydrogen ions do not exist in free form. It introduced to modern concept of oxidation and reduction reactions, by defining them as removal of electrons from a species is termed as oxidation whereas addition of electrons is called reduction. Almost all the elements and their compounds have tendency to undergo oxidation-reduction reactions. The unit also explained other redox chemical reactions of the environment like photosynthesis, respiration, corrosion that we experience in our day to day life.

Unit 3 deals with the **Environmental Chemistry Part 3**. This unit discussed about the concept of solubility, solubility product and solubility of gases governed by Henry's law. How carbonate cycle dominates pH control in most of aquatic system and regulation of various biogeochemical processes in environment. The role of Chemical speciation in assessing the toxicity associated with different forms of chemical element and their oxidation state illustrating it with arsenic Sources and toxicity of heavy metals such as Pb, Cd, As, Hg in the environment Different types of radionuclides and its effects. Impact of hydrocarbon (saturated and unsaturated) on environment. Mechanism, functions and classification of lubricants into lubricating oil, greases and solid lubricants. It concluded by explaining about degradation of biological waste into biogas.

Unit 4 deals with the **Developments in Environmental Chemistry**. The unit describes about the scientific methodologies/solutions showing phenomenal capability to protect human health and environment and these innovative solutions have been possible through green chemistry. We have had fantastic accomplishments as the solutions were at the molecular level since green chemistry preached the prevention of waste at source rather than treating it up after generation. So, quite a few of the existing processes were redesigned using the basic tenets of green chemistry. Progressively, industries also started adopting the path of green synthesis.

INTRODUCTION TO BLOCK 2

This block focuses on the sources and types of pollutants in air, water and soil. This block explains chemistry of air, water and soil, their composition. Distribution of chemical species in air and water, their properties, – Cation exchange capacity, pH, NPK in soils, soil organic matter, water soluble organic carbon and its significance. This block contains three units.

Unit 1 deals with **Atmospheric Chemistry**. It focusses on atmosphere, the gaseous envelope of the earth, nurtures life on earth and protects it from the harsh environment of the outer space. It serves as a source of oxygen for the sustenance of life on earth and carbon dioxide for plant's photosynthesis, provides nitrogen which is consumed by nitrogen fixing bacteria in producing nitrogen-based molecules, an essential component of life molecules. The emphasis was given on various chemical species, including oxides of nitrogen, carbon and sulphur as well as their particulate matter, present naturally and due to man's activities undergo photochemical reactions, and are responsible for several environmental episodes.

Unit 2 deals with the **Water Chemistry.** This chapter, highlighted the importance of water, distribution and unique properties of water. Apart from various other factors, the properties of water also depend on the salt content in it. Thus, the physical and chemical properties of fresh water differ from those of sea water of. It also discussed the presence / addition of chemical species, gases, and organic matter in the water not only alters its physical properties such as colour, odour etc., but also changes its chemical and biological characteristics.

Unit 3 deals with the **Soil Chemistry.** In this unit we have elaborated upon the importance of soil, its nature, chemical properties and composition along with various processes that are involved in the formation of soil. The study of soil is based on two approaches, pedology that deals with studying soil as a natural body while edaphology deals with the study of soil properties with respect to plant production

INTRODUCTION TO BLOCK 3

This block focuses on the **Pollution Chemistry.** The block explains air, water, soil pollution and pollution parameters. The emphasis has been made on various organic and inorganic air pollutants, photochemical smog and their reactions, consequent health effects are addressed. This block contains four units.

Unit 1 deals with **Chemistry of Air Pollution – I.** It explained about gaseous inorganic air pollutants, Carbon monoxide, carbon dioxide, sulphur dioxide, nitrogen oxides, ozone. You will be interested to know about detailed account of various effects of air pollution and the chemistry of acid rain.

Unit 2 deals with the **Chemistry of Air Pollution Part II.** It focusses on the classification of organic air pollutants and their sources. It described in detail about the ozone layer depletion and its impact on biosphere. This unit make you to understand the organic reactions that causing photochemical smog and it concluded by explaining the effects of photochemical smog in humans

Unit 3 deals with the **Parameters of Water Pollution.** It explained basic concept of physico chemical and biological parameters of that are responsible for the constituents found in water systems. This unit also elaborated about the significance of chelating agents in aqueous system, describe the properties of colloidal substance present in water bodies, and the ion exchange method of water management.

Unit 4 deals with the **Chemistry of Hazardous Substances and Wastes.** It explained about various physical and chemical hazards and the effects of physical and chemical hazards. It provided the knowledge of identifying various types of combustible chemicals and make you to understand the effects of hazardous substance

INTRODUCTION TO BLOCK 4

This block focuses on the **Analytical Techniques in Environmental Chemistry**. In block 3you have studied about chemistry of air, water and soil pollution. This block deals with the classical method of analysis. This block contains four units.

Unit 1 deals with **Basic Analytical Techniques**. The unit gives a detailed account of the various types of analytical techniques available and used by the analysts. The basic principles involved in the process of analysis will be explained. The unit also deals with the criteria for evaluating that includes sampling and measurement. A brief about evaluation of the analytical data would be given along with an introduction to the concepts of errors, accuracy, precision and significant figures. The unit ends with the ways of reporting the results of analysis. Optical methods based on absorption and emission are discussed in this unit.

Unit 2 deals with **Spectrophotometry.** In this introductory unit on spectrometry we have discussed about the principle and applications of three important analytical tools. Two of these, viz., UV-VIS spectrophotometry and IR spectrometry being techniques based on spectroscopy. We started the unit by defining spectroscopy as an important domain that concerns study of interaction of electromagnetic radiation with matter. It is also discussed the difference between spectroscopy, spectrometry and spectrophotometry before taking up UV-VIS spectrophotometry.

Unit 3 deals with **Chromatographic Techniques.** This unit mainly focused on chromatographic techniques which are used for the quantitative and qualitative analysis in various pharmaceuticals, food and dyes industries. Column chromatography has various disadvantages including manual setup, time consuming and accuracy. It explained various advanced techniques like gas liquid chromatography, HPLC Super critical liquid chromatography and their application which are being employed not only in industries but also for monitoring the environment, GLC is an advanced technique compared to conventional chromatography.

Unit 4 deals with **Radiochemical Techniques.** In this unit we have studied about the basics of radiochemical techniques like isotopes radioactive decay and various units used to measure the radioactivity. Apart from that we have discussed about the terminology that is frequently used while studying radioisotopes and the techniques associated with it. The concept of carbon dating is highlighted that is widely used by anthropologist and paleontologist to estimate the age of dead or decaying material obtained from fossils. At the end of this unit we have discussed about techniques that are popularly used to measure the radiation, principles and brief working mechanisms along with their applications with special reference to environment.

CONTENTS

BLOCK 1 : FUNDAMENTALS OF ENVIRONMENTAL CHEMISTRY

BLOCK 2 : CHEMISTRY OF AIR, WATER AND SOIL

BLOCK 3 : POLLUTION CHEMISTRY

BLOCK 4 : ANALYTICAL TECHNIQUES IN ENVIRONMENTAL CHEMISTRY

BLOCK 1
Fundamentals of Environmental Chemistry

UNIT 1 : ENVIRONMENTAL CHEMISTRY-I

Structure

1.1 INTRODUCTION

Environmental Chemistry is traditionally outlined as "*The study of the origin roots, reactions, impacts, transport, and fortunes of chemical species, in water, soil, and air environments, and the human activity influence on these*". So according to this definition, the environmental chemistry is generally focusing on the pollutants in the various environments.

However, an updated definition would be "*The study of the chemical balance in natural systems and how the disturbance in this chemical balance happens by the anthropogenic activities by releasing of chemicals into the environment which change the natural ecosystem levels*". Environmental chemistry encompasses all branches of chemistry that are influenced on a regional basis by the loss of stratospheric ozone or global warming by urban air pollution or hazardous compounds that emerge from a chemical waste site, or on a global scale. Our courses concentrate on gaining a basic understanding of the nature of these chemical processes, so that the actions of humanity can be measured accurately.

1.2 OBJECTIVES

After studying this unit, you will be able to:

- Understand the concept and scope of environmental chemistry
- Explain chemical equilibrium
- Understand the concept of rate of reaction
- Explain chemical reactions involved in environments

1.3 CONCEPT AND SCOPE OF ENVIRONMENTAL CHEMISTRY

Environmental chemistry should never be synonymous with "green chemistry." The **green chemistry** aim is to implement chemical processes that use smaller amounts of harmless chemicals as well as less energy use to reduce their impact on the environment. While, the goal of *environmental chemistry* is to understand the chemical reactions and processes that regulate the environmental systems and how the introduction of anthropogenic chemicals affects them. This approach allows environmental chemistry to be constructive instead of reactive.

Formerly, environmental chemistry has had a more reactive approach, to precure and recognizing problems after the occurrence of a tragedy or a clearly obvious impact on the environment with loss of life, damage to plants and animals, or radical changes in environmental conditions – such as the loss of visibility in air and water systems or the depletion of the stratospheric ozone layer. So, "Almost everything that happens in world around us could come under the general heading **Environmental Chemistry".** Such as-

- Chemical reactions of all kind occur continuously in the atmosphere, in oceans, lakes and rivers, in all living things and even underneath the earth's crust.
- These reactions take place quite independently of human activities. In order to understand environmental concerns, we need to have awareness not only about what products are intentionally released into the atmosphere, but also of the process under way.
- The normal concepts must be understood so that accurate assumptions can be made about the potential effects of new but related substances.

Indeed, although convenient to segment topics for study purposes, **Barry Commoner's first law of the environment** should always keep in mind: **"Everything is related everything else."**

Some topics that are rather distant from equations and reactions should be discussed. That is-

- Knowledge in biological, meteorological, oceanographic, and other fields is equally important to the overall understanding of the environment.
- Conservation of environmental resources along with biological diversity.
- Pollution prevention.
- The social problems associated to growth and environment.
- Design of a renewable energy network that is not polluting.
- Everybody, whatever profession, is impacted by environmental problems such as global warming, ozone depletion, declining forests, energy resources, global biodiversity loss, etc.
- It authorizes us to create a secure, balanced and clean natural environment.
- Large-scale heavy metal land pollution by factories. So this can be passed to waterways and taken up by living beings.
- It also discusses important issues such as safe and fresh clean water, hygienic living standards, fresh and clean air, soil quality, nutritious food and development.
- Nutrients draining from farmland into waterways can contribute to algae blooms and topsoil erosion
- Organometallic compounds
- As new career prospects for environmental protection and sustainability, sustainable environmental law, business administration, environmental health, sustainability and environmental engineering are emerging.
- During rainstorms metropolitan drainage of pollutants rinsing off resistant surfaces (streets, rooftops and parking lots). Typical contaminants include gasoline, motor oil, and other hydrocarbon, metals, chemicals, and soil.

1.4 FUNDAMENTALS OF ELEMENTAL STOICHIOMETRY

A chemical reaction written in form of equation provides the idea of both qualitative and quantitative features. Where qualitative is more a theoretical approach but quantitative is explicitly a practical approach summarizing basic principle of chemistry viz. principle of mass conversion to produce the resultant drawn in product side.

$$CH_4(g) + O_2(g) \longrightarrow CO_2(g) + H_2O \quad \text{---------------} \quad (i)$$

$$CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O(l) \quad \text{-------------} \quad (ii)$$

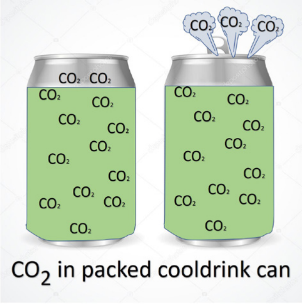
In equation (i) the reaction describes the formation of CO_2 and water from methane via the combustion process, but the quantity of elements in the reactant side and the product side are not equal, hence requires to poise the element proportion on the either side to determine the amount of each compound involved, which is defined as **Stoichiometry**.

Stoichiometrically balanced equation can be achieved via trial-and-error approach. As in the eqn. (i) **1 C-atom** approaches on the both side of the equation, but on the left side, there are **4H** atoms and only **2H** on the right, and to equalise the stoichiometry as in eqn. (ii), we have to double the number of water molecules so that we can get **4H** atoms in either sides. Now, while adjusting the elemental stoichiometry of hydrogen we by-chance changed the atom count for **oxygen** on left-hand side (LHS) and right-hand side (RHS) of the arrow. To normalize this issue, we can multiply the amount of molecular oxygen with 2 to get equal amount of oxygen atom on both the sides.

The **quantitative** idea of a balance equation is measured in either moles= Weighed Mass/ Molecular Weight (unit less quantity); Molarity = moles/volume (mol/L) and Weight by Volume (W/V): unit mg/L = Molarity (mol/L) × Molecular weight (g/mol) × 10^{-3} (mg/g).

1.5 CHEMICAL EQUILIBRIUM

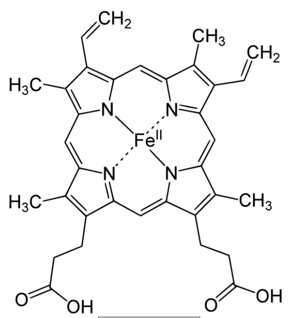
Chemical equilibrium defined as "A reaction is in chemical equilibrium if the forward reaction rate step equivalent to the reverse reaction rate step. a reaction exists in equilibrium, when the concentration of reactants and products are constant."



There are several examples of chemical equilibrium across us. One example is a fizzy cooldrink tin can or bottle. In the tin-can there is CO_2 thawed in the liquid. There is also CO_2 gas in the state between the liquid and the cap. There is a continual switch of CO_2 from liquid to gas phase, and from gas phase to liquid form. Though, if we glance at the tin can there does not seem to be any difference. The system is in equilibrium.

$$CO_2(g) + H_2O(l) \rightleftharpoons H_2CO_3(aq)$$

Without the chemical equilibrium it would not be possible to exist as we know it. Another sign of equilibrium in our daily lives occurs within our own bodies. Hemoglobin is a macromolecule which carries oxygen all over our bodies. We wouldn't survive without it. The hemoglobin must be capable of consuming oxygen, while also releasing it, and this is managed by adjustments in the steadiness of this reaction in distinct locations within our bodies.



$$\text{hemoglobin(aq)} + 4O_2(g) \rightleftharpoons \text{hemoglobin}(O_2)_4(aq)$$

Hemoglobin binds with oxygen within the red blood cells of the lungs. This *oxyhemoglobin* flows through the blood stream to cells in the body along with the red blood cells.

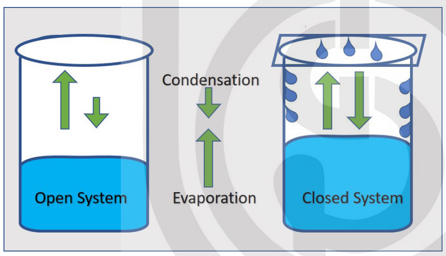
1.5.1 Open and Closed System

An **open system** allows the **energy** and **matter** to move within and outside of the system.

A **closed system** is one where only energy can travel inside and outside the system. Matter can't be obtained from the system or removed from it.

Liquid-gas phase equilibrium

Apparatus



2 beakers, glass sheet, water

Method

Fill two beakers with water in half, and in each case mark the water level.

One beaker covered with a glass sheet.

Leave the beakers for 2 -3 days, watch how the water levels in both beakers change.

Note: You could accelerate this experiment by putting the two beakers over a Bunsen burner, or by heating the water in direct sunlight.

Discussion

In the open beaker, due to evaporation, liquid water become vapors after heat, and the water level decreases. A small proportion of gas molecules may condense again, but condensation is far lower than evaporation, so, the vapors will escape to the atmosphere. The first beaker was an example of an open device in the liquid-gas demonstration, as the beaker could be heated or cooled (a variation in energy), and water vapor (the matter) would evaporate from the beaker.

Evaporation happens in the second beaker too. But at covered condition, the vapour reaches the glass cover sheet, where it cools where condenses again to become liquid water. It returns the water to the beaker. After start of condensation, the rate at which the water volume falls may begin to decrease. There will be no change in the water level at any stage, because the rate of evaporation will be equivalent to the rate of condensation. This represents **the closed System.** This can be indicated as-

Liquid $\rightleftharpoons$ vapour

Observations

Evaporation is the phenomena when a liquified constituent goes from the liquid stage to the gaseous stage. Condensation is the phenomena when transition of gaseous substance occurs from gas to liquid phase

You will find that; the water level falls faster because of evaporation in the open beaker than in the covered beaker. There is an initial drop in the water volume in the closed beaker, but evaporation tends to stop after a while and the water level in this covered beaker is higher than that in the open one. The reaction in this example will go on in both directions. There is a transition of liquid to water vapor in the forward direction. There may also be a reverse transition, as vapor condenses again to form liquid.

So, conclusion from above discussion is that, in a **closed system** it is apparent for reactions to be reversible and to reach at **equilibrium**.

1.5.2 Reversible Reactions

There are certain reactions which can occur in two directions. First direction is **forward reaction** in which reactants come together to shape the products, while other direction is **reversible direction** and vice versa in which the products again form the reactants. To display this form of reversible reaction, a special double-headed arrow ($\leftrightarrows$) is used.

$XY+Z \leftrightarrows X+YZ$

So, in reversible reaction-

$H_2(g)+I_2(g) \leftrightarrows 2HI(g)$ is the **forward** reaction.

The forward reaction always written **left** to **right** from the reaction.

$2HI(g)\rightarrow H_2(g)+I_2(g)$ is the **reverse** reaction.

The reverse reaction always written **right** to **left** of the given equation.

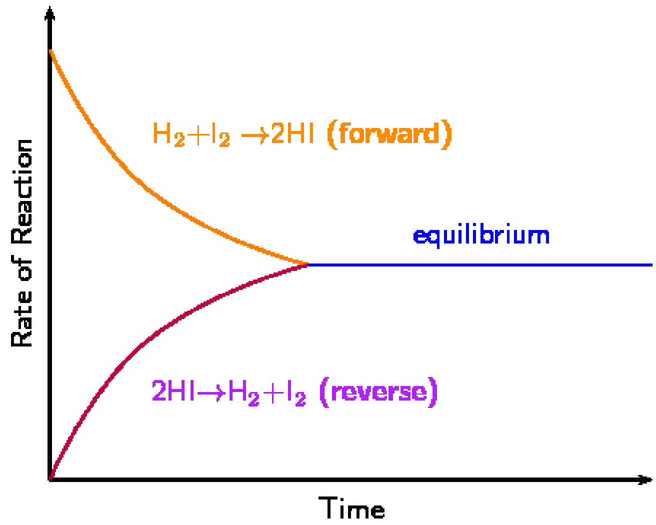


Figure 1.1 The change in rate of forward and reverse reactions in a closed system

In any reaction, the product is formed more slowly as reactant amount decreased. In a reversible reaction, the reactant is formed rapidly as the product quantity increases. Ultimately, forward reaction rate (reactants → products) equals to reverse reaction rate (products → reactants). at this moment, there are still reactants present, but no more progresses seem to be happening. The reaction is said to be in chemical equilibrium.

Although no macroscopic changes can always be detected, this doesn't mean the reaction has ended. The forward and reverse reactions tend to occur and so there are still microscopic changes in the system. This state is called **dynamic equilibrium**.

Reactions which are going to be completed are *irreversible*. However, in some reactions the reactants form products (in a forward reaction), and the products can change back into reactants (in a reverse reaction).

SAQ 1

What is the traditional definition of environmental chemistry?

1.6 CHEMICAL POTENTIAL

In the year 1875 J. Gibbs coined the word "chemical potential". Chemical potential is a concept of thermodynamics, common to all, not only in the study of materials but also in physical, chemical engineering and biology background. It is a key concept because it is possible to achieve all the thermodynamically impressions of a substance at a given temperature and pressure from knowledge of its chemical potential. Chemical potential establishes the stability of materials under the constant pressure and temperature, such as chemical products, and solutions, and their capacity to react chemically to form new materials, to turn into another physical states, or to move from one space position to another.

There are few important points related to Chemical Potential. Such as-

- Chemical potential energy mainly depends on the strength of the bonds. Strong bonds have low energy and weak ones have high energy. Large quantities of heat and/or light energy can be released during strong bonds are forming. In other terms, chemical potential is correlated with the system's electronegativity.

  Relation between chemical potential and electronegativity:

  $$\mu = -\chi \text{ - (1)}$$

  where, μ = Chemical potential
  χ = Electronegativity

- The stored energy in food molecules are preserved in form of **chemical potential energy**, that can-do work in the future. When our body breaks molecules for catabolism, the released energy can be used as work. But potential energy can't preserve only in food molecules.

- The value of the chemical potential of a substance depends upon both temperature and pressure. Additionally, the potential will increase when the pressure increase.

- Intensive properties are substance independent and depend entirely on material. Some intensive properties are like chemical potential temperature, density, refractive index, etc.

- Chemical energy is the capability of a chemicals to undergo a chemical reaction to turn it into other substances. Examples include food, batteries, fuel, etc. Breaking or creating of chemical bonds requires energy, which may be either evolved or absorbed from a chemical system.

- Chemical energy is a type of potential energy; however, kinetic energy is the energy in motion. Chemical energy deals with the energy stored in the bonds between atoms.

- Chemical potential increases with the density "n" of particles. Particles flow from high 'n' systems to low 'n' systems. The chemical potential of an ideal gas is always negative.

1.7 CHEMICAL KINETICS

Chemical kinetics is the evaluation of the rates and the mechanisms of chemical processes. This is of great practical importance. It is also important to know under what conditions a slow but valuable reaction can be made to proceed rapidly to produce a desired high yield product. For environmental chemistry, it has great predictive value.

By observing the kinetics and environmental reaction mechanisms, we can at least estimate approximately the residence times of both polluting and non-polluting organisms. Our knowledge of various pollutants and atmospheric phenomena such as depletion of ozone, acid rain, photochemical smog etc. is focused primarily on the study of the kinetics of individual rate constants of thermal and photochemical reactions.

1.7.1 Kinetics of Reactions of Different Orders

Zero order reactions occur at a constant rate, and the magnitude of the rate does not change with time. The concentration plot against time is linear, and the slope of this line is equal to the zero-order rate constant. The differential of rate equation and its integrated forms is in Eq1.

$$-d[A]/dt = k = xt \quad -(2)$$

where k is the zero-order rate constant, and x is the amount reacted in time, t.

For a **first order reaction***, the rate depends on the first powe*r *reactant concentration*. Consider the rate of a first order reaction, defined by Eq-3.

$$-\ d[A]/dt = k[A] \quad -(3)$$

On integrating Eq. 3, we obtain the following forms of rate law for a first order reaction.

(i) $\log \{[A]o/[A]t\} = 2.303kt$

(ii) $[A]t = [A]o\ e\text{-}kt$

(iii) $\log \{a/(a-x)\} = 2.303kt \quad -(4)$

where $[A]_o$ = initial concentration of the reactant, A = a and $[A]_t$ = concentration at time t = (a-x), and concentration of the reactant reacted in t time = x.

The unit of the first order rate constant is s^{-1}. It is independent of the unit of concentration used.

A **second order reaction** is one in which the rate depends on *(i) the concentration of one reactant raised to power two* or (ii*) on the concentrations of two reactants each raised to the first power* as shown in the rate laws (5) and (6).

$$\text{Rate} = k\ [A]^2 \quad -(5)$$

$$\text{Rate} = k[A][B] \quad -(6)$$

The Eq.6 changes into Eq. 5, if the initial concentrations of both A and B are taken equal. So when [A] = [B], Eq. 6 becomes same as Eq. 5: rate = $k[A][A] = k[A]^2$.

On integrating Eq. 5, we get

$$k = (1/[A]_t - 1/[A]_o)$$

$$k = (1/t)(\ x/\{a(a-x)\} \quad -(7)$$

In Eq. 7, the symbols have same meaning as in case of first order reaction. The unit of a second order rate constant is $L.mol^{-1}.s^{-1}$.

In ***third order reactions***, the rate expressions are of the type:

$$\text{Rate} = k\,[A]^3$$

$$\text{Rate} = k[A][B]^2 \text{ and}$$

$$\text{Rate} = k\,[A]\,[B]\,[C].$$

Under the condition, [A] = [B] = [C], rate laws (ii) and (iii) will also reduce to rate law (i). The integrated form of the rate law: rate = k $[A]^3$ is Eq. 8.

$$k = (1/2t)\,(1/[A]t^2 - 1/[A]_o^{\,2}) \quad \text{-(8)}$$

The atmospheric oxidation of nitrous acid by O_2 ($2HNO_2 + O_2 \rightarrow 2HNO_3$) obeys the third order rate law: $-d[O_2]/dt = k\,[HNO_2][O_2]^2$. The unit of third order rate constant is $L^2 mol^{-2} s^{-1}$.

For **Pseudo-order reactions**, because of the experimental conditions used, the order of the chemical reaction tends to be less than the true order. Pseudo-orders arise when there is a significant excess of one or more reactants relative to others. For example, the atmospheric reaction: $NO + O_3 \rightarrow NO_2 + O_2$, obeyed the rate law:

$$-d[O_2]/dt = k\,[NO]\,[O_3]. \quad \text{-(9)}$$

Under the condition $[O_3]$ in large excess (≥10 times) over [NO], up to the end of reaction there would hardly be any significant change in $[O_3]$ and so $[O_3]$ would remain virtually constant. The kinetics will then obey the rate law:

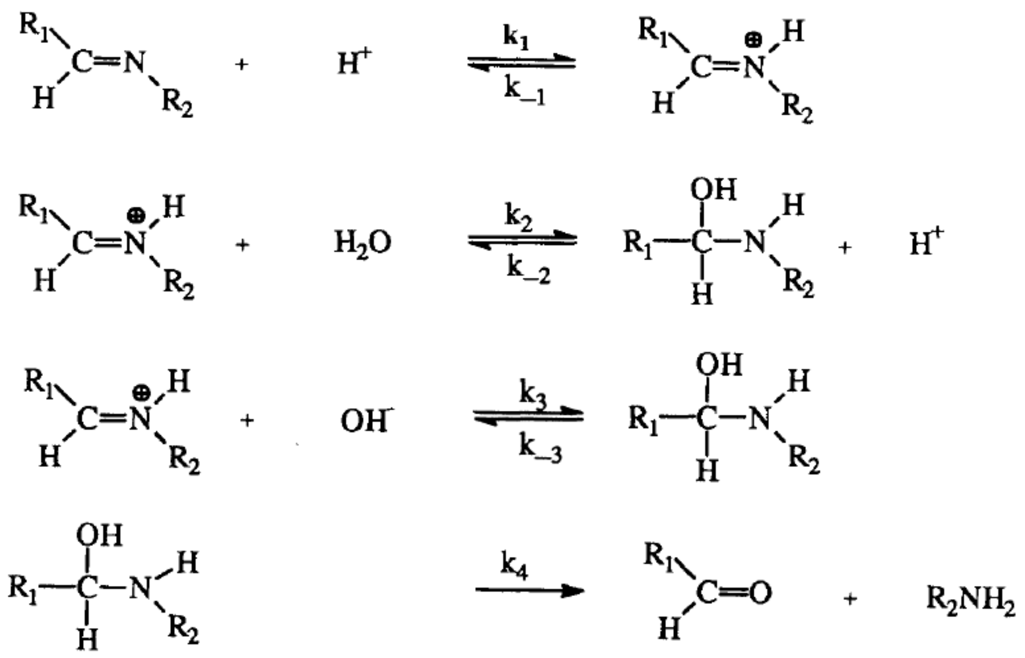
$$-d[O_2]/dt = k_1[NO] \quad \text{-(10)}$$

where k_1 = k[NO]. So, the apparent order in O_3 during the reaction will be zero. The reaction will, therefore, will follow a first order kinetics in [NO].

1.8 SIMPLE REACTION MECHANISMS

The transformation of organic chemicals most often occurs in several molecular events referred to as ***elementary reactions***. An elementary reaction is described as a process in which reacting chemical species pass via a single transition state without the intervention of an intermediate. A sequence of individual elementary reaction steps constitutes a ***reaction mechanism***. For example, the overall reaction for the hydrolysis of a Schiff base is written:

The reaction mechanism for this seemingly simple reaction is in fact quite complex and is composed of four elementary reactions (Cordes and Jencks, 1963).



A reaction mechanism is a hypothesis or model that has been conceived from experimental evidence. As new experimental evidence is obtained, changes in the proposed reaction mechanism may be required. Chemical kinetics is probably the most powerful tool for the investigation and development of reaction mechanisms. The consistency of a reaction mechanism can be verified in the laboratory by determining the dependence of reaction rates on concentration.

It is useful to classify elementary reactions as per their molecularity and described as the sum of the exponents that appear for one single elementary reaction in a rate equation.

1.9 ORDER AND MOLECULARITY OF CHEMICAL REACTIONS

1.9.1 Reaction Order

The **reaction order relating to a chemical species** *is defined as the order of the concentration in the differential form of that reactant species of the rate law,* and **the order of a *reaction*** *is defined as sum of the powers of all the concentration terms that occur in such a rate law.* The experimental kinetics rate law for the important atmospheric reaction (11) is Eq. 8.

$$2NO + O_2 = 2NO_2 \quad \text{-(11)}$$

$$- d[O_2]dt = k[NO]^2[O_2] \quad \text{-(12)}$$

The Eq. 12 shows the order with respect to NO is two and with respect to O_2 is one, and the order of reaction to be three (i.e., the aggregate of the orders of NO and O_2).

Consider the aqueous phase atmospheric Fe^{3+}-catalyzed oxidation of dissolved sulfur dioxide, generally written as sulfur (IV), by O_2. It obeys the rate law:

$$dS(IV)dt = k[Fe(III)][S(IV)] / [H^+]. \quad -(13)$$

the reaction is first order in each of Fe(III) and S(IV) and of negative first order in $[H^+]$. So, order can be negative, and fractional also zero.

1.9.2 Molecularity of the Reaction

Reactions typically occur in more than one step, so each reaction is comprised of a series of elementary reactions. *The molecularity is the number of reactant molecules, ions, radicals or any other chemical species in an elementary step.* A reaction may have two or more steps and then the molecularity of each step would depend on the number of reactant molecules involved in that step.

Unimolecular Reactions require only one molecule of reactants in the reaction. Example-The photodissociation of ozone in troposphere.

$$O_3 + h\nu\ (290\text{ nm} < \lambda < 350\text{ nm}) \rightarrow O + O_2 \quad -(14)$$

Bimolecular Reactions involve two reactant species. These are two-body reactions. Examples-The formation of the hydroxyl radical, (OH-radical is most import in atmospheric chemistry).

$$O(^1D) + H_2O \rightarrow 2OH^{\cdot} \quad -(15)$$

Termolecular (or Trimolecular) Reactions involve three reactants. The formation of carbon dioxide, formation of ozone in troposphere are classical examples of this type.

$$2CO + O_2 \rightarrow 2CO_2 \quad -(16)$$

SAQ 2

What is the source of electrons in natural reducing environments?

....

....

What is the kinetic of chemical reaction in different orders?

....

....

1.10 CHEMICAL REACTIONS

There are several chemical reactions involved in environments. Among them, highly involved reactions are given below-

1.10.1 Hydrolysis

Hydrolysis is an example of nucleophilic displacement reactions in which a **nucleophile** (*an electron-rich species containing an unshared pair of electrons*) attacks an **electrophilic** atom (*an electron-deficient reaction center*).

Hydrolytic processes are not limited to the bodies of water such as rivers, streams, lakes, and oceans usually associated with the term aquatic ecosystems. Hydrolysis of organic chemicals also can occur in fog water, biological systems, groundwater systems, and the aqueous microenvironment associated with soils and sediments.

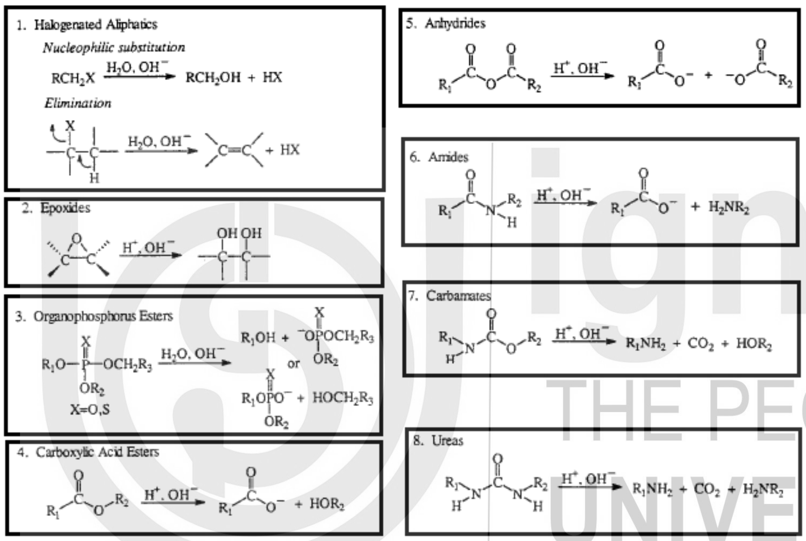


Figure 1.2 Hydrolysis reactions involved in our ecosystems

Hydrolysis is described as a chemical transition in which an organic molecule, RX, reacts with water, resulting in the forming of a new covalent bond with OH and a cleavage in the original molecule of the covalent bond with X (the leaving group). The resulting reaction is that X is substituted by OH~ (Harris, 1981; Mill and Mabey, 1988)

$$RX + H_2O \rightarrow ROH + X^- + H^+ \quad -(17)$$

Hydrolysis Kinetics depends on-

1. Specific Acid and Base Catalysis
2. pH Dependence

1.10.2 Reduction

Reducing environments abound in nature (soils and waters, aquatic system, sewage sludge, hypolimnia of stratified lakes, and oxygen-free areas of eutrophic rivers);

until recently, however, very few investigations of organic reactions characteristic of these systems have been undertaken.

The realization that pollutants, which are normally considered persistent in aerobic environments, may not be nearly as persistent in a reducing environment has generated a great deal of interest in the behaviour of organic chemicals in anaerobic environments. The elucidation of reductive transformation pathways may lead to the development of remediation technologies for the removal of organic pollutants from contaminated ecosystems.

Reduction, defined as a gain of electrons, occurs when there is a transfer of electrons from an electron donor "or reductant" to an electron acceptor "or oxidant". The oxidant in this case is the organic chemical or pollutant of interest. Reductive (and oxidative) transformations can be distinguished from other processes (e.g., hydrolysis) by determining if a change in the oxidation state of the atoms involved in the reaction process has occurred.

For example, the transformation of 1,1,2,2-tetrachloroethane can result in the formation of two reaction products, 1,1,2-trichloroethylene and 1,2-dichloroethylene. Although our understanding of reductive transformations in the environment has progressed to the point that we can identify the types of functional groups that will be susceptible to reduction in the environmental systems, our **limited** considerate of reaction mechanisms for such transformations currently is a barrier to the prediction of absolute reduction rates, and how reaction rates will vary from one environmental system to the next.

The noticeable question arises here is, **"What is the source of electrons in natural reducing environments?"** The cumulative knowledge in this area suggests that naturally occurring reductants are not limited to two or three reactive species, as is the case for hydrolysis, but that a complex array of species is involved, ranging from chemical or "abiotic" reagents such as sulfide minerals, reduced metals, and natural organic matter, through extracellular biochemical reductants such as iron porphyrins, corrinoids, and bacterial transition-metal coenzymes, to biological systems such as microbial populations. Furthermore, the relationship between these various reductants in natural systems is most likely quite complex. For example, chemical species such as reduced metals and sulfide ion may be the direct result of microbial metabolism.

Reductive transformations are most suitably classified according to the type of functional group that is reduced. General schemes explaining the reductive transformations that are known to occur in natural reducing environments are summarized-

1.10.2.1 Reductive Dehalogenation

The reduction of halogenated aliphatic and aromatic compounds chemicals has been observed in a variety of environmental and laboratory systems, including anaerobic sediments, soils, anaerobic sewage sludge, groundwaters, aquifer materials, reduced iron porphyrin systems, bacterial transition-metal coenzymes, and mineral sulphides, and various chemical reducing agents. Reductive dehalogenation is a detoxification process. The motivation to understand the reaction pathways of halogenated chemicals in these types of systems results from their widespread contamination of sediments, soils and groundwaters due to their frequent use as agrochemicals, solvents, and synthetic intermediates.

1.10.2.2 Nitroaromatic Reduction

The reduction of nitroaromatics in anaerobic systems has received considerable attention because of their importance as agrochemicals, munitions, textile dyes, and dye intermediates. The reaction products resulting from the reduction of nitroaromatic compounds are aromatic amines. The formation of aromatic amines occurs through a series of electron transfer reactions with nitroso compounds and hydroxylamines as even-electron intermediates. Pentachloronitrobenzene (PCNB) is another example of a nitroaromatic agrochemical that is known to undergo facile reduction in anaerobic systems. PCNB has been identified as a pollutant in river water and groundwater. The reduction of trifluralin has been observed in sediments and flooded soils. Reduction of nitroaromatics can occur through abiotic pathways. The reduction of TNT and DNT, as well as other polynitroaromatics, has been studied in some detail in anaerobic microbial enzyme systems. The reduction of 2-bromo-4,6-dinitroaniline, an important intermediate in the preparation of textile dyes that has been detected in river waters, in anaerobic sediment-water systems.

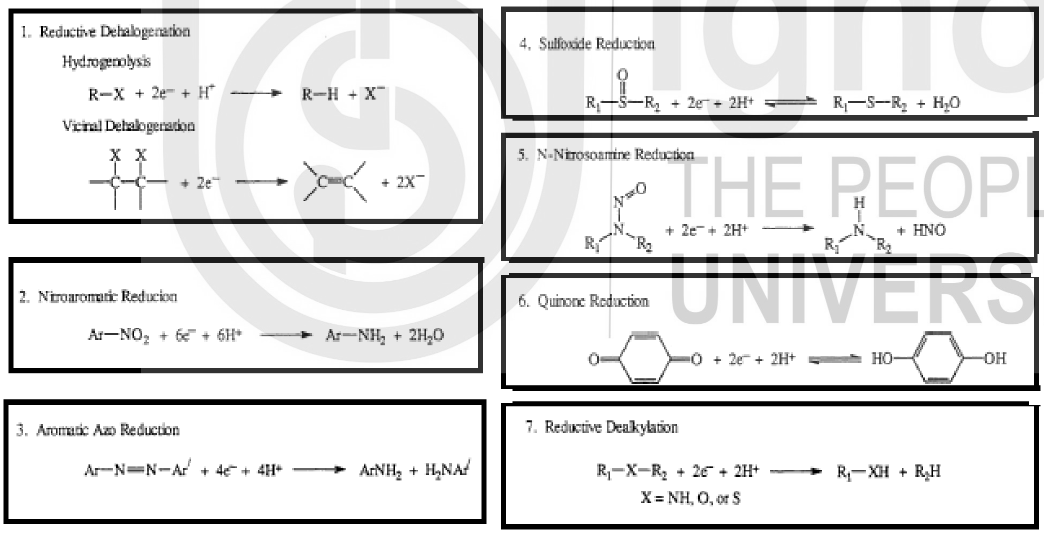


Table 3.1. Reductive Transformations Known to Occur in Natural Reducing Environments

Figure 1.3 Reduction reactions involved in environment systems

1.10.2.3 Aromatic Azo Reduction

Concern over the environmental status of aromatic azo compounds derives largely from their significance in the textile dye industry. Around 50 % of the dyes available on the market today are azo compounds. About 12 % of the synthetic fibre dyes produced each year are expected to be lost in waste streams by production and recycling processes, and 20 % of such loses would reach the atmosphere by wastewater treatment system effluents. The reductive cleavage of the azo linkage of the dyestuffs in aquatic ecosystems represents a possible pathway for the entry of aromatic amines into these systems. The aromatic amines are considered a hazardous class of compounds because of their mutagenic and carcinogenic properties.

1.10.2.4 *N*-Nitrosamine Reduction

The N-nitrosoamines are also a highly studied and important class of chemicals due to their mutagenic and carcinogenic properties. Much of this work has suggested that the formation of N-nitrosoamines may occur in crop soils and in foodstuffs that have been treated with agrochemicals containing secondary nitrogens (e.g, atrazine) and heavy applications of nitrogen fertilizers. Several N-nitrosoamines also have important uses in industrial processes. For example, N-nitrosodiphenylamine is extensively applied in the rubber industry to delay vulcanization.

1.10.2.5 Sulfoxide Reduction

The environmental significance of organic sulfoxides results primarily from their importance as agrochemicals; specifically, the organophosphorus and carbamate insecticides containing sulfoxide moieties. The reduction of sulfoxides is a two-electron transfer process that results in the formation of thioethers. Because the reduction of sulfoxides to thioethers is reversible, reoxidation of the thioether to the sulfoxide may occur in aerobic environments. Although oxidation of sulfoxides to sulfones (RSO_2R) has been observed in aerobic systems, no evidence exists that suggests reduction of the sulfone functional group to the sulfoxide moiety will occur in anaerobic systems. The chemical reduction of sulfones in the laboratory also is known to be quite difficult.

1.10.2.6 Quinone Reduction

Quinones are reduced by a one-electron transfer reaction to the radical. Transfer of the second electron results in formation of the hydroquinone. Because this reaction is readily reversible, the equilibrium between the quinone, semiquinone, and hydroquinone will be dependent on the reduction potential of the system of interest.

1.10.2.7 Reductive Dealkylation

Reductive dealkylation is the replacement of an alkyl group on a heteroatom with hydrogen. Although dealkylation may be considered a relatively minor structural change for many organic chemicals, the "unmasking" of a reactive functional group, such as an amino or hydroxyl group, can significantly alter a chemical's behaviour in environmental systems. Dealkylation appears to be a common transformation pathway for agrochemicals with the requisite functionality.

1.10.3 OXIDATION

Oxidation is defined as a loss of electrons. Oxidizing agents gain electrons and are electrophiles. In organic chemistry, oxidation can either be associated with the introduction of oxygen into a molecule or the conversion of a molecule to a higher oxidation state. For example, in Equation 18.

$$RCH_{\cdot}\text{-->}\ R\text{-}CH_{\cdot}\text{-}OH\ \text{->}R\text{-}CHO\ \text{-(18)}$$

The first step in the sequence is the incorporation of oxygen by a formal "oxygen atom donor," whereas the second is formally a dehydrogenation, or oxidation of the carbon atom to a higher formal oxidation state. Almost all oxidations are kinetically second-order reactions rate is proportionate to the concentrations of both the oxidizing agent. Ox, and the substrate, A (Equation 19).

$$-dA/dt = k [Ox] [A] \quad -(19)$$

For reactions in water, there are few values for the rate constant, k, and few data on the concentrations of potential oxidants.

In environmental chemistry, there are many sources of oxidants whose importance is highly variable due to changes in concentration or reactivity as one moves from one region of the environment to another. A potent oxidant such as the hydroxyl radical, *OH, that is of critical importance for organic transformations in the gaseous stage or during combustion, may not be important at all in another environmental compartment such as the soil. This could either be due to inadequate production rates for the species or to rapid side reactions that diminish its steady-state concentration. For the oxidation reaction, Oxygen is present as molecular oxygen (like autooxidation, superoxides, singlet oxygen, or in the form of ozone), Hydrogen peroxides decay products (like hydroxyl radical, peroxy radical, alkoxy and phenoxy radicals), clay and metal oxides, and thermal oxidation in the environments. These considerations will be addressed as each oxidizing species is discussed in turn for the remainder of the chapter.

A. Molecular Oxygen
   a. Autooxidation
   b. Superoxide
   c. Singlet oxygen
   d. Ozone and Related Compounds: Photochemical Smog

B. Hydrogen peroxide and its decay products
   a. H2O2
   b. Hydroxyl radical
   c. Peroxy radical
   d. Alkoxy and Phenoxy radical

C. Surface Reactions
   a. Clay
   b. Silicon oxides
   c. Aluminium oxides
   d. Iron oxides
   e. Manganese oxides

D. Thermal Oxidations
   a. Combustion
   b. Incineration

REACTIONS WITH DISINFECTANTS

Water disinfection has been quite successful in practically eliminating many acute waterborne diseases from the countries where it has been widely practiced, but its possible roles in the development of chronic health disorders, such as cancer, have not been fully scrutinized. Potential biohazards arise during drinking water treatment and disinfecting agents are used for the treatment of industrial wastewaters to remove undesirable colors, odors, and toxic organic compounds as well as to reduce the number of potentially harmful microorganisms in the effluent. Because of the higher concentrations of organic matter typically found in these waters, the likelihood of organic-disinfectant reactions occurring at significant rates is enhanced.

A. Free aqueous chlorine (HOCI)
   1. Chlorine in Water
   2. Oxidation Reactions
   3. Substitution and Addition Reactions
   4. Enolizable Carbonyl Compounds: the Haloform Reaction

B. Combined aqueous chlorine (chloramines)

C. Ozone

D. Chlorine dioxide

ENVIRONMENTAL PHOTOCHEMISTRY

The uptake of light energy (quanta) by organic compounds may cause subsequent photophysical or photochemical events to occur. Photophysical processes include emission of radiant energy (light or heat), whereas photochemical changes produce new compounds by transformations that include isomerization, bond cleavage, rearrangement, or intermolecular chemical reactions. In the environment, photochemical reactions have been reported to occur in the gas phase (troposphere, stratosphere), aqueous phase (atmospheric aerosols or droplets, surface waters, land-water interfaces) and in the solid phase (plant tissue exteriors, soil and mineral surfaces). All these possibilities need to be considered when fates of organic compounds in nature are considered. Although in many cases the contribution of photochemical processes is negligible, in others photolysis is by far the dominant pathway for loss of a chemical. There are several definitions to understand about photochemistry. Such as-

Internal conversion, thermal reversion of the excited singlet to the ground state with the release of heat to surrounding molecules such as the solvent. Because these two states are of like multiplicity, the transformation is allowed in terms of quantum theory and is often a very favourable process with a rate constant close to diffusion control.

Fluorescence, emission of visible or ultraviolet radiation. The emitted photon is always of a longer wavelength than the absorbed photon. This is also a quantum mechanically allowed process and usually occurs rapidly.

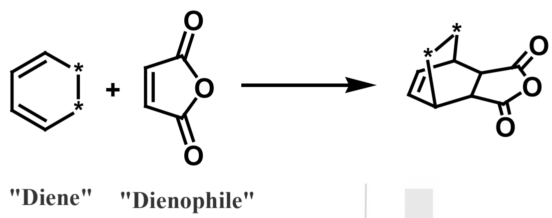
Intersystem crossing, a quantum-forbidden transition that produces a new excited state with unpaired electrons, a triplet. These transitions are usually 100 or more times slower than diffusion control. Triplet states have some degree of diradical character.

Quantum Yield: The quantum yield, of any photochemically induced process is defined as that fraction of the species converted to an excited state that undergoes a fate. Conversion of a triplet to the corresponding ground-state singlet (without the intervention of a quencher) is another quantum-mechanically forbidden and usually inefficient ISC process, called **phosphorescence** when it is accompanied by emission of radiation. Phosphorescence and fluorescence are examples of luminescence phenomena.

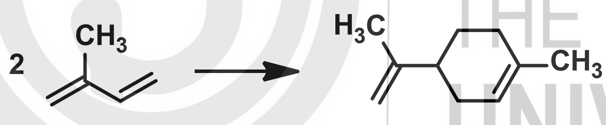
Chemical reaction involved. Because of the low instantaneous concentrations of most excited singlets, bimolecular chemical transformations involving them (such as reaction with a solvent) are not usually important unless they are kinetically very fast. Normally, one of the above processes predominates.

The Diels-Alder Reaction

The Diels-Alder reaction, a classic of synthetic organic chemistry, is theoretically a 4 + 2 cycloaddition in which a diene and a “dienophile” with very different electron abundances combine to form a new six-membered ring. The reaction has great synthetic usefulness and has attracted many researchers interested in its practical applications and mechanistic aspects.



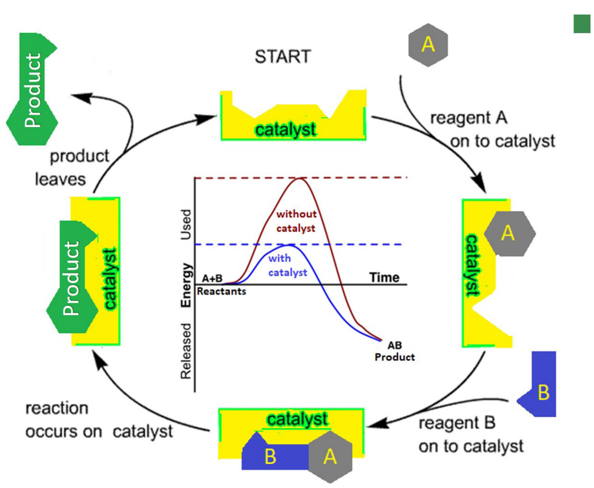
It is not at all certain whether any Diels-Alder reaction occurs under environmental conditions, although it has been suggested as a mechanism for the formation of *dl*-limonene from polyisoprene in the pyrolysis of waste rubber tire material (Equation 7.2; Pakdel et al., 1991).



Berzelius coined the term “catalysis” as early as 1836 to describe different reactions to decomposition and transformation that involve the catalyst and its environmental aspects.

1.11 CATALYSIS

Catalysis is the process where a very little quantity of foreign material, called a catalyst, raises a chemical reaction rate without being absorbed by itself. Catalysis is environmentally beneficial and plays a essential role in the production of new and existing chemicals. In the catalysed reactions, less energy is needed for production, and higher efficiency is observed in



generating less by-products, co-products and other waste substances. Catalysts can be configured to be safe to the environment. Currently, a variety of catalysts are used in industrial manufacturing, and several have advantageous impacts. The green synthesis uses catalysts that can be recycled and reused to drive the reaction rather than the additional reagents used for the original process, which affected waste to accumulate at each stage. Any catalytic reaction that generates no waste will sustain the atoms. A catalyst allows the reaction rate to increase, and we can suggest that the catalyst speed up the chemical reaction. The catalysts are the substances that expedite a reaction process without being absorbed and undergoing any chemical changes. The catalysts though participate in the reaction and act by providing a route of lower activation energy. The catalysts are regenerated at the end of the reaction. For example, the stratospheric ozone depletion is due to catalytic action of species such as Cl, NO etc., as in the reactions:

$$Cl + O_3 \longrightarrow ClO + O_2$$
$$ClO + O \longrightarrow Cl + O_2$$

In the above cycle, a single Cl can destroy thousands of O_3 molecules. Another important example is the iron-catalyzed oxidation of dissolved SO_2 in rainwater.

Another Example: When $KClO_3$ is heated at very high temperatures, the emitted oxygen gas will escape very gradually, but if MnO_2 is mixed up in tiny quantities then oxygen gas will be produced rapidly even at normal temperatures. In the first case Cl is serving as a catalyst and in another case MnO_2 is serving as a catalyst.

Types of catalysis

There are two types of catalysis:

(i) **Homogeneous catalysis:** catalyst and reactants are in the same physical state

(ii) **Heterogeneous catalysis:** catalyst and reactants both are in different physical states

Types of catalyst

There are various types of catalyst but usually, there are four types which are considerable and important-

(i) **Positive catalyst**: The catalyst which stimulates the rate of reaction

(ii) **Negative catalyst**: The catalyst which turn off the rate of reaction

(iii) **Auto catalyst**: If any substance is produced during any chemical reaction, will work as catalyst for the *same* chemical reaction.

(iv) **Induced catalyst**: If any substance is produced during any chemical reaction, will work as catalyst for *some other* chemical reaction.

Surface and Aqueous Catalysis

Clay Catalyzed Reactions or rearrangements:

Chemicals play an important role in contributing pollution; hence retrievable and reusable catalysts are prime demand where clays and zeolites are noteworthy. Clay were reintroduced in multiple reaction strategies like hydroalkoxylation in ether formation, esterification etc. Modified clays have also been used in oxidation and formylation reactions. The divergence of clay catalyst was identified when they acted superbly with rearrangement reactions starting from pinacole -pinacolone, to isomerization, to Claisen rearrangement.

1.12 ADSORPTION IN CATALYSIS

A global environmental issue raises due to the toxic effects of chemicals, and heavy metal contamination. At varying concentrations these contaminants are found dissolved in water and wastewater. Because of the large quantities of coloring required in clothing, paper product, dyes are identified as one of the heavy pollutants of water bodies. paints and plastics. Potential risks in drinking water caused by heavy metals and other anions have raised public concern. Generally, the intake of metal ions and anions causes chronic effects, and humans can experience long-term exposure to such pollutants without knowing it.

Various physical and chemical approaches have been used to remove contaminants from polluted water. Adsorption was proposed as cheaper and more effective than chemical or physical techniques between all these approaches. Due to their reasonably easy design, activity, cost efficiency and energy savings these are favoured over other methods. Many components were effectively used in means of staining of water contaminants, such as microspheres, clays, composites clay-alginate, activated carbon chitosan, metal oxides, etc. The main classes of adsorbents include mesoporous silica, activated charcoal, bagasse, zeolite, metal oxides, alumina, carbon nanotubes, clay, activated carbon, and polymeric resin, etc.

These catalysts demonstrate the value of adsorption as catalytic properties. This type of catalysis is ecofriendly as it escalates the reaction rate without being utilized by itself and therefore the chemical and energy industries depend heavily on catalysis as a result. The popular Haber-Bosch method for ammonia generation uses metal-based catalyst for adsorption of Nitrogen and hydrogen gas and synthesis of ammonia industrially.

Figure 1.4 Different types of adsorbents

Classification of Adsorbent

Adsorbent can be categorized into three groups:

(a) Nonpolar solids:
Most of the molecular solids are non-polar. Such non-polar molecular solids do not dissolve in water. Where the adsorption is mainly physical e.g., activated charcoal.

(b) Polar solids:
These materials adsorb both polar and nonpolar molecules. However, they show preference with polar molecules. When the adsorption is chemical e.g. silica and aluminium oxides.

(c) Chemical Adsorbing surfaces which adsorb the molecules and subsequently release them after reaction, which may be either show catalytic approach by leaving the surface unchanged, or non-catalytic by replacing the surface atoms.

Adsorbent can also be classified as conventional or non-conventional adsorbents. Activated carbon, polymeric carbon, aluminous, silica gel, bauxite etc categorized as **Convention adsorbents,** while **Non-conventional adsorbents** include, hardwood, lignite, fly-ash, sawdust etc.

Mechanism of Adsorption
Adsorption is a simplistic process in which adsorptive molecules (gas or liquid) bind to a solid surface. In fact, adsorption is implemented as an operation in a column filled with porous sorbents, whether in batch or continuous mode. In these cases, the consequences of mass transfer are inevitable. The three common steps include: Film diffusion (external diffusion), Pore diffusion [intraparticle diffusion (IPD)] and Surface reaction.

Factors affecting adsorption
Some of the significant variable affecting adsorption are nature of the adsorbate and adsorbent, diffusion through the passage with in the catalyst particles (either macropores or micropores), surface area of adsorbent, Activation of adsorbent, rate of air flow, vapour concentration, humidity, temperature, degree of regeneration, contact time, processing method adopted etc. The slowest of these steps will determine the rate for the whole process and is called rate determining step.

Equilibrium Adsorption Isotherms
The equilibrium properties of a gas-solid system are defined by a curve of the concentration of adsorbed gas on the solid at constant temperature as a function of the equilibrium partial pressure of the gas. Such a curve is considered as an adsorption isotherm. Table shows the most well-known Isotherms which indicate the adsorption heat dependence on surface coverage implied by increasing isotherm. There are several more isotherms available according to individual requirements.

Isotherm	Isotherm equation	Uses
Henry's	$q_e = K_{HE} C_e$	One-Parameter Isotherm Simplest adsorption isotherm

Langmuir	$\frac{C_e}{q_e}=\frac{1}{q_m k_L}+\frac{C_e}{q_m}$ $R_L=\frac{1}{1+k_L C_o}$	Homogeneous binding sites (same affinities), alike sorption energies, and no interactions between adsorbed species. RL is separation aspect which decides whether the adsorption is un-favorable when its value is >1, linear (=1), favourable (0 < RL < 1), and irreversible (RL = 0)
Freundlich	$q_e = k_f c_e^{1/n}$	Heterogonous surfaces (varied affinities), 1/n (adsorption intensity) that specifies the energy and the heterogeneity of the adsorbent sites (1/n < 1= Langmuir, 1/n > 1 cooperative adsorption
Temkin	$q_e=\frac{RT}{b}(k_T c_e)$	Effective only for an intermediate range of adsorbate concentrations and gives information for adsorbate/adsorbate interactions, where b (J/mol) is Temkin isotherm constant k_T (L/g)—Temkin isotherm equilibrium binding constant
Flory-Huggins	$\ln\left(\frac{\theta}{C_o}\right)=n\ln(1-\theta)+\ln k_{FH}$ $\Delta G^o=-RT\ln(k_{FH})$ $\ln\left(\frac{1000*q_e}{C_e}\right)=\frac{\Delta s^o}{R}-\frac{\Delta H^o}{RT}$	Account for the characteristic surface coverage of the adsorbed adsorbate on the adsorbent and the spontaneity of the process using ΔGo value obtained from K_{FH}, where, n is number of adsorbates occupying adsorption sites, and K_{FH} is Flory-Huggins equilibrium constant ($L.mol^{-1}$)
Elovich	$\ln\frac{q_e}{c_e}=\ln k_E q_m-\frac{q_e}{q_m}$	Define the kinetics of chemisorption and Multilayer adsorption, where KE is equilibrium constant ($L.mg^{-1}$) and q_m is maximum adsorption capacity ($mg.g^{-1}$)

SAQ- 3

What is the difference between adsorption and absorption? Explain with example?

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Explain the Catalytic cycle with figure.

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1.13 LET US SUM UP

Environmental chemistry focuses on the nature and effect of chemicals in groundwater, surface water and soil. We will learn in the Environmental Chemistry how chemicals-generally pollutants-pass through the environment. We are also studying the impacts of these pollutants on habitats, animals and human health. Environmental chemistry advises on the movement and effect of soil and groundwater pollutants, evaluate long-term threats to ecological and human health, apply for environmental permits to carry out corrective strategies, identify polluted soils as hazardous waste and control their disposal, and monitor on-site remediation. What type of reactions are happening during chemical processes and its effect on environment? After reading contents of this unit, we will observe how chemical reactions are involved in our surroundings. What type of reaction order and molecularity, chemical kinetics and potential are involved? Reactions are going on one direction or reversible.

1.14 KEY WORDS

Anthropogenic Pollution

It is defined as the introduction of harmful substances or the creation of harmful impacts in the environment that are directly tied to man's activities, including: agriculture, industry, and energy production and use.

Acid rain

Acid rain, or acid deposition, is a specific term that encompasses any type of precipitation of acidic elements, including rain, snow, fog, hail or even acidic particles, such as sulphuric or nitric acid dropping from the atmosphere onto the earth in wet or dry forms.

Ozone layer depletion

Ozone shield, carries high concentration of trioxygen, in Earth's stratosphere is a special region that absorbs most of the UV radiation coming from Sun. **Ozone depletion**, steady diminishing of Earth's ozone layer in the stratosphere caused by the release of chemical gaseous chlorine or bromine from industry and other human activities.

Global warming

Global warming is a human caused phenomenon, in which average temperature of Earth climate increases. *Carbon dioxide* plays a vital role in elevating greenhouse gas concentration promoting severe illness to climate in form of global warming.

Nuclear explosion

Anything excess is a poison, the intense and rapid release of energy from a high-speed nuclear reaction or explosion more precisely cause a huge damage to the life. Apart from heat energy nuclear explosions can introduce radioactive particles in the stratosphere causing global fallout.

1.15 REFERENCES AND SUGGESTED FURTHER READINGS

1. Environmental chemistry Tenth edition. / Manahan, Stanley E | New York: CRC Press, [**2017**]; ISBN 9781498776936. (785 pages)

2. Reaction mechanisms in environmental organic chemistry, Richard A. Larson and Eric J. Weber [**1994**]; ISBN 0-87371-258-7, (448 pages)

1.16 TERMINAL QUESTIONS

1. Define the following
   a. Chemical equilibrium
   b. Dynamic equilibrium
   c. Chemical potential
   d. Order of reaction

2. What is the difference between Green chemistry and Environmental Chemistry?
3. Explain open and closed systems in equilibrium with suitable examples?
4. Explain the following in detail.
a. First order reaction b. pseudo order reaction
5. Explain the mechanism of adsorption?
6. Explain the catalytic cycle in detail?

UNIT 2 : ENVIRONMENT CHEMISTRY-II

Structure

2.0 INTRODUCTION

Most of the chemical reactions take place in solutions, thus the study of such solutions forms an important branch of chemical sciences especially physical chemistry. If the solubility of solutes is analyzed in different solvents, we observe that the solutes follow the common principle of solubility which is '*like dissolves like*' means polar solvents dissolve polar solutes rapidly and to a great extent, while non-polar solutes are more soluble in non-polar solvents such as NaCl quickly dissolves in water while it remains insoluble in CCl_4 as water possesses high dielectric constant and polar properties that makes it suitable for most of ionic substances like sodium chloride. Depending upon the relative values of conductivities in aqueous solutions, the solutes may be categorized in following classes: (i) Strong electrolyte: highly conducting, (ii) Weak electrolyte: mildly conducting and (iii) Non electrolyte: non conducting but this classification also has a serious limitation i.e. a particular electrolyte might behave differently when it is dissolved in some other solvents. For instance:- if we dissolve acetic acid and sodium chloride in ammonia, both exhibit strong electrolytic behavior, whereas acetic acid ionizes slightly in water therefore, another classification is given which majorly focuses on the characteristics of solute rather than solvent. According to it, solutes can be classified as true and potential electrolytes. True electrolyte shows conductivity even in the pure liquid state while potential electrolyte does not display conductive behavior in pure liquid state but on dissolution in an ionic solvent it may provide conducting solution. When a weak electrolyte is dissolved in a polar solvent, equilibrium is established between its ions and unionized molecules and this phenomenon is known as ionic equilibrium. In order to get more information regarding this, we are going to discuss some important concepts in detail.

2.1 OBJECTIVES

After studing this unit you will be able to

- Understand the concepts of acid-base and their reactions,
- Understand the ionization of water,
- Describe the theory of hydrolysis with examples of salt hydrolysis
- Discuss the origin of pH scale and pOH
- Define buffer solutions
- Define common ion effect and
- Explain fundamentals of oxidation- reduction with examples of redox reactions.

2.2 ACID-BASE REACTIONS

Acid

Acids are the substances that undergo dissociation when dissolved in water and form hydrogen ions as the only positive ion species but in aqueous solutions the hydrogen ions (H^+) or protons do not exist in free state. Every hydrogen ion readily combines with one water molecule to form hydronium ion (H_3O^+).

$$HCl + H_2O \longrightarrow H_3O^+ + Cl$$

$$HNO_3 + H_2O \longrightarrow H_3O^+ + NO_3^-$$

The driving force for this ionization is the great affinity of free hydrogen ions to combine with water molecules to form hydronium ions. The above two acids dissociate completely in aqueous solutions. Hydrochloric and nitric acids are the examples of monoprotic acids i.e. they liberate only one proton or hydrogen ion in water whereas sulphuric and phosphoric acids have two and three hydrogen atoms therefore they are called as diprotic and triprotic acids respectively.

$$H_2SO_4 + H_2O \longrightarrow H_3O^+ + HSO_4^-$$

$$HSO_4^- + H_2O \rightleftharpoons H_3O^+ + SO_4^{2-}$$

Ionization of phosphoric acid

$$H_3PO_4 + H_2O \rightleftharpoons H_3O^+ + H_2PO_4^- \text{ Primary ionization}$$

$$H_2PO_4^- + H_2O \rightleftharpoons H_3O^+ + HPO_4^{2-} \text{ Secondary ionization}$$

$$HPO_4^{2-} + H_2O \rightleftharpoons H_3O^+ + PO_4^{3-} \text{ Tertiary ionization}$$

These ionization do not always occur to same extent. They follow the order of primary ionization > secondary > > tertiary.

Some acids like carbonic acid, boric acid nitrous acid, ethanoic acid etc. do not ionize completely. Their extent of dissociation is comparatively less hence, known as weak acids while those which dissociate almost completely are called strong acids such as hydrochloric, hydrobromic, hydroiodic, nitric and perchloric acid.

Base

Originally, base is a substance which undergoes dissociation when dissolved in water and forms hydroxide ions (OH^-) as only negative ion species. Some bases like hydroxides of sodium potassium and certain bivalent metals dissociate almost completely in aqueous solution therefore, they are called strong bases, whereas those which produce small concentration of hydroxide in aqueous solution are termed as weak bases such as ammonia NH_3.

$$NaOH + H_2O \longrightarrow Na^+ + OH^-$$

$$Ba(OH)_2 + H_2O \longrightarrow Ba^{2+} + 2\ OH^- \text{ Strong Base}$$

$$NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^- \text{ Weak Base}$$

The Bronsted-Lowry theory of acid and bases

A general theory of acids and bases was proposed in 1923 independently by J. N. Bronsted and T. M. Lowry that defines the concept of acid and base in all solvents. As described by these two scientists, a species that has the tendency to lose proton is called acid while bases possess characteristic feature to accept proton.

Acid = proton + conjugate base

$$A \rightleftharpoons H^+ + B$$

According to this theory, a conjugate base is formed after donation of proton from a Bronsted-Lowry acid. Similarly, a conjugate acid is produced when a Bronsted –Lowry base receives a proton.

$$\begin{array}{lcl} A_1 = B_1 + H^+ & \quad\text{and}\quad & B_2 + H^+ = A_2 \\ \text{Acid} = \text{conjugate base} + \text{proton} & & \text{base} + \text{proton} = \text{conjugate acid} \end{array}$$

$$\begin{array}{ccccccc} HCl & + & H_2O & \rightleftharpoons & H_3O^+ & + & Cl^- \\ \text{Acid}_1 & & \text{base}_2 & & \text{acid}_2 & & \text{base}_1 \end{array}$$

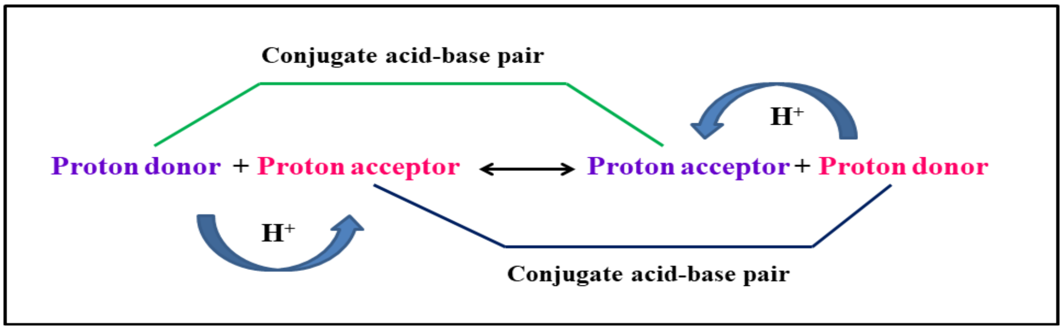

Figure 2.1: Representation of conjugate acid-base pair

It is also mentioned that strength of a conjugate base varies inversely as the strength of its respective acid. Further, the properties of an acid or base are significantly influenced by the solvent used.

Lewis Concept

In 1938, another definition of acids and bases was introduced by G.N. Lewis. Based on his investigations, he revealed that both acids and bases are capable of forming a covalent bond. During this phenomenon, an acid accepts a pair of electrons, whereas a base donates it. For example: BF_3, $AlCl_3$, SO_2 etc. have ability to attract an electron pair therefore termed as Lewis acids or electron acceptors. On the other hand, NH_3, pyridine etc. possess a lone pair of electrons that can be donated easily hence called as Lewis base or electron donor.

$$\underset{\text{Acid}}{BF_3} + \underset{\text{Base}}{NH_3} \longrightarrow \underset{\text{Adduct}}{F_3BNH_3}$$

$$\underset{\text{Acid}}{Cu^{2+}} + \underset{\text{Base}}{4CN^-} \longrightarrow \underset{\text{Adduct or Salt}}{[Cu(CN)_4]^{2-}}$$

Acid Base Reactions

Acid and base reaction is a type of chemical reaction wherein acid and base reacts together in stoichiometric ratio to neutralize each other as acid counteracts the base and vice versa. Therefore, this type of reactions is generally called as neutralization reaction.

$$\underset{\text{Acid}}{HNO_3\,(aq)} + \underset{\text{Base}}{NaOH\,(aq)} \longrightarrow \underset{\text{Salt}}{NaNO_3\,(aq)} + \underset{\text{water}}{H_2O\,(l)}$$

In the above reaction, a proton is transferred to a water molecule to form hydronium ion H_3O^+ that further reacts with hydroxide ion OH^- to produce water and the positive portion of base combines with negative part of acid to yield salt.

$$H_3O^+\,(aq) + OH^- \longrightarrow (aq)\; 2H_2O\,(l)$$

These reactions can also be classified in the category of double displacement reactions. In the case of polyprotic acid, all the acidic hydrogen atoms of acid are replaced by the base if it is added in sufficient amount.

$$H_2SO_4(aq) + NaOH(aq) \longrightarrow Na_2SO_4(aq) + 2H_2O(l)$$

$$H_3PO_4(aq) + NaOH(aq) \longrightarrow Na_3PO_4(aq) + 3H_2O(l)$$

Similarly, reaction of a diacidic base with monobasic acid

$$Ba(OH)_2(aq) + HCl(aq) \longrightarrow BaCl_2(aq) + 2H_2O(l)$$

As we know when an acid reacts with a base they neutralize each other and during this process some amount of heat is liberated that depends on the nature of acid, base and their number of gram equivalents. The energy released when one gram equivalent of acid reacts with one gram equivalent of base is known as enthalpy or heat of neutralization.

$$NaOH(aq) + HCl(aq) \longrightarrow NaCl(aq) + H_2O(l) + 13.7\ \text{Kcal/mole} \quad \text{(Exothermic Reaction)}$$

1 gm eq. 1 gm eq. 1 mole ($\Delta H = -13.7$ Kcal)

In the cases of weak acid- strong base, weak acid- weak base and strong acid-weak base, heat of neutralization will be lesser than 13.7 Kcal/mole as weak acids and bases do not dissociate completely. Some examples of weak acid-base reactions are given below:

$$\underset{\text{Weak base}}{NH_3(aq)} + \underset{\text{weak acid}}{CH_3COOH(aq)} \longrightarrow \underset{\text{salt}}{CH_3COONH_4(aq)} + H_2O(l)$$

$$\underset{\text{Weak base}}{NH_3(aq)} + \underset{\text{strong acid}}{HCl(aq)} \longrightarrow \underset{\text{salt}}{NH_4Cl(aq)} + H_2O(l)$$

$$\underset{\text{Strong base}}{NaOH(aq)} + \underset{\text{weak acid}}{CH_3COOH(aq)} \longrightarrow \underset{\text{salt}}{CH_3COONa(aq)} + H_2O(l)$$

Acid-Base equilibria in water

To understand acid-base equilibrium in water, dissociation of a weak electrolyte i.e. ethanoic acid in dilute aqueous solution is given below:

$$CH_3COOH + H_2O \rightleftharpoons H_3O^+ + CH_3COO^-$$

In conventional manner the above can be written as:

$$CH_3COOH \rightleftharpoons H^+ + CH_3COO^-$$

Here, H^+ represents the hydrated hydrogen ion. On applying law of mass action:

$$K = \frac{[CH_3COO^-][H^+]}{[CH_3COOH]}$$

Here, K denotes equilibrium constant at a given temperature. It is also called ionization constant or dissociation constant of acid that can also be symbolized by K_a in aqueous solutions.

If V liters of a solution contain 1 mole of weak electrolyte ($V = 1/c$, where c indicates the concentration in mol L^{-1}) and suppose α represents degree of ionization at equilibrium, then α will be the amount of each ions and $(1-\alpha)$ will be the amount of unionized electrolyte. In terms of concentration, $(1-\alpha)/V$ and α/V show the concentration of unionized ethanoic acid and each of ions respectively. On putting these values in equilibrium equation:

$$K = \frac{\alpha^2}{(1-\alpha)V} \quad \text{or} \quad K = \frac{\alpha^2 c}{(1-\alpha)}$$

This expression represents **Ostwald's dilution law**.

Similarly, for polyprotic acids dissociation constant can be given for each ionization as follows:

$$H_2A \rightleftharpoons H^+ + HA^-$$
$$HA^- \rightleftharpoons H^+ + A^{2-}$$

If acid is weak electrolyte then applying law of mass action on the above equations:

$$K_1 = [H^+]\,[HA^-]/[\,H_2A]$$
$$K_2 = [H^+]\,[A^{2-}]/[\,HA^-]$$

Likewise, consider the dissociation of a weak base in aqueous media

$$NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$$

On applying law of mass action

$$K_b = \frac{[NH_4^+]\,[OH^-]}{[NH_3]}$$

Here, K_b is dissociation constant of a particular base that depends on temperature, nature of solvent.

Relative strength of weak acids and weak bases

According to Ostwald dilution law, dissociation constant of a weak acid

$$K_a = \frac{\alpha^2 c}{(1-\alpha)}$$

If degree of dissociation (α) is very less then we can neglect α in comparison to unity so finally we get

$K_a = \alpha^2 c$ or $K_a = \alpha^2/V$

If dissociation constants of two different weak acids are Ka_1 and Ka_2 at a same concentration c then

$$\alpha_1/\alpha_2 = \sqrt{K_{a1}/K_{a2}}$$

Here, α_1 and α_2 represent the respective degree of dissociation of two acids

Similar expression can be used for weak bases with the substitution of dissociation constant of acid with ionization constant of base i.e.

$$\alpha_1/\alpha_2 = \sqrt{K_{b1}/K_{b2}}$$

In words it can be stated as, for any two slightly ionized electrolytes (weak acid or base) at equal dilution, the degrees of dissociation are directly proportional to the square roots of their dissociation constants.

Significance of Acid-Base reactions

Neutralization or acid-base reactions play a critical role in maintaining the necessary balance in our ecosystem as well as in our body. These reactions are especially very useful in pharmaceutical, food and fertilizers manufacturing industries. Our day to day life is governed by these reactions such as digestion of food in our stomach. Similarly, use of an antacid to cure heartburn is also an example of neutralization reaction wherein stomach acid is neutralized either by $Mg(OH)_2$ or by $Al(OH)_3$ that are present in antacid. Besides, to resolve the problem of acidic water and soil in mining land some basic oxides or carbonates such as CaO, $CaCO_3$ or $MgCO_3$ are added to reduce their acidity.

SAQ 1

What are the fundamental theories to explain acidity and alkalinity of the substances and how they are different from one another?

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2.3 IONIC PRODUCT OF WATER

Dissociation of water

In 1894, Kohlraush and Heyweiller observed that even highly purified water behaves like a very weak electrolyte and possesses definite conductivity. It ionizes in the following manner:

$$H_2O + H_2O \rightleftharpoons H_3O^+ + OH^-$$

To simplify, it can be written as:

$$H_2O \rightleftharpoons H^+ + OH^-$$

Applying law of mass action to the above equation on a particular temperature

$$K_i = \frac{[H^+][OH^-]}{[H_2O]}$$

Here, K_i represents the ionization constant of water. As we know that degree of dissociation of water is very less, so the concentration of undissociated water molecules remain almost constant during this ionization process and this concentration can be combined with K_i (ionization constant of water) to give a new constant (K_w) that is known as **ionic product of water**.

$K_w = K_i [H_2O] = [H^+][OH^-]$

In pure water, concentration of hydrogen ion $[H^+]$ will be equal to the concentration hydroxyl ion $[OH^-]$ i.e. 1×10^{-7} M at 25 ºC.

$K_w = (1.0 \times 10^{-7}) (1.0 \times 10^{-7})$ $[H^+] = [OH^-] = 1.0 \times 10^{-7}$ M

$K_w = 1.0 \times 10^{-14}$

The ionic product of water primarily depends upon temperature; it increases with increase in temperature.

$[H^+] = K_w/[OH^-]$ and $[OH^-] = K_w/[H^+]$

Acidic or alkaline nature of a solution relies upon the relative concentration of hydrogen ions to that of hydroxyl ions. When the concentration of hydrogen ions is exactly equal to the concentration of hydroxyl ions, the solution is called neutral solution. At ordinary temperatures, if the concentration of hydrogen ions $[H^+]$ is greater than 10^{-7} mol L^{-1}, solution will show acidic behavior and if less than 10^{-7} mol L^{-1}, alkaline nature will be exhibited. Similarly, if concentration of hydroxyl ions is greater than 10^{-7} mol $L^{-1,}$ solution will be basic and if less than 10^{-7} mol L^{-1} acidic properties will be displayed.

Neutral solution: [H+] = [OH-] = 10-7

Acidic solution: $[H^+] > [OH^-]$ or $[H^+] > 10^{-7}$

Alkaline Solution: $[H^+] < [OH^-]$ or $[OH^-] > 10^{-7}$

SAQ 2

What do you understand by ionic product of water and explain its significance?

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2.4 pH and pOH

While dealing with dilute solutions, it is very difficult to represent the concentrations of hydrogen and hydroxide ions in terms of moles per liter (mol L^{-1}). Therefore; a very convenient method was introduced by S. P. L. Sorensen in 1909. He proposed a logarithmic scale to express hydrogen ion concentration as given below:

$pH = \log_{10} 1/[H^+] = -\log_{10} [H^+]$ or $[H^+] = 10^{-pH}$

Thus pH is the logarithm of hydrogen ions with negative sign. In this method, numbers between 0 to 14 are used to denote all the possible states of acidity and alkalinity from 1 mol L^{-1} hydrogen ions to 1 mol L^{-1} hydroxide ions.

As we know in neutral solution

$[H^+] = [OH^-] = 10^{-7}$ therefore, pH = 7

For acidic solution pH < 7 and $[H^+] > 10^{-7}$ mol L^{-1}

For alkaline solution pH > 7 and $[H^+] < 10^{-7}$ mol L^{-1}

The concentration of hydroxyl ions can be expressed in similar manner:

$pOH = 1/\log_{10} [OH^-] = -\log_{10} [OH^-]$ or $[OH^-] = 10^{-pOH}$

As we know

$[H^+][OH^-] = K_w = 10^{-14}$

Taking logs of both sides

$\log [H^+] + \log [OH^-] = \log K_w = -14$

then $pH + pOH = pK_w = 14$

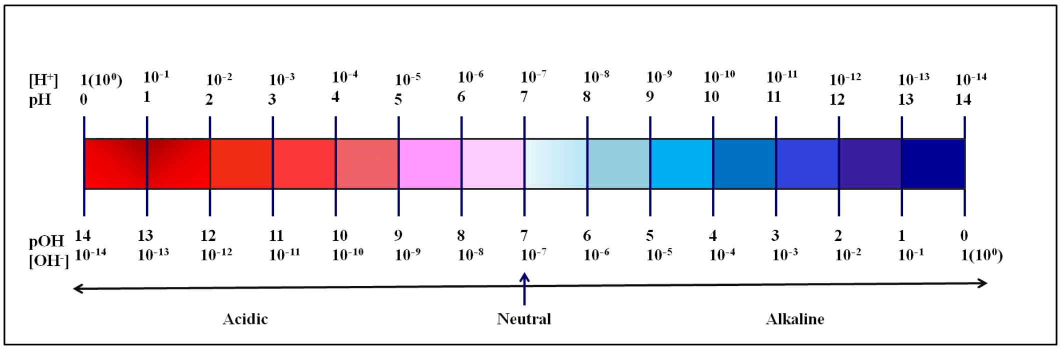

Figure 2.2 The pH scale

In quantitative analysis, small quantities are usually expressed in logarithmic or exponential form. Dissociation constants, ionic concentrations and solubility products may also be represented in this form for convenience.

SAQ 3

How pH and pOH are related to each other?

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2.5 HYDROLYSIS

The term hydrolysis is derived from ancient Greek words (Hydro-water+ Lysis-break down) and it means dissolution or destruction by water. Hydrolysis is a type of chemical reaction in which chemical bonds of a substance are decomposed by addition of water molecules. This phenomenon plays a quite significant role in biological systems as well as in chemical sciences. The process of hydrolysis is used to break down a variety of polymers such as carbohydrates, proteins, fats, nucleic acids etc. These decomposition reactions are usually catalyzed by acid, base or enzymes (hydrolases). Hydrolysis of salts is one of the most common examples of such reactions wherein the interaction between cations or anions of a salt and water takes place. During this process, a salt breaks down into its ions completely or partially depending upon its solubility factor and dissolution of salt into water may alter pH of solution that primarily relies upon its constituent ions. According to the nature of their fragmentary ions, these salts are generally classified into four major categories:

(i) Salts of strong acids and strong bases, e.g. NaCl

(ii) Salts of weak acids and strong bases, e.g. CH_3COONa

(iii) Salts of strong acids and weak bases, e.g. NH_4Cl

(iv)Salts of weak acids and weak bases, e.g. CH_3COONH_4

Dissolution of salts of strong acids and strong bases in water results in neutral solution as their cations and anions have no tendency to combine with hydrogen and hydroxyl ions of water therefore, the equilibrium hydrogen ions and hydroxyl ions remains undisturbed that maintains the neutrality of the solution whereas the salts of other three groups may interact with the ions of water and change their equilibrium and ultimately the pH of solution.

Degree of hydrolysis

Case 1: Salts of a weak acid and a strong base

The salts of this class give alkaline solutions on hydrolysis as the anionic part of a weak acid interacts with hydrogen ions of water to form weak acid which dissociates feebly. In order to maintain the constant value of ionic product (K_w), water further ionizes but hydrogen ions are again taken up by the negative charge bearing species of salt. Thus, the decrement in concentration of hydrogen ions with respect to hydroxyl ions is observed and due to high concentration of hydroxyl ions, solution becomes alkaline. Consider the hydrolysis of CH_3COONa:

$$CH_3COO^- + Na^+ + H_2O \rightleftharpoons CH_3COOH + Na^+ + OH^-$$

As Na^+ ions are present on the both side of equation therefore, they may be left out and the equation can be written in following manner:

$$\underset{\text{Unhydrolysed salt}}{CH_3COO^-} + H_2O \rightleftharpoons \underset{\text{acid}}{CH_3COOH} + \underset{\text{base (strong)}}{OH^-}$$

On applying law of mass action and considering the concentration of water as constant because it is present in excess, we obtain

$$K_h = \frac{[OH^-][CH_3COOH]}{[CH_3COO^-]} \quad \text{or} \quad K_h = \frac{[\text{base}][\text{acid}]}{[\text{Unhydrolysed salt}]}$$

Here, K_h is called hydrolysis constant. Since generated acid is weak in nature hence, it ionizes slightly while free strong base and unhydrolysed salt dissociate completely.

'The fraction of each mole of anion (CH_3COO^-) that is hydrolysed at equilibrium is known as degree of hydrolysis."

Assume 1 mole of salt is dissolved *V* litre of solution and *x* represents the degree of hydrolysis then the concentrations in mol L^{-1} are

$[CH_3COOH] = [OH^-] = x/V$ and $[CH_3COO^-] = (1 - x)/V$

Putting these values in the above equation

$$K_h = \frac{(x/V)\,(x/V)}{(1 - x)/V} = \frac{x^2}{(1 - x)/V}$$

It is evident from the above expression on increasing the dilution, the degree of hydrolysis x is also increased.

With hydrolytic equilibrium ($CH_3COO^- + H_2O \rightleftharpoons CH_3COOH + OH^-$),
two equilibria must be coexist i.e.

$$H_2O \rightleftharpoons H^+ + OH^- \text{ and } CH_3COOH \rightleftharpoons H^+ + CH_3COO^-$$

We know that

$K_w = [H^+]\,[OH^-]$, $K_a = [H^+]\,[CH_3COO^-]/[CH_3COOH]$ and $K_h = [OH^-][CH_3COOH]/[CH_3COO^-]$

$$\text{But } \frac{K_w}{K_a} = \frac{[H^+]\,[OH^-]\,[CH_3COOH]}{[H^+]\,[CH_3COO^-]} = \frac{[OH^-]\,[CH_3COOH]}{[CH_3COO^-]} = K_h$$

Thus, $K_h = Kw / K_a$ or $pK_h = pK_w / pK_a$

The above expression shows the relationship between hydrolysis constant, ionic product of water and dissociation constant of water. As we know variation in temperature affects K_a slightly but K_w is significantly dependent on temperature therefore, the degree of hydrolysis will be majorly governed by temperature changes.

As a result of hydrolysis, the equal amounts of acid (CH_3COOH) and base (OH^-) will be produced. If the concentration of salt dissolved in water is c mol L^{-1} then

$$K_h = \frac{[OH^-]\,[CH_3COOH]}{[CH_3COO^-]} = \frac{[OH^-]^2}{c} = \frac{K_w}{K_a}$$

and $[OH^-] = \sqrt{c\,K_w/K_a}$

since $[H^+] = K_w/[OH^-]$, substituting the values

$$[H^+] = \sqrt{K_w K_a / c}$$

$$pH = \frac{1}{2}pK_w + \frac{1}{2}pK_a + \frac{1}{2}\log c$$

The above equation gives hydrogen ion concentration of hydrolyzed salt (weak acid and strong base) solution.

Case 2: Salts of a strong acid and a weak base

The salts of this category produce acidic solution on hydrolysis as cationic part of salt participates actively in hydrolysis and forms weak base which ionizes poorly therefore, concentration of hydrogen ions increases and solution exhibits acidic nature. Consider the dissociation of ammonium chloride:

$$NH_4^+ + Cl^- + H_2O \rightleftharpoons NH_4OH + Cl^- + H^+$$

Neglecting Cl^- ions as they are conjugate base of strong acid thus will not take part actively in hydrolysis

$$\underset{\text{Unhydrolysed salt}}{NH_4^+} + H_2O \rightleftharpoons \underset{\text{free base}}{NH_4OH} + \underset{\text{free acid}}{H^+}$$

Applying law of mass action (Similar to case 1)

$$K_h = \frac{[H^+][NH_4OH]}{[NH_4^+]} = \frac{K_w}{K_b} = \frac{x^2}{(1 - x)V}$$

Here, K_b represents dissociation constant of base and as we know the concentrations of formed base and acid are equal therefore, substituting values in the above equation

$$K_h = \frac{[H^+]^2}{c} = \frac{K_w}{K_b} \quad \text{and} \quad [H^+] = \sqrt{c\,K_w\,K_b}$$

or $$pH = \frac{1}{2}pK_w - \frac{1}{2}pK_b + \frac{1}{2}\log c$$

The above expression gives pH of hydrolysed salt of strong acid and weak base.

Case 3: Salts of a weak acid and weak base

The salts of this class may give acidic or alkaline or neutral solution on hydrolysis depending upon the strengths of their constituent acid and base e.g. ammonium acetate

$$\underset{\text{Unhydrolysed salt}}{CH_3COO^-} + \underset{\text{unhydrolysed salt}}{NH_4^+} + H_2O \rightleftharpoons \underset{\text{free acid}}{CH_3COOH} + \underset{\text{free base}}{NH_4OH}$$

In this case both cation and anion of salt interact with hydrogen and hydroxyl ions of water i.e. hydrolysed.

Applying law of mass action

$$K_h = \frac{[NH_4OH][CH_3COOH]}{[CH_3COO^-][NH_4^+]} = \frac{[\text{base}][\text{acid}]}{[\text{unhydrolysed salt}]^2}$$

We know that

$$K_a = \frac{[CH_3COO^-][H^+]}{[CH_3COOH]} \qquad K_b = \frac{[NH_4^+][OH^-]}{[NH_4OH]} \quad \text{and} \quad K_w = [H^+][OH^-]$$

$$K_h = \frac{K_w}{K_a \times K_b} \quad \text{or} \quad pK_h = pKw - pKa - pK_b$$

The individual concentrations can be represented as given below if 1 mol salt is dissolved in V L of solution and x denotes degree of hydrolysis

$[CH_3COOH] = [NH_4OH] = x / V$ and $[CH_3COO^-] = [NH_4^+] = (1 - x/V)$

Substituting the values

$$K_h = \frac{(x/V)(x/V)}{\{(1-x/V)\}\{(1-x/V)\}} = \frac{x^2}{(1-x)^2}$$

It is evident from the above expression that degree of hydrolysis does not depend on the concentration of solution thus the pH of solution.

The concentration of hydrogen ions of hydrolysed salt solution can be calculated as given below

$$[H^+] = \frac{K_a[CH_3COOH]}{[CH_3COO^-]} = K_a\{(x/V)/(1-x)V\} = K_a(x/1-x)$$

Since $x/(1-x) = \sqrt{K_h}$ therefore $[H^+] = K_a\sqrt{K_h} = = K_a\sqrt{K_w K_a / K_b}$

$$pH = \frac{1}{2}pK_w + \frac{1}{2}pK_a + \frac{1}{2}pK_b$$

It is clear from the above equation if $K_a = K_b$ then solution will be neutral and if $K_a > K_b$ and $K_a < K_b$ then the solution will be acidic and alkaline respectively.

SAQ 4

What is degree of hydrolysis? Derive the expression for salts of weak acid and strong base.

2.6 BUFFER SOLUTIONS

A particular pH is required for the proper functioning of various chemical, pharmaceutical, biological and biotechnological processes; therefore we need to have such solutions whose pH does not change appreciably even when small amounts of strong acid or strong base are added to them. These solutions are called buffer solutions. "Buffers are the chemical substances that resist changes in the concentration of hydrogen or hydroxyl ions and their resistance to change in pH upon introduction of small quantities of acid or alkali is known as buffer action."

Generally buffer solutions are comprised of a mixture of weak acid and its salt with strong base (usually sodium or potassium) or they can be a combination of weak base and its salt. In other words, buffers are the mixture of a weak acid and its conjugate base or weak base and its conjugate acid. The buffers have ability to reverse both acidity and alkalinity to some extent.

Case 1: A mixture of weak acid and its salt

A very common example of this class is a mixture acetic acid and its sodium salt. This buffer solution contains CH_3COOH molecules, CH_3COO^- and Na^+ ions as acetic acid dissociates feebly whereas sodium acetate ionizes almost completely. The buffer action of acetic acid- sodium acetate is given below:

On the addition of strong acid to this buffer solution, the introduced hydrogen ions will be immediately consumed by acetate ions to form acetic acid that dissociates in very slight amount.

$$H^+ + CH_3COO^- \longrightarrow CH_3COOH$$

Thus the neutralization of added hydrogen ions takes place. Contrary to this when a strong base is introduced in to the solution, the hydroxyl ions are taken up by acetic acid to maintain the neutrality.

$$OH^- + CH_3COOH \longrightarrow CH_3COO^- + H_2O$$

Case 2: A mixture of weak base and its salt

The most common example of this category is an equimolar mixture of aqueous solutions ammonium hydroxide and ammonium chloride. The buffer solution contains unionized NH_4OH, NH_4^+ and Cl^- ions. Let us consider the buffer action of this mixture:

When acid is added to this solution, the increased concentration of hydrogen ions are neutralized by ammonium hydroxide

$$NH_4OH + H^+ \longrightarrow NH_4^+ + H_2O$$

On addition of strong base, the neutralization of excess amount of hydroxyl ions is attained by ammonium ions

$$NH_4^+ + OH^- \longrightarrow \underset{\text{Feebly dissociates}}{NH_4OH}$$

Calculation for pH of buffer

The hydrogen ion concentration of weak acid and its salt can be expressed by following equation:

According to law of mass action $[H^+] = K_a\,[acid]/[salt]$

$$\text{or} \quad pH = pKa + \log \frac{[salt]}{[acid]}$$

The above equation is called as Henderson-Hasselbalch equation that gives pH of buffer solution which is a mixture of weak acid and it’s conjugate.

Similar expression can be given for weak base and its salt

$$[OH^-] = K_b[base]/[salt] \qquad \text{or} \quad pOH = pK_b + \log \frac{[salt]}{[base]}$$

and pH can be calculated by following relationship

$pH = pK_w - pOH$ or $pH = 14 - pOH$

Buffer Capacity: The capacity of a buffer to withstand the change in pH is known as buffer capacity. A buffer solution that contains equimolar concentrations of acid or base and its salt exhibits the maximum buffer capacity.

SAQ 5:

What are buffers and how can we calculate the pH of buffer solution containing a weak base and its conjugate acid?

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2.7 COMMON ION EFFECT

If a strong electrolyte is introduced to a solution of feebly dissociated electrolyte that has at least an ion common with strong electrolyte, suppression in the ionization of weak electrolyte is observed and this phenomenon is known as common ion effect. E.g. reduction in dissociation of acetic acid (weak electrolyte) on addition of sodium acetate (strong electrolyte) in its solution

$$CH_3COOH \rightleftharpoons CH_3COO^- + H^+$$

slightly dissociated

$$CH_3COONa \longrightarrow CH_3COO^- + Na^+$$

completely dissociated

Introduction of sodium acetate to the solution of acetic acid increases concentration of CH_3COO^- in the medium and to overcome the addition of salt equilibrium shifts to backward direction hence, the degree of dissociation of CH_3COOH decreases.

In the same manner, ionization of weak base (NH_4OH) is suppressed in the presence of NH_4Cl (salt) or NaOH (strong base) as ammonium chloride supplies NH_4^+ ions, while NaOH furnishes OH^- ions and both ions cause reduction in the dissociation of ammonium hydroxide.

Common ion effect has prime importance in qualitative analysis wherein it has been used in salt as well as gravimetric analysis. The other applications of this effect include softening of water, purification of common salt and salting out of soap.

SAQ 6

Explain common ion effect with example and its applications.

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2.8 OXIDATION AND REDUCTION

Oxidation and reduction, these two fundamental terms have been used in chemistry since beginning. Oxidation includes gain of oxygen, loss of hydrogen or electron and increase in oxidation number or positive charge, whereas reduction deals with loss of oxygen, gain of electron or hydrogen and decrease in oxidation number or positive charge or increase in negative charge of any atom or molecule or species. The phenomena of oxidation and reduction are just opposite to each other and they must occur simultaneously. These processes are integral parts of chemical, physical, biological and biotechnological sciences.

Oxidation

$CH_3CH_3 + O_2 \longrightarrow CH_3CH_2OH$ (addition of oxygen)

$CH_3CH_2OH \longrightarrow CH_3CHO$ (loss of hydrogen)

$Ag \longrightarrow Ag^+ + e^-$ (loss of electron)

$Cu^+ \longrightarrow Cu^{2+} + e^-$ (increase in oxidation number)

$Fe + S \longrightarrow FeS$ (addition of electronegative element)

$KI + Cl_2 \longrightarrow KCl + I_2$ (removal of electropositive element)

Reduction

$CH_3COOH \longrightarrow CH_3CHO$ (loss of oxygen)

$CH_2CH_2 + H_2 \longrightarrow CH_3CH_3$ (addition of hydrogen)

$Cl_2 + 2e^- \longrightarrow 2Cl^-$ (gain of electrons)

$Cu^{2+} + e^- \longrightarrow Cu^+$ (decrease in oxidation number)

$FeCl_3 + H_2S \longrightarrow 2FeCl_2 + HCl + S$ (removal of electronegative element)

A substance (atom, ion or molecule) that has ability to oxidize other substances is called **oxidizing agent** and while oxidizing other species the substance itself gets reduced whereas **reducing agent** are those substances which have tendency to reduce other molecules or atom or element or species and during this process they themselves get oxidized. In other words, oxidizing agents are known as electron acceptor while reducing agents are termed as electron donor.

As we mentioned earlier that oxidation and reduction take place concurrently therefore, these processes cannot be separated from each other and studied together.

Redox reactions

The chemical reactions in which the oxidation numbers of participating species (atoms, molecule or ions) are changed by the transfer of electrons from one species to another species are termed as redox reactions (**red**uction/**ox**idation).

Reduction

$$Fe_{(s)} + CuSO_{4(aq)} \longrightarrow FeSO_{4(aq)} + Cu_{(s)}$$

Oxidation

Redox Reaction

Here, oxidation state of iron changes from 0 to +2 while copper is converted from +2 to 0.

Redox reactions are prevalent in nature and almost one third reactions that have been taking place in our surrounding are oxidation reduction reactions. These reactions can occur abruptly, slowly or rapidly depending upon their environmental conditions. Few examples of redox reactions are given below:

1. Respiration: As we know that respiration is the ultimate energy yielding process in living organisms wherein consumed or stored food is transformed into energy through a series of redox reactions. During this process, oxidation of glucose and reduction of oxygen take place simultaneously.

$$C_6H_{12}O_6 + 6O_2 \longrightarrow 6CO_2 + 6H_2O + \text{Energy}$$

2. Photosynthesis: In this process green plant and certain micro-organisms convert light energy to chemical energy wherein carbon dioxide and water are transformed into energy rich organic compounds such as glucose or other sugars and oxygen in the presence of sunlight and some specific pigments like chlorophyll.

$$6CO_2 + 6H_2O \xrightarrow[\text{Chlorophyll}]{\text{Sun light}} C_6H_{12}O_6 + 6O_2$$

3. Photography: In classical approach, photographic films containing light sensitive silver halides are treated with solution of a reducing agent such as

hydroquinone, *N*-methyl-*p*- aminophenol, ascorbic acid etc. which reduces silver halides to metallic sliver. Besides, fixation of image is carried out by using the solution of sodium thiosulphate.

$$AgBr \xrightarrow[\text{Reducing agent}]{\text{Light}} Ag(s) + Br_2(g)$$

4. Corrosion: It is a natural galvanic process wherein deterioration of refined metals takes place through oxidation.

$$2Fe + O_2 + 2H_2O \longrightarrow 2Fe(OH)_2 + O_2 \longrightarrow \underset{\text{rust}}{Fe_2O_3.H_2O}$$

5. Breath analyzer: In a chamber of breath analyzer, an aqueous solution containing potassium dichromate, sulfuric acid and silver nitrate is filled. When an alcoholic breath sample is bubbled into this solution, sulfuric acid helps in transferring of ethanol from air to liquid where it reacts with potassium dichromate in presence of sliver nitrate to yield acetic acid and chromium sulfate (green colour). This process is accelerated by $AgNO_3$ which acts as a catalyst for the above redox reaction.

$$3C_2H_5OH + 2K_2Cr_2O_7 + 8H_2SO_4 \xrightarrow{AgNO_3} Cr_2(SO4)_3 + 3CH_3COOH + 2K_2SO_4 + 11H_2O$$

Oxidation

Reduction

6. Cells and batteries: Generally, in cell and batteries either chemical energy is converted into electric energy or electric energy is transformed into electrochemical energy through redox reactions.

$$Zn(s) + CuSO_4(aq) \longrightarrow ZnSO_4(aq) + Cu(s) \longrightarrow (\text{galvanic cell})$$

Another type of redox reaction is disproportionation reaction that sometimes also called as dismutation in which a compound or molecule is converted into two or more dissimilar products through simultaneous oxidation and reduction of same species.

$$Hg_2Cl_2 \xrightarrow{\text{UV light}} Hg + HgCl_2$$

reduction

oxidation

$$2H_2O_2 \longrightarrow 2H_2O + O_2$$

reduction

oxidation

SAQ 7

What are oxidizing and reducing agents? Explain with suitable examples.

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2.9 LET US SUM UP

The brief description of some fundamental points of this unit is given below:

The chemical reactions that occur between an acid and base and result into formation of salt and water are generally called acid- base reactions. These reactions play a critical role in maintaining balance in our body as well as our environment.

Highly purified water also behaves like a weak electrolyte and shows conductivity. It ionizes in hydrogen and hydroxyl ions but hydrogen ions do not exist in free form. They are found in the form of hydronium ions [H_3O^+]. The product of active concentrations of hydronium ion and hydroxyl ions at a particular temperature is known as ionic product of water and its value at 25ºC is $K_w = 1 \times 10^{-14}$.

Basically, pH scale is used to express the acidity or alkalinity of a solution. Mathematically, it is defined as negative logarithm of hydrogen ion concentration at a given temperature. pH > 7 represents alkaline nature of solution whereas solutions having pH < 7 show acidic behavior and pH = 7 indicates neutrality of the solution.

The term hydrolysis is used to describe dissolution by water molecules. In simple words we say it includes chemical breakdown of a compound because of its reaction with water. In salt hydrolysis, reaction between salt and water takes place that results in to formation acidic, alkaline or neutral solution depending upon the salt used.

A buffer solution is an aqueous mixture of either weak acid and its conjugate base or weak base and its conjugate acid that resists in alteration of pH when small amounts of strong acid or base are added to it.

The common ion effect refers to decrease in dissociation of weak electrolyte when a strong electrolyte having a common ion is introduced to the solution of feebly ionized electrolyte. This effect has fundamental importance in quantitative analysis.

According to modern concept, removal of electrons from a species is termed as oxidation whereas addition of electrons is called reduction. Almost all the elements and their compounds have tendency to undergo oxidation-reduction reactions.

Photosynthesis, respiration, corrosion etc are some common examples of redox reactions that we experience in our day to day life.

2.10 GLOSSARY

Conductivity: A capacity of a substance to conduct electricity.

Electrolyte: A substance that conducts electricity in molten state or when dissolved in polar solvent.

Equilibrium: It is a state of a system when the rate of forward and backward reaction becomes equal and concentration of reactants and products remains constant.

Polyprotic acid: An acid which is capable of donating more than one hydrogen ion per molecule to aqueous solutions.

Solute: A substance (solid, liquid or gas) which is dissolved in solvent to form solution.

Solubility: It is the maximum amount of a solute that can be dissolved in a solvent at particular temperature.

2.11 SUGGESTED READINGS

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4. K. L. Kapoor, 2009, A Text Book of Physical Chemistry, 3rd.
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2.12 TERMINAL QUESTIONS

1. Describe general theories of acids and bases.
2. What do you mean by hydrolysis? Explain salt hydrolysis by giving an example with equations.
3. What are buffers? Derive the formula for pH of mixture of weak acid and its salt type buffer.
4. Give detailed description of oxidation-reduction with suitable examples.

5. What are neutralization reactions? Illustrate the process of neutralization with examples.

ANSWERS

Self-Assessment Questions

1. Fundamental theories of acids and bases are given in section 2.2 Acid-Base reactions that include general principle (Arrhenius theory), Bronsted-Lowry and Lewis concept. These theories are different from each other at the basic level as they define the concept of acidity and alkalinity on libration of hydrogen and hydroxyl ions, conjugate acid-base pair and donation as well as acceptance of electrons respectively.

2. The product of concentration of hydrogen ions and hydroxyl ions at a particular temperature is known as ionic product of water. As it remains constant at a given temperature and represents the neutrality of water therefore, it plays critical role in gathering information about reaction medium and conditions.

3. pH and pOH are inversely proportional to each other. Moreover, pH + pOH = pK_w = 14.

4. The fraction of each mole of anion or cation that hydrolysed at equilibrium is known as degree of hydrolysis. For the derivation of expression of salts of weak acid and strong base refer section 2.5 Hydrolysis (case 1: salts of weak acid and strong base).

5. Buffers are the chemical substances that resist changes in the concentration of hydrogen or hydroxyl ions when small quantities of acid or alkali are added to it. For calculation of pH of weak base and its conjugate acid visit section 2.6 Buffer solutions (Case 2: A mixture of weak base and its salt).

6. If a strong electrolyte is introduced to a solution of feebly dissociated electrolyte that has at least an ion common with strong electrolyte, suppression in the ionization of weak electrolyte is observed and this phenomenon is known as common ion effect. E.g. reduction in dissociation of acetic acid (weak electrolyte) on addition of sodium acetate (strong electrolyte) in its solution.

7. Common ion effect has prime importance in qualitative analysis wherein it has been used in salt as well as gravimetric analysis. The other applications of this effect include softening of water, purification of common salt and salting out of soap.

8. A substance (atom, ion or molecule) that has ability to oxidize other substances is called **oxidizing agent** and while oxidizing other species the substance itself gets reduced whereas **reducing agent** are those substances which have tendency to reduce other molecules or atom or element or species and during this process they themselves get oxidized. In other words, oxidizing agents are known as electron acceptor while reducing agents are termed as electron donor.

For the examples of oxidizing and reducing agents refer section 2.8 Oxidation and Reduction (Redox reactions).

Terminal Questions

1. General theories of acids and bases are described in section 2.2 Acid-Base reactions. Please refer this section for more information.

2. Hydrolysis is a type of chemical reaction in which chemical bonds of a substance are decomposed by addition of water molecules. This phenomenon plays a quite significant role in biological systems as well as in chemical sciences. For example of salt hydrolysis refer section 2.5 Hydrolysis.

3. The detailed description of buffers is provided in section 2.6 Buffer Solutions. Please refer that section for answer.

4. Oxidation and reduction is explained thoroughly in section 2.8 Oxidation and Reduction. Please visit that section for details.

5. Acid and base reaction is a type of chemical reaction wherein acid and base reacts together in stoichiometric ratio to neutralize each other as acid counteracts the base and vice versa. Therefore, this type of reactions generally called neutralization reaction. For example refer subsection Acid-base reactions in sectional heading 2.2 Acid-Base reactions.

UNIT 3 : ENVIRONMENTAL CHEMISTRY III

Structure

3.0 INTRODUCTION

Environment chemistry is an interdisciplinary field which encompasses the fundamental principles of chemistry, biochemistry, soil, toxicology and ecology. It deals with chemical composition, properties, reaction rate, transport , effects and ultimate fate of chemical species encountered in the environment. It is pertinent to identify and understand the problems associated in the environment and to find the mechanism by which it can be minimized or completely eliminated from the environment. The main objective behind environmental chemistry is to create awareness about its importance and need to restrain human activities which leads to indiscriminate release of pollutant into the environment.

3.1 OBJECTIVES

After studing this unit you will be able to

- describe concept of solubility and solubility product
- understand the aquatic chemistry of carbonate system

- explain about distribution, mobility and biological availability of chemical elements in natural system
- Classify atmospheric pollutants pertaining to heavy metals, radionuclides and saturated and unsaturated hydrocarbons encountered in the atmosphere
- Understand the chemistry of gasoline and additives used
- Explain the strategies to prevent knocking in gasoline
- Describe the mechanism, classification and properties of lubricants

3.2 SOLUBILITY AND SOLUBILITY PRODUCT

Solubility product of the saturated solution of a sparingly soluble salt in water can be defined as the product of concentrations of ions raised to a power equal to number of times ions occur in the equation representing the dissociation of electrolyte.

When a saturated solution of an electrolyte is formed, a dynamic equilibrium is maintained between the excess of the solute and ions furnished by the already dissolved electrolyte.

Consider a sparingly soluble salt say, A_xB_y when added to water, a very small amount of it dissolves and rest of it remains in the water in the solid form. Here, the undissolved A_xB_y is in equilibrium with A and B ions formed in the solution.

This can be represented as:

A_xB_y (s, saturated solution) $\rightleftharpoons xA^{y+}$ (*aq*) + yB^{x-} (*aq*)

According to the law of chemical equilibrium, equilibrium constant can be represented as:

$$K = \frac{a_{xAy+} \times a_{yBx-}}{a_{AxBy}}$$

where a_{Ay+}, a_{Bx-} and a_{AxBy} denote the activities of the furnished ions A^+, B^- and the undissolved electrolyte A_xB_y respectively. Now, the activity of solid is taken as 1 by convention. Therefore, the equation can be represented as:

$$K_{sp} = a_{xAy+} \times a_{yBx-}$$

Where, K_{sp} is known as the solubility product of electrolyte A_xB_y.

In common practice, concentration terms are more preferred as compared to the activities. In such a case, solubility product K_{sp} becomes concentration solubility product K'_{sp} and can be written as:

$$K'_{sp} = [A^{y+}]^x [B^{x-}]^y$$

In case of sparingly soluble salts, ionic concentration of the dissolved salts is very low so that the concentrations and activities can be taken as almost same indicating that solubility product is equal to the concentration solubility product. Therefore,

$K_{sp} = K'_{sp}$

Consequently,

Solubility product of a sparingly soluble salt A_xB_y can be represented in the equation form as:

$K_{sp} = [A^{y+}]^x [B^{x-}]^y$

Interestingly, one can find the solubility of a sparingly soluble salt if its solubility product is known and vice-versa. Solubility and solubility product are corelated as follows. When a saturated solution of sparingly soluble salt AB is prepared, its solubility equilibrium can be represented as:

AB (s, saturated solution) $\rightleftharpoons$ A^+ (*aq*) + B^- (*aq*)

Let s denote the solubility of the salt. Then,

$[A^+]$ = s mol L^{-1} and similarly,

$[B^-]$ = s mol L^{-1}

$K_{sp} = [A^+][B^-] = (s\ mol\ L^{-1})(s\ mol\ L^{-1}) = s^2\ mol^2\ L^{-2}$

Therefore, Solubility, $s = \sqrt{K_{sp}}\ mol\ L^{-1}$

Thus, molar solubility is defined as the number of moles of solute that can be dissolved in 1 litre of the solution.

The concept can be smartly employed to find out if a solution is saturated or not. If the ionic product $[A^{y+}]^x [B^{x-}]^y$ is found to be less than the solubility product, K_{sp} the solution is termed as unsaturated and it has the capacity to dissolve more of the electrolyte, whereas if the ionic product and the solubility product are found to be the same then the solution is perfectly saturated at that particular temperature.

3.3 SOLUBILITY OF GASES

Solubility of gases in water is governed by Henry's law. This law states that solubility of gas at constant temperature is directly proportional to partial pressure of gas in contact with the liquid. Consider

A (g) $\rightleftarrows$ A (aq)

Where A(aq) is aqueous concentration of gases in $molL^{-1}$; p_x is partial pressure of gas in atm; K is henry's constant

Units of k = $molL^{-1}atm^{-1}$

Table illustrates the Henry's law constants for common dissolved gases in water at 25ºC.

Gas	K($molL^{-1}atm^{-1}$)
O_2	1.3 x10^{-3}
CO_2	3.4 x10^{-2}
H_2	7.8 x10^{-4}
N_2	6.5 x10^{-4}
CH_4	1.4 x10^{-3}

In order to calculate gas solubility considering an example where concentration of O_2 in water is in equilibrium with air at 1 atmospheric pressure and temperature, a correction must be made for partial pressure of water by subtracting it from total pressure of gas (0.0313 atm) at 25 ^oC. Henry's constant for O_2=1.3 x10^{-3} $molL^{-1}atm^{-1}$ and considering that dry air is 20.95% by volume of O_2 yields the following equation"

PO_2 = (1.0000 atm − 0.0313 atm) x 0.2095 = 0.2029 atm

[O (*aq*)]=k P_{O2} =1.3 x 10^{-3} $molL^{-1}atm^{-1}$ x 0.2029atm

[O_2(aq)]= 2.6 x10^{-4} $molL^{-1}$

Molar mass of oxygen is 32, so concentration of dissolved oxygen in water at 25^{o}C is 8.3mgL^{-1} or ppm.

The solubility of gases decrease with increasing temperature given by Clausius-Clapeyron equation:

$$\log \frac{c1}{c2} = \frac{\Delta H}{2.303\ R}\left[\frac{1}{T1} - \frac{1}{T2}\right]$$

c_1 and c_2 denotes gas concentration at absolute temperatures of T_1 and T_2.

ΔH = heat of solution and its units are in cal mol^{-1}

R = gas constant where R = 1.987 cal $degree^{-1}$ mol^{-1}

At high temperatures, decreased solubility of O_2 combined with increased respiration rates by aquatic organisms results in depletion of oxygen making survival difficult for aquatic species.

3.4 CARBONATE SYSTEM

The major portion of carbon is present in the oceans in form of carbonate. The carbonate system is primarily responsible for controlling the pH and regulates various biogeochemical processes occurring in ocean. Carbon dioxide emitted through natural and anthropogenic sources are taken up by the oceans. This results in limiting the increase in CO_2 concentration and greenhouse warming but increasing acidification of sea water. Rapid exchange of CO_2 occurs between the atmosphere and aquatic system through equilibrium of CO_2(g) and dissolved CO_2

$CO_2(g) \leftrightarrow CO_2(aq)$

CO_2 rapidly reacts with water owing to its high solubility resulting in formation of carbonic acid.

$CO_2\,(aq) + H_2O \leftrightarrow H_2CO_3$ …………………….. (1)

$H_2CO_3 \leftrightarrow HCO_3^- + H^+$ …………………….. …. (2)

$HCO_3^- \leftrightarrow CO_3^{2-} + H^+$ ………………………… (3)

Series of proton transfer reaction takes place resulting in bicarbonate and carbonate ion. Reactions 2 and 3 produces H^+ and lowers the pH of sea water. Dissolved CO_2 occurs in three inorganic forms: free aqueous CO_2, HCO_3^-, and CO_3^{2-}which becomes an integral part of carbonate system. When CO_2 dissolves in sea water, it is not fully dissociated to CO_3^{2-} and number of H^+ are smaller due to natural buffering capacity of sea water as shown below

$CO_2\,(aq) + H_2O + CO_3^{2-} \leftrightarrow 2\,HCO_3^-$

The sum of three aqueous carbonate species $[CO_2\,(aq)]+ [HCO_3^-] + [CO_3^{2-}]$ is defined as total CO_2. Most of the surface water of ocean is approx. pH 8 where HCO_3^- is the dominant species and this represent 90% of DIC. A very small portion of H_2CO_3 is present hence concentration of CO_2 (aq) + H_2CO_3 is combined and represented as $[CO_2\,(aq)]$

Ocean acidification limits the coral life all marine organisms to properly synthesize their carbonate skeletal via the formation of biogenic calcium carbonate, $CaCO_3$.

Due to increase in rate of anthropogenic emissions of CO_2, calcification rate have declined and reduced the alkalinity of the oceans.

SAQ 1

Explain the role of carbonate system in oceans.

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3.5 CHEMICAL SPECIATION

The term chemical speciation refers to specific chemical forms of elements which may be inorganic, organic or organometallic species in the environment. It also refers to the oxidation state in which a particular element exists in the environment.

Metals are non-biodegradable in nature. When they enter in the environment they get partitioned into various physico-chemical forms such as colloidal, chelated, free ionic combined with other chemical forms. The physical properties and chemical behaviour vary amongst different species in the environment. The phenomena such as chemical and biological transformation, metal mobility, bioavailability, toxicity and persistence in the environment depends on chemical form or speciation of a given metal ion. Example: Arsenic is extremely toxic in its inorganic form but arsenobetaine is non-toxic Similarly elemental mercury is non-toxic but alkylmercurials are highly toxic.

The toxicity and mobility of metal ions depends upon:

- Concentration of metal ions which decreases on complexation.
- Redox environment. eg. toxicity associated with As(V) is low in oxidising environment while reduced form As(III) is highly poisonous.
- Oxidation number Cr(III) is essential element as glucose tolerance factor while Cr(VI) is highly toxic.

Because of the above reasons, the determination of the total concentration of a metal provides inadequate information. That is why speciation has gained considerable importance.

Speciation studies involves estimating individual physico-chemical forms of elements that make up the total concentration of a particular element in order to assess its impact on organism/environment.

But at times, the determination of specific concentration of species becomes quite a difficult task than determination of total element because of the following reasons:

1. Isolating the compound of particular interest from complex matrix is difficult.
2. Speciation techniques available disturb the equilibria existing between the various chemical species present in the system under study.
3. Determination of species at ultra-trace level requires analytical procedures of high degree of sensitivity.
4. Suitable standard reference materials are often unavailable.

The speciation analysis of metal ions can be performed by sequential extraction, whereby each of the fractions obtained shows a different form of association of the metal in the soil. The accepted modern procedure is:

Step I. Extraction of the acid soluble fraction (i.e., carbonates) with CH_3COOH

Step 2. Extraction of the reducible fraction (i.e., iron/manganese oxides) with hydroxylamine hydrochloride

Step 3. Extraction of the oxidizable fraction (i.e., organic matter and sulfides) by H_2O_2 oxidation followed by extraction with $NH_4(CH_3COO)$

Chemistry of Cr(III) , Cr (VI) speciation

In natural water, chromium exhibits two common oxidation states +3 and +6 exhibiting different chemical behaviour, toxicity and bioavailability. Cr(III) is associated with lesser toxicity because of limited solubility and strong complex formation. On the contrary, Cr(VI) is highly toxic and much more bioavailable to aquatic living organisms since it is easily transported into cells as sulphate analogues. It is also recognised as human carcinogen and mutagen.

Cr(III) speciation in sea water is dominated by Cr(III) hydrolysis products- $Cr(OH)^{2+}$; $Cr(OH)_3$, $Cr(OH)^{2+}$ and $Cr(OH)_4^-$. $Cr(OH)_3$ is dominant species in the pH range 7-11, while $Cr(OH)_4^-$ becomes dominant above pH 11.3. Chromium is present as Cr(VI) under oxidizing conditions as anionic species- CrO_4^{2-}, $HCrO_4^-$, and dimer $Cr_2O_7^{2-}$

In natural systems, Cr(III) does not migrate since its minerals precipitate in alkaline or neutral range . It remains as dissolved Cr(III) in low concentration. In slightly acidic conditions, it can be is removed by adsorption. Phenomenon of precipitation and adsorption can be inhibited by complexation of Cr(III) with dissolved ligands. Dissolved Cr(III) concentration becomes higher and migrates when pH reaches around 5. Oxidants such as MnO_2 can solubilise Cr(III) in form of Cr(VI) compounds.

Cr(VI) enters in to aquatic system in two ways:

(a) reduction of Cr(VI) to Cr(III). Reduction is facilitated by Fe(II) which may be synthesized by photo-oxidation of organic matter in sea water surface.

(b) Removal of Cr(VI) by sorption process where there is high content of Mn or Fe oxide under oxidizing conditions. But sorption is not strong enough to keep moving Cr(VI) through soil or sediments under alkaline medium.

SAQ 2

What is meant by the term metal "speciation" and why is it important in aqueous systems?

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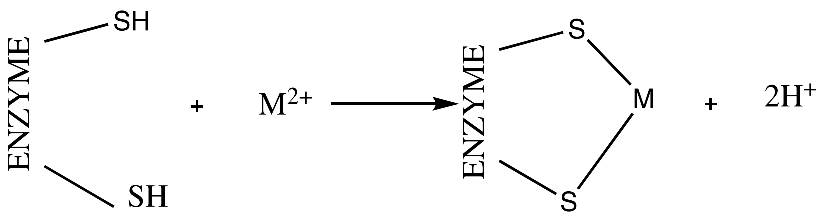
3.6 CHEMISTRY OF HEAVY METALS

The term heavy metal refers to those elements which have density greater than 6 gcm^{-3}.

Heavy metals occur naturally in soils, sedimentary deposits and water bodies. They find their way into the environment through soil, and sediments in form of dust, smoke and aerosol by anthropogenic sources. They are also added to the agricultural land by extensive uses of fertilizer and water bodies through industrial wastes, sewage sludges. Mainly, four heavy metals such as lead, mercury, cadmium and arsenic are of greatest concern because of their extensive use, their toxicities and pervasiveness in the to the environment.

Heavy metals exert their toxic action by binding to the active site of the enzyme thereby inhibiting the enzymatic action. They have strong affinity for the sulfhydryl group (-SH) of the enzyme. Certain enzyme contains metal ions in their structures called metalloenzyme. Their action is inhibited when the essential metal ion of the enzyme is replaced by the toxic meatal ion having same size or charge. Example: substitution of Zn by Cd in Zn^{2+}containing metalloenzyme like carbonic anhydrase, carboxypeptidase, alcohol dehydrogenase.



Schematic representation of toxicity mechanism of heavy metals

Cadmium

Cadmium occurs in a very low concentration in the earth's crust. It is not an essential element and no biological function till date is reported in humans. It is used in industries to manufacture nickel cadmium batteries, control rods in nuclear power industry, pigments, semiconductor devices, stabilizers for PVC and other plastics. Other applications include electroplating metals for protection against corrosion and in non-ferrous alloys. A small amount of Cd occurs in form of impurity in zinc containing ores.

Cadmium in water arises mainly from industrial discharge and from waste mining. It is extremely dangerous to plants and animals because it absorbs efficiently and gets accumulated in the tissues. Rate of Cd absorption in humans increases particularly if they are on low Calcium diet. Majority of the cadmium ingested gets trapped in kidneys and is eliminated. A small portion gets bound to protein metallothionein present in kidneys. Effects of Cd poisoning in humans results in anemia, high blood pressure, damage to testicular tissues.

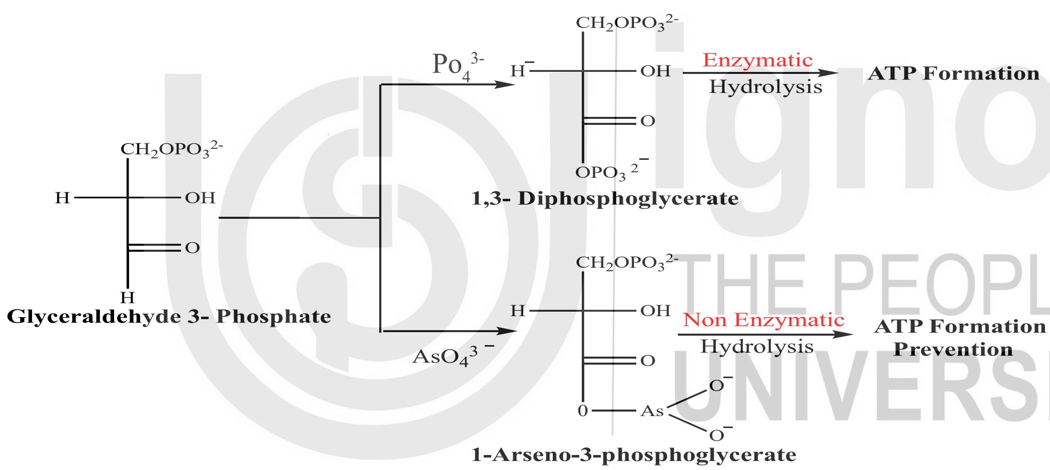
Cadmium poisoning caused itai -itai or ouch -ouch disease that was first recognized in in Jintzu river basin of Japan. Water in the Jintzu basin became contaminated with cadmium from mining activities. This contaminated water was used to irrigate rice fields, a staple food in that area. The bones of the victims become fragile and broke easily from loss of Ca. Many people suffered from abdominal pain, diarrhea, vomiting, liver damage and kidney failure.

Chemically, cadmium is similar to zinc since it occurs in association with zinc mineral and found in similar oxidation states (+2). It is proposed that Cd poisoning involves substitution of Zn by Cd in many metalloenzymes thereby altering the stereostructure of the enzyme and impairing the catalytic activity.

Arsenic

Arsenic is commonly used in pesticides (fungicide, insecticide, herbicide), alloys and in wood preservatives. Arsenic is a by-product used in the manufacture of some chemicals and mining operations. Many subsoils naturally contain arsenic compounds and seep into the groundwater and contaminate the source. In some parts of India and Bangladesh wells were dug up used for irrigation purpose have been epidemics of acute arsenic poisoning. Presence of arsenic in drinking water cause keratoses, skin discolouring which subsequently leads to cancer and damages liver and kidneys.

Arsenic exerts its toxic action by binding it strongly to the sulphudryl groups on protein. The enzyme pyruvate dehydrogenase which generates cellular energy in citric acid cycle , is deactivated on complexation with As^{3+} thus preventing the formation of ATP. Arsenic is similar to phosphorus, so it is reduced from +5 to + 3 oxidation state, so arsenic inhibits all biochemical reactions of phosphorus. Arsenic interferes in synthesis of ATP producing 1-arseno-3-phosphoglycerate instead of 1,3-diphosphoglycerate which on hydrolysis gives arsenate and 3-phosphoglycerate as shown in below diagram.



The three major biochemical effects of arsenic are :

1. Protein coagulation
2. Complexation with co-enzyme
3. Uncoupling of phosphorylation

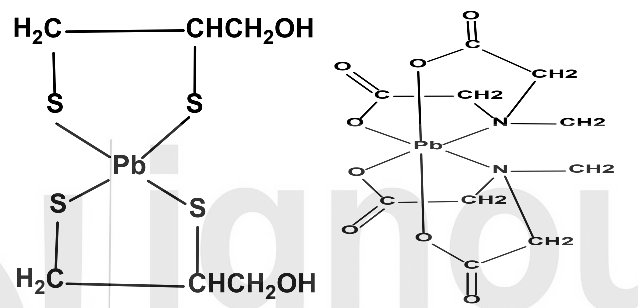
As(III) [AsO_3^{3-}] is more toxic than As(V) [AsO_4^{3-}) because it it undergoes methylation by bacteria that produces species that are highly reactive and damages DNA. Antidotes for As poisoning includes chelating agents like British anti Lewsite (BAL)which possess -SH capable of binding strongly with As(III)

Lead

The sources of inorganic lead include industries, mining and from leaded gasoline. Lead is most abundant heavy metal in the atmosphere. The major source of atmospheric lead comes from Et_4Pb which is added to gasoline to inhibit knocking and improve fuel efficiency. Lead enters the human system by inhaling lead and swallowing lead salts. When it enters the blood stream as Pb^{2+} and it becomes biologically active highly and potentially more toxic.

Lead when present interferes in the heme synthesis which leads to hematological damage. Presence of lead disrupts heme synthesis by inhibiting the conversion of δ-aminolevulinic acid to prophobilinogen catalyzed by δ-aminolevulinic acid dehydrogenase. It does not permit utilization of oxygen and glucose for energy production. High levels of lead in blood results in anemia due to deficiency of hemoglobin. Chronic effects include causes kidney malfunction, increased blood pressure and interference in vitamin D metabolism. Lead exposure is highly dangerous to unborn and young children. It crosses the placenta barrier and causes premature births and mental impairements. Chemically, lead resemble calcium, so it accumulates in the body replacing calcium.

Lead poisoning can be treated by administering chelating agents which bind strongly to Pb^{2+}.Ca-EDTA complex is given to patients poisoned by lead, it displaces Ca^{2+} from chelated complex and resulting complex can be excreted in urine. Other chelating agents used to treat lead poisoning includes BAL, d-penicillamine.



BAL= 2,3-dimercaptopropanol Pb-BAL Chelate

Mercury

The main source of mercury is the principal ore cinnabar i.e. HgS. It is used industrially in three different forms: elemental mercury, organic mercury compounds and inorganic mercury. The greatest use of mercury is production of electrical apparatus. It forms amalgams with various metals like zinc, lead, tin, cadmium and widely used as fungicide. Mercury(II) chloride is used to make organo-metallic compounds, as a wood preservative, and as an antiseptic for sterilising surgical instruments. Mercury(II) sulphide is used in the red pigment, vermilion.

A s result, mercury and its compounds enters the natural bodies from manufacturing or sources where mercury is mined. The major sources of mercury emissions into the atmosphere is through burning of fossil fuels, electric power plants, waste incineration or weathering of rocks

All forms of mercury are highly toxic but toxicities vary considerably. Elemental mercury is non-toxic but its vapours when inhaled are highly toxic because it can diffuse through lungs into the blood and then brain. Hg^{2+} is fairly toxic but does not cross the biological membrane though it has strong affinity for the sulphur containing amino acids and forms bonds with haemolglobin through sulphydryl groups. Hg^{1+} is non-toxic when ingested because it forms insoluble Hg_2Cl_2 in the stomach in acidic medium. Alkyl mercurial RHg^+ are highly toxic forms of mercury compounds, because they are highly soluble in fat and brain tissues. Hg-C bond is not easily disrupted stable and remains in tissue cell for prolonged period. They have the ability to easily cross the blood-brain and placental barrier.

Chemical species of mercury and their chemical and biochemical properties

Species	Chemical and biochemical properties
Hg	Non-toxic, but vapours when inhaled highly toxic
Hg_2^{2+}	Not very soluble, low toxicity
Hg^{2+}	Toxic but not transported across biological tissues
CH_3Hg	Highly toxic, highly liophilic i.e easily transported across biological membrane
$(CH_3)_2Hg$	Low toxicity but easily transformed to CH_3Hg

Biological methylation

The transformation of inorganic form of mercury(Hg^0, Hg^{1+}, Hg^{2+}) to methyl and dimethyl derivatives by the action of anerobic bacteria in sediments and lakes is referred to as methylation. This transformation is mediated by methylcobalamin, Co(III) containing vitamin B_{12} co-enzyme. $-CH_3$ group bonded to co-enzyme is enzymatically transferred by methylcobalamin to Hg^{2+} giving CH_3Hg^+ or $(CH_3)_2Hg$.

Under acidic conditions, conversion of dimethylmercury to methylmercury occurs which is soluble in water. It is the methyl mercury form which enters into the food chain through phytoplankton. In this way concentration of mercury is build up at every level. In alkaline medium, methyl mercury can be converted to volatile dimethylmercury, $(CH_3)_2Hg$, may be formed which evaporates relatively quickly both from soils and water bodies.

The deleterious effects of the mercury came into limelight after the incidence of Minamata Bay in Japan during the period 1953-1960. The non-toxic inorganic mercury was discharged as effluent into the Bay by Minamata chemical plant manufacturing acetaldehyde using mercuric sulphate as catalyst . In the sediments, inorganic mercury was converted into methyl mercury which got accumulated into the fish which was eaten up by local inhabitants. This resulted in death of hundred people and thousands were permanently paralysed . Congenital defects were in new born babies whose mothers fed themselves with sea food contaminated by methyl mercury. Other effects includes impairment of vision, hearing and neurological damage. This was the first natural case of bioaccumulation (in fish) of a toxic species (methylmercury) which killed hundreds of people and caused genetical damage.

Chronic effects caused by mercury poisoning mercury includes neurological damage including increased excitability, irritability, chromosomal damage and birth defects and kidney problems.

SAQ 3

Explain the toxic effects of heavy metals on enzymes?

SAQ 4

Explain the biochemical effects of Heavy Metals

a) Arsenic b) Lead c) Mercury with reference to their source , species, pathways in environment and impact on humans.

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3.7 RADIONUCLIDES

Radionuclides are species of atoms that emit radiation when they undergo radioactive decay through emission of alpha particles, beta particles and gamma rays. Table provides the characteristics of defined decay processes:

Radiation	Symbol	Charge	Transformation
Alpha (α)		+2	Mass number decrease by 4, atomic number decreases by 2
Beta (β)		-1	Atomic number increases by 1, mass remains unchanged
Gamma (γ)	$^{0}\gamma_{0}$	0	No change in atomic number or mass

When α or β particles are emitted , the identity of elements is changed

e.g. decay of U-238 to Th-234,

$$^{238}_{92}\mathrm{U} \rightarrow {}^{234}_{90}\mathrm{Th} + {}^{4}_{2}\mathrm{He}$$

In contrast to a emission, β decay results in increase in atomic number by 1, while mass number remains the same.

$$^{14}_{6}C \rightarrow {}^{14}_{7}N + {}^{0}_{-1}e$$

Gamma rays are highly energetic electromagnetic radiations similar to X-rays having no charge or mass.

^{238}U undergoes radioactive decay by forming chain of products through - variety of alpha and beta emissions ultimately leading to stable Lead-206 . The transformation of Ra-226 to Rn-222 is of particular environmental significance. Radon is a colorless, odorless, tasteless gas that naturally occurs in rocks, soil, water, and air. Radon in the ground, groundwater, or building materials invades working and living spaces and disintegrates into its decay products. It is second leading cause of lung liver cancer. Decay of radionuclides follow first order reaction

Radiation damages living organisms by initiating harmful chemical reactions in tissues. In case of acute radiation poisoning, bone marrow is destroyed and concentration of RBC is decreased. Radionuclides having very short life are highly dangerous but decay fast to affect the environment. In contrast radionuclides having very long half-life are quite persistent but may cause very little damage. But radionuclides having intermediate half-lives are highly dangerous and can pose threat. Sr-90 is the waste product of nuclear testing is interchangeable with Ra and K-40 are found in natural sources like water. Radionuclides are formed in large quantities as waste products in nuclear power plants. Artificially produced radionuclides find their widespread use in industrial and medical applications particularly as tracers.

SAQ 5

What characteristic features of radionuclides make them hazardous ?

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3.8 SATURATED AND UNSATURATED HYDROCARBONS

Majority of the hydrocarbons have their widespread use in fuels and therefore represents the most important organic pollutant of the atmosphere. The anthropogenic source of hydrocarbon pollutant encountered in atmosphere comes from incomplete combustion of gasoline (petroleum product). Hydrocarbons are broadly classified as

- **Saturated Hydrocarbon**: comprising of alkanes (also referred to as paraffins)
- **Unsaturated Hydrocarbons**: comprising of a. *alkenes* (olefinic compounds with double bonds between adjacent carbon atoms) b. *Alkynes* (compounds with triple bonds between adjacent carbon atoms) c. *aryl compounds* (aromatic)

Alkanes have single bond between the carbon atoms and represented by the general formula: C_nH_{2n+2}, Cyclic alkanes :C_nH_{2n}. examples include: methane, 2,2,3-trimethylbutane

The most stable hydrocarbons present in the atmosphere are alkanes. Straight chain alkanes with carbon atoms 1-30 and branched hydrocarbons with 6 carbon atoms are present in polluted atmosphere either as gas or as aerosol or may adsorbed to atmospheric particles.

The hydroxyl radicals present in atmosphere react with alkanes to form alkyl radical which then immediately picks up a molecule of oxygen leading to the formation of alkylperoxy radical. This radicals acts as oxidant by losing an atom of oxygen to form alkoxy radical. These reactions of alkanes leads to the oxidized products that are precipitated from atmosphere with particulate matter and finally undergoes biodegradation in soil.

$$C_xH_{2x+2} + HO\bullet \rightarrow C_xH_{2x+1}\bullet$$
$$C_xH_{2x+1}\bullet + O_2 \rightarrow C_xH_{2x+1}O_2\bullet$$
$$C_xH_{2x+1}O_2\bullet \rightarrow C_xH_{2x+1}O\bullet$$

Alkenes have atleast one C=C double bond per molecule. The simplest alkene is ethylene. They are widely used to synthesize polymers for plastics (polyethylene, polypropylene, polystyrene), synthetic rubber (polybutadiene,) and various applications. They are emitted into the atmosphere through internal combustion engines, petroleum refinery, industrial applications.

Alkynes have triple bond between the carbon atoms and they are less commonly present in the atmosphere than alkenes. acetylene which is used as fuel for cutting and welding and butyne used in manufacture of synthetic rubber are sometimes found in raceable amounts in atmosphere. Chemically, alkenes are more reactive than alkanes since they possess electron available to form additional bonds with other atoms. In the presence of NO_x and sunlight, alkenes (e.g. propylene)react readily with $OH^{.}$ radicals and subsequent addition of O_2 leads to formation of peroxyl radical. These reactions are quite significant in the atmosphere as they participate in photochemical smog formation and contributes to atmospheric pollutants.

Aromatic (aryl)hydrocarbons consists of rings of six carbon atoms having alternative single and double bond. Aryl hydrocarbons find their widespread application as raw materials for the manufacture of monomers and plasticizers.

Simplest example is benzene. It is emitted from the exhaust engine of all petrol and diesel vehicles as well as from chemical industries. Aromatic compounds with multiple rings referred to as polycyclic aromatic hydrocarbons are of greatest concern. Most important example is benzo (∝) pyrene, compound that the body can metabolize to a carcinogenic form.

Features

- They are persistent organic pollutants because they do not break down easily.
- They are volatile making them highly mobile throughout the environment.
- These compounds are the most stable form of hydrocarbons having low hydrogen-to- carbon ratios They are formed by the combustion of petrol and diesel hydrocarbons under oxygen- deficient conditions.
- They are found deposited on surface of particulate matter present in smoke.

The most potent carcinogenic PAH's include : benzo-α-anthracene, dibenzo-a-anthracene and benzo-α-pyrene.

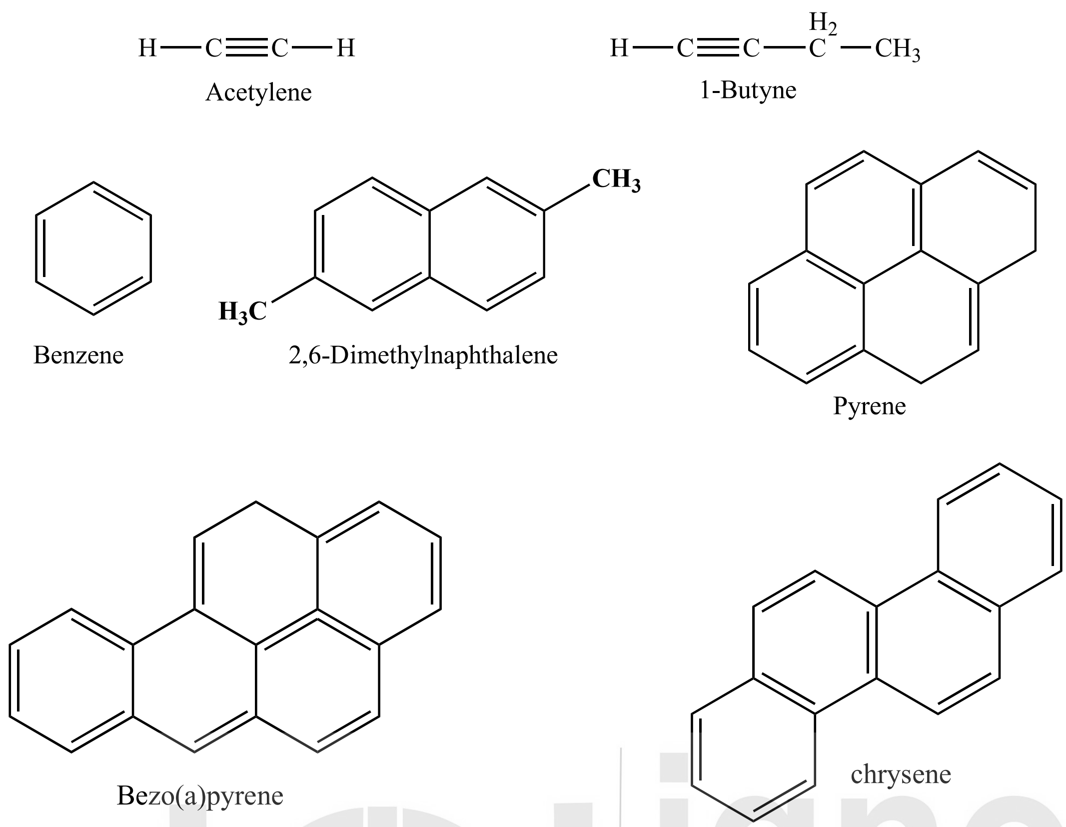

Figure 3.1 : Structural formulas of different types of hydrocarbon pollutants present in environment

SAQ 5

Explain how aromatic hydrocarbons act as environmental pollutants

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3.9 CHEMISTRY OF FUELS

Gasoline

The first product obtained in distillation of crude oil is gasoline. Generally, the boiling range of gasoline is 25-150°C. Gasoline are sometimes classified as motor gasoline for use in automobile and light trucks, aviation gasoline for use in piston engine aircrafts. Hydrocarbons having 3-8 carbon atoms condensed from natural gas are composed of natural gasoline. Natural gasoline does not provide good combustion performance desirable for automobile engine. Consequently, it is now used mainly as feedstock for petrochemical industry.

The gasoline produced from distillation of crude oil without any chemical treatments is straight run gasoline. It comprises primarily of straight chain hydrocarbons and is of limited value as automobile fuel. A high content of straight chain hydrocarbons causes fuel to begin burning before the spark plug ignites it. This pre-mature ignition produces an acoustic sound called knocking. Knocking is combustion phenomenon which takes place when air fuel mixture in the gasoline does not burn smoothly.

When the fuel in a gasoline engine is ignited by the spark plug, the flame tends to move away from the spark plug, and the heat and pressure is generated. Chemically, combustion is a multistage oxidation process where the organic hydroperoxides are generated initially. These decompose to form free radicals that start branch chain reactions. These chain reactions start local auto ignitions/detonations, ahead of the flame front in the unburned hydrocarbon air mixture due to the high temperature and pressure generated in the last portion of the fuel. This violent detonation creates pressure fluctuations and noise. This noise is called knock, and damages engine parts nd the fuel efficiency of the engine is decreased.

The tendency of gasoline to cause knocking is rated according to a scale called octane number. It measures fuel's ability to avoid knocking. Branched hydrocarbon 2,2,4- trimethylpentane (isooctane)was found to be superior fuel that burned without causing knocking and is assigned an octane number of 100. Straight chain hydrocarbon n-heptane produced severe knocking and was assigned octane rating 0.

The octane rating of gasoline is determined by burning it in a special single- cylinder engine and comparing its anti-knocking properties with the standard mixtures of isooctane and heptane.

Table: Octane numbers of various hydrocarbons

Hydrocarbon	**Formula**	**Octane Number**
n-butane	C_4H_{10}	94
n-pentane	C_5H_{12}	62
2-methylbutane	C_5H_{12}	94
n-hexane	C_6H_{14}	25
2-methylpentane	C_6H_{14}	73
Heptane	C_7H_{16}	0
2-methylhexane	C_7H_{16}	42
2,3-dimethylpentane	C_7H_{16}	90
2-methylheptane	C_8H_{18}	22
2,3-dimethylhexane	C_8H_{18}	71
2,2,4-trimethylpentane	C_8H_{18}	100
Benzene	C_6H_6	106
Toluene	C_7H_8	118
o-xylene	C_8H_{10}	107
p-xylene	C_8H_{10}	116

From table, it is concluded that straight run gasoline has octane rating between 20-55 which is too, low to be used in automobile engine.

Octane number can be increased by three ways

- Cracking
- Catalytic reforming
- Anti-knocking compounds /octane boosters

Cracking is the process in which long chain hydrocarbons are converted to short chain hydrocarbon. The octane number of alkanes decreases as the carbon chain is lengthened but increases as chain branching increases. (Table)

Catalytic reforming is the process in which hydrocarbon vapours of straight chain gasoline are converted into branched or aromatic hydrocarbon in presence of catalyst to increase octane rating .

Chemistry of Gasoline Fuel Additive

The fuel additives in very small dosage are added to enhance the performance of the fuel which otherwise cannot be obtained through refining process.

Reasons for using additives in fuels include

- Reduction in emission from combustion of fuel
- Improvise combustion properties of fuel
- Provides cleanliness and protection to engine
- Increases stability of fuel.

There are six types of gasoline fuel additive:

1. **Antioxidants:** During oxidation of gasoline, free radical chain reaction initiates which results in formation of hydroperoxides ultimately leading to the formation of insoluble gum as the end product. Anti-oxidants minimizes the oxidative degradation of gasoline fuel and prevents the gum forming tendency of the fuel. Antioxidants consists of hindered phenols, and aromatic diamines. Examples: 2,6-Di-tertiary butyl-para-cresol and 2,6-di-tertiary butyl phenol Phenylenediamine

2. **Metal deactivators/Passivators:** The presence of soluble metal salts in gasoline provides instability to fuel by catalysing oxidation reaction which forms gum and reduces fuel stability. Most commonly used metal deactivator is N,N-disalicylidene-1,2-propanediamine. This compounds can form chelate with copper metal in gasoline.

3. **Octane Booster:** It provides cost effective improvement in gasoline octane quality by reducing knocking. Examples : teraethyl lead (TEL), methylcyclopentadienyl tricarbonyl (MMT)' oxygenate additives such as ethanol, methyl tertiary butyl ether (MTBE) considered as blending components rather than octane boosters.

4. **Anti-icing additive**: Anti-icing additives minimizes engine stalling and power loss due to formation of ice crystals which may restrict the flow of fuel to engine. It also provides protection against microbiological growth in fuel.It

comprises of freezing point depressants in particular glycol ethers example: ethylene glycol methyl ether.

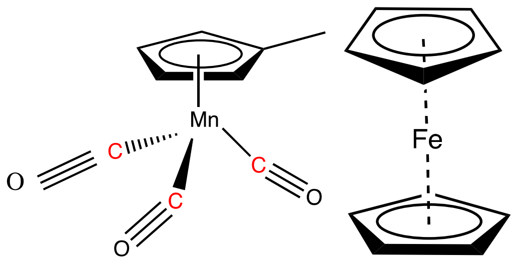
5. **Combustion improvers**: are used to heat the fuels and contains ash-less and metal additives. Examples: tertiary butyl acetate enhances antiknock characteristics of TEL; metal additives e.g. Compounds of boron and Phosphorus increase the ignition point of engine. Incomplete combustion leads to soot deposits in the combustion chambers, and impairs heat transfer and may result in unwanted fires. Combustion improver additives minimizes misfire and improves efficiency of the engine through reduced soot deposition,.

6. **Deposit control additives/detergent dispersants**: It consists of polar hydrophilic and non polar hydrophobic group in its structure. and which is derived from oxygen or nitrogen molecules and a which The nonpolar group enables enables fuel to be soluble in additive. The polar group attaches to the deposit particles and removes them from the metal surface through the detergent action. Control Additives controls the build up of soot deposit in carburettor and fuel injection equipment and inlet system by forming protective films. This will provide performance benefits such as reduced exhaust Emissions, improved fuel economy and reduced maintenance cost. Examples include amides, amines, polybutene succinimides, polyether amines, polyolefin amines and mannich amines.

Anti-knocking Agents

Knocking in the internal combustion engines affects the efficiency of the vehicle and fuel economy. It can be minimized by engine design and adjusting operating conditions or by using high octane gasoline. Most widely used additives to control knocking includes alkyl lead compounds such as tetraethyl lead (TEL) , tetramethyl lead (TML) and hence improves the performance of the engine. Addition of lead additives deactivates the chain branching reactions in the pre-ignition phase . When air-fuel mixture becomes compressed and heated, the weak bonds between the alkyl lead breaks releasing the free lead atom. This then reacts with free O2 to form PbO and PbO_2 which tends to deposit in the combustion chamber This problem can be solved by using scavengers such as Ethylene dibromide and ethylene dichloride by converting non-volatile PbO_2 to volatile PbX_2 In this way lead is removed from the engine and released into the atmosphere. Gradually leaded gasoline was phased out by unleaded gasoline. The reason was that catalyst like Rh, Pt used in catalytic convertors were poisoned by lead and rendering them inactive. Several alternative antiknock agents were proposed and developed as substituent of lead-containing fuels, for example: methyl-cyclo-pentadienyl-manganese- tricarbony (MMT], ferrocene iron pentacarbonyl.

Methylcyclopentadienyl manganese tricarbonyl (MMT) an organometallic was used as an antiknock additive. This compound releases Mn atom which is then converted to Mn3O4 particles inside the combustion chamber which capture hydrocarbon radicals and inhibits knocking. Toxicity associated with Mn is low once it is released into the environment. But it was banned by USEPA because this additive was found to be toxic when ingested or inhaled. Secondly, fouling of spark plugs and of on board sensors had been reported. Recently it has been permitted in US at level equivalent to 1/32 g of Mn gal-1of gasoline. MMT is suspected to be a strong neurotoxin and respiratory toxin. Additionally, MMT impairs the effectiveness of automobile emission control and increases pollution of vehicles. Knocking can be reduced by altering the composition of gasoline by reducing the fraction of low octane components and increasing the fraction of high octane components. In United Nations, removal of alkyl lead compounds was initially compensated by increasing the content of aromatic compounds particularly benzene, toluene, and xylene. Increasing the aromatic components in gasoline boosted octane rating.



Methylcyclopentadienyl manganese tricarbonyl (MMT) Ferrocene

The cause of BTX has decreased though it has high octane rating because

- Xylenes react with hydroxyl radicals and have greater potential to smog formation than alkanes
- Benzene is a carcinogen though having low photochemical activity.

This prompted the need to develop alternative fuel additives for improving octane rating of gasoline.

Focus has been laid on use of oxygenated compounds containing one or more oxygen atoms. These include ethanol, methanol, methyl tertiary butyl ether (MTBE), ethyl tertiary butyl ether (ETBE) whose octane rating >100. MTBE is made by reacting methanol and isobutene in presence of acid catalyst.

It is synthesized in refineries in which isobutene is generated during catalytic cracking process. However, MTBE contaminates ground water storage tanks through gasoline leak. Its presence has unpleasant odour and is unacceptable in water supplies. Environment Protection Agency recommended reduction its use as gasoline oxygenate. Hence, ethanol and methanol emerged as gasoline oxygenate Ethanol is added in amounts upto 10% of total volume to produce gasoline blends known as gasohol. This mixture has been used as fuel without any modification of automobile.

The advantage of using MTBE and ethanol is that they have low OH· reaction rates and vapour pressure. It is because of this reason that oxygenates in RFG was mandated upto level of 2% oxygen by weight according to 1990 Clean Air Act amendments which requires reduction in ozone forming VOC. Oxygenated fuels provide good combustion performance along with reduced emission of CO and NOx.

Ethanol separates out from gasoline when in contact with water because of its extreme solubility. Vapour pressure of ethanol is twice than MTBE and requires special adjustments in gasoline components to lower volatile organic compounds .By removing lighter hydrocarbons from blend lowers gasoline volatility . In case of MTBE, vapour pressure requirements is met by removing butane fraction but replacing MTBE with ethanol requires removing pentane thereby increasing the cost. With introduction of RFG smog forming precursors has been reduced.

Disadvantages of alcohols, especially methanol, produce aldehydes as by-products during combustion. Since aldehydes are not major components of gasoline emissions, their emission from combustion sources is not regulated. Aldehydes can be highly irritating to the mucous membranes of the respiratory tract, and thus are undesirable in air. This problem can be remedied by using catalytic converters in the exhaust system to insure the complete oxidation of the aldehydes to carbon dioxide and water.

SAQ 6

Why lead components are added to gasoline and why was it removed with the advent of catalytic convertors ?

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SAQ 7

What advantages do oxygenates offer when added to gasoline in comparison to lead compounds.

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3.10 LUBRICANTS

Lubricant is as substance which is introduced between two sliding surfaces to reduce friction and wear rate. This process of reducing friction by application of lubricant is referred to as lubrication.

Functions of lubricants

- Reduces friction and wear and tear of surfaces
- Acts as coolant by dissipating heat
- Prevents entry of moisture

- Minimizes power loss in internal combustion engine
- Reduces maintenance cost

Mechanism of Lubrication

The phenomenon of lubrication can be explained by three types of mechanism :

1. **Thick Film Lubrication or hydrodynamic lubrication**: In hydrodynamic lubrication, the sliding surfaces are separated from each other by a thick fluid film of 1000A to avoid surface to surface contact. The thick film covers irregularities of the surfaces a so that metal surfaces do not come in contact with each other which finally reduces the friction. The lubricant selected has minimum viscosity to reduce internal resistance and should remain in place and separate the surface. Hydrocarbon or mineral oils (having low molecular carbon atoms) are preferred lubricants for this type lubrication.

2. **Thin film or boundary Lubrication**: Boundary lubrication occurs when continuous fluid film cannot persist either due to low speed , high load or low viscosity of lubricating oil. The clearance space between the sliding surfaces is lubricated with a material that gets adsorbed on metallic surface by physical or chemical means which keeps the metal surfaces apart . Vegetable or animal oils and their soaps can be used for this type of lubrication. Though vegetable oils possess good adhesion property but they tend to break down at elevated temperatures.

3. **Extreme Pressure Lubrication:** When sliding surfaces are under high pressure and speed liquid lubricants fail to stick and decompose. To sustain such extreme pressure conditions, additives are added to mineral oils . These additives form strong films capable of sustaining high loads at elevated temperatures. The additives include organic compounds such as sulphur in sulphurized oils or phosphorus in tricresyl phosphate.

3.10.1 Classification of Lubricants:

Based on shear strength lubricants are classified into three types:

1. Lubricating oils or Liquid lubricants
2. Greases or semi-solid lubricants
3. Solid Lubricants
4. **Lubricating Oils or Liquid Lubricant:** They are classified into three categories:

   i. **Animal and vegetable oils:** Vegetable oil is extracted from castor or rapeseed oil while animal oil extracted from animal fats such as fish comprising of fatty acids and alcohols. They have good sticking phenomenon and can withstand high loads at high temperatures and pressures. But they suffer from following drawbacks :

      - Expensive
      - Undergo oxidation to form gummy product
      - Hydrolyse easily in presence of moist air

ii. **Mineral or Petroleum oil:** They are obtained by fractional distillation of petroleum comprising of lower molecular weight hydrocarbons having 12-50 carbon atoms. They are mostly used lubricants due to its high stability, abundance, and less costly. But oiliness of mineral oil is less, hence e higher molecular weight compounds such as oleic acid and stearic acid are added.

iii. **Blended Oils:** Vegetable/animal or mineral oil suffer from drawbacks, hence additives are added to such oil to increase their performance . Such additives added to lubricating oils are referred to as blended oil. Examples :stearic acid, palmitic acid, coconut or castor oil.

5. **Semi-Solid Lubricants or Grease**: A semi-solid lubricant obtained by combining lubricating oil of low viscosity which is thickened by addition of finely dispersed solids called thickeners. It consists of followings:

- **Base oil:** It is the principal component and can be either petroleum oil or a synthetic hydrocarbon of low to high viscosity.
- **Additives:** are added to grease to increase their performance. Extreme pressure and friction modifiers are commonly used examples include: zinc alkyl dithiophosphate, dibenzyl sulphide
- Thickeners: It comprises of two basic types: inorganic and organic thickeners. Organic thickeners are either soap based or non-soap based while inorganic thickeners are non-soap based. Simple soap based thickeners are formed by combing fatty acid or an ester with an alkali or alkaline earth metals like Na, K, Ca, by applying heat and pressure through process known as saponification.Non soap thickeners include synthetic polymers and silica gel etc.

Advantages of grease

- Reduces noise and vibration
- Water resistant and reduces oil vapour problem
- Prolong the life of worn parts
- Support much heavier load at lower speed.

Disadvantages

- Cannot effectively dissipate heat from the bearings, so work at relatively lower temperature
- Contaminants cannot be separated

6. **Solid Lubricants**: Solid lubricants are used in powder form or with binders so that they firmly stick to the metal surface. Also available in dispersion in non-volatile carriers such as soap, waxes, fats etc.

They are used under following conditions:

- Where high loads are required
- Operating temperatures and pressures are too high

- lubricating film cannot be protected by grease
- Contamination by grease

Classification of solid lubricants

- Inorganic lubricants with lamellar structure: The crystal lattice of these lubricants has layered structure with hexagonal rings that forms parallel layers. Each atom is bonded to another atom by weak Vander walls thus allowing sliding movements of plane. Examples include : ***Graphite*** is most commonly used solid lubricant in powder or suspension form. The high thermal and oxidation stabilities at temperature of 500-600° C makes it effective to be used at high temperatures and high sliding speeds. It has layered structure which enables the parallel layers to slide over one another thus making it an efficacious lubricant Also the layer of graphite has a tendency to absorb oil and to be wetted of it.
- Oxides: ZnO, MoO_2, Re_2O_7, B_2O_3 etc.
- Soft metals: Some soft metals possess lubrication properties due to low shear strength and high plasticity. Eg. Pb, Sn, Bi, Cd. They are used in pure form or as alloys and applied as coatings.
- Organic lubricants with chain structure of polymeric molecules: examples: polytetrafluoroethylene (PTFE) . The structure consists of long chain molecules parallel to each other held by weak forces so that they may slide past one another at low shear stress.
- Soaps: metal salts (Li, Na, K, Ca) of fatty acids.

3.10.2 Properties of Lubricants

1. Viscosity: It is defined as the resistance offered due to the motion of the forces between the fluid molecule. It is one of the most important property of lubricating oil. The formation of fluid film of lubricant between the metal surfaces depends upon the viscosity. High viscosity of lubricating oil results in excessive friction while because of low viscosity, film cannot be withstand between the sliding surfaces. Viscosity of liquid decreases with increasing temperature, hence as operating temperature rises, lubricating oil becomes thinner due to low viscosity. Viscosity of lubricating oil should such that it should not change with temperature changes.

2. Corrosion Stability: Steel or copper corrosion test are carried out to estimate the corrosive stability of lubricant. Corrosive substances like hydrogen sulphide , polysulphides are present in petroleum products which are removed by refining process. In copper corrosion test, a piece of copper strip is placed in lubricant for stipulated period of time and defined temperature to examine for tarnishing of strip. Steel corrosion test is carried out to determine the ability of oil to prevent corrosion of ferrous parts in water. Inhibitors like zinc dithiophosphate is added to prevent corrosion of lubricating oil.

3. Flash point and fire point: One of the important characteristic of good lubricant is that it should not volatilise under operating temperatures. Flash point is

defined as minimum temperature at which oil vapourizes to ignite momentarily when flame is brought near to it. Fire point is the minimum temperature at which lubricant oil burns continuously for atleast five seconds when flame is in contact with oil. The temperature of fire point is 10-40 C higher than flash point. The requirement for good lubricant is that it should have flash point above the operating temperature thus ensuring the safety against any fire hazards.

4. Aniline Point: Aniline point is defined as minimum equilibrium solution temperature for equal volumes of aniline and lubricant oil. It indicates deterioration of oil when in contact with rubbing sealings. Aromatic hydrocarbons have ability to dissolve natural and synthetic rubbers. A high value of aniline point indicates high paraffinic hydrocarbons and low percentage of aromatic hydrocarbons. Low content of aromatic hydrocarbon is preferred criteria for good lubricant oil. Aniline point is determined by mixing equal proportion of lubricating oil and aniline in a test tube and heating till a homogeneous solution is obtained. The test tube is then cooled till two phases separates. This temperature is recorded as aniline point.

5. Cloud point and pour point: Petroleum products are complex hydrocarbons and do not exhibit fixed freezing point. When they are cooled, they become solid due to congealing of hydrocarbons. Cloud point is the temperature at which crystallization of solids occur in form of haziness, while temperature at which oil just ceases to flow when cooled is called pour point. Cloud and pour point determines suitability of lubricant under cold environment. Examples include refrigerators plants and air craft engines which operates at sub-zero temperatures requires lubricants having low pour point which otherwise cause solidification of oil and jamming of machine parts.

SAQ 8

What are different types of mechanisms for lubrication?

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SAQ 9

What are important characteristics for a good lubricant?

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3.11 BIOGAS

Increasing production of organic waste is one of the major environmental concerns of the present scenario. Sustainable waste management as well as waste prevention have become major goals to reduce pollution and greenhouse gas emissions to mitigate global climate changes. Uncontrolled waste dumping or incineration of organic waste are now a days not recommended practices for waste disposal from environmental point of view. This problem can be circumvented by production of biogas by degrading biological waste (manure, sewage, sludge, solid waste) in the presence of anaerobic bacteria.

Biochemical reaction of biomass can be represented as :

$$2[-CH_2O] \rightarrow CH_4 + CO_2$$

When micro-organisms have acted upon the biomass under anaerobic conditions, it results in formation of digestate and biogas. Digestate refers to decomposed substrate which is rich in nutrients and used as potent plant fertilizer.

Biogas is a combustible gas chiefly comprising of methane (50-70%) and carbon dioxide CO_2 (25-30%) as primary constituents while others include H_2 (0-1%), N_2(0-10%), H_2S (0-3%) O_2(0-2%).

Examples of biogas include :

- Natural gas produced by decaying animal and vegetable waste buried under the earth's surface.
- Gobar gas produced by anaerobic fermentation of cattle dung.
- Sewage gas-produced from municipal solid waste including human and animal faeces

The main advantage of using biogas through anaerobic digestion is that biodegradable waste is treated thus destroying disease causing pathogens and reducing the volume of the sludge generated. Secondly, value of the waste is conserved so that it serves as fertilizers for the crops.

Gobar Gas

Gobar gas is produced by degrading cow dung anaerobically. The plant consist of a digester covered with dome shaped roof made up of concrete fitted with a outlet pipe with a gas valve. There is a sloping inlet on one side and a outlet chamber on the other side. Manure along with wate slurry is directed to the digester through the inlet. In the presence of anaerobic bacteria, cattle dung undergoes fermentation resulting in the evolution of biogas. Pressure is built up on the slurry due to the evolution of biogas which forces the spent slurry to the overflow tank via outlet chamber. The residue (digestate) left after is withdrawn periodically and used as biofertilizer.

The digesters used are generally located in close proximity to dwellings or sources of the waste.

Biogas is widely used in villages owing to its simplicity in implementation and work up procedure and use of inexpensive raw material.

Advantages:

- It burns without smoke causing no pollution to the environment
- Possess high calorific value
- Cheap
- Clean and convenient
- Involves no storage problem

Disadvantages:

- Dependence on cattle dung from which it is produced through anaerobic fermentation
- Gas stove or burner should be in vicinity of 10 m of the plant.

SAQ10

Give reasons why production of biogas should be encouraged ?

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3.12 LET US SUM UP

This unit explained about

- Chemistry of the concept of solubility, solubility product and solubility of gases
- governed by Henry's law.
- How carbonate cycle dominates pH control in most of aquatic system and regulation of various biogeochemical processes in environment.
- Role of Chemical speciation in assessing the toxicity associated with different forms of chemical element and their oxidation state illustrating it with arsenic at
- Sources and toxicity of heavy metals such as Pb, Cd, As, Hg in the environment
- Different types of radionuclides and its effects.
- Impact of hydrocarbon (saturated and unsaturated) on environment.
- Chemistry of gasoline, additives added to gasoline to increase performance of engine, and how knocking can be averted by adding antiknock compounds.
- Mechanism, functions and classification of lubricants into lubricating oil, greases and solid lubricants.
- Degradation of biological waste into biogas.

3.13 GLOSSARY

Additive: The substance added to increase the performance properties

Antiknock: Resistance to pinging in sparked engines

Anti-knocking agent: Substance which is added to internal combustion engine to reduce knocking

Aniline point: Temperature above which equal proportion of petroleum products and aniline are miscible

Biogas: Combustible gas synthesized by decomposing biological waste under anaerobic conditions

Cloud Point: Temperature at which solid substance crystallises from the solution and imparts haziness

Cracking: Process to break down high molecular weight hydrocarbons to lighter ones

Octane number: Number which denotes antiknock characteristic of gasoline

PAH: Polyaromatic hydrocarbons comprises of two or more fused aromatic rings

Gasoline: Mixture of hydrocarbons with small amount of additives added to internal combustion engine.

Oxygenates: Oxygen containing compound used as gasoline blending agent eg. ETBE, MTBE, TAME

Pour Point: Lowest temperature at which oil flows when it is chilled under definite conditions

Solubility product: Equilibrium constant for a chemical reaction in which solid ionic compounds dissolves to yield its ions in solution

3.14 SUGGESTED READINGS

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3.15 TERMINAL QUESTIONS

1. Define solubility product.
2. Name the predominate carbonate species in natural waters of oceans.
3. Name the antidotes that can be used for heavy metal poisoning.
4. For each of the following pairs of hydrocarbons, indicate which has the higher-octane rating.
   a) Pentane or octane
   b) 2-Methylpentane or 2,2-dimethylbutane
   c) Octane or 2-methylheptane
5. List four compounds that are used as octane enhancers.
6. Why C2H4Br is added to antiknocking agent tetraethyl lead?
7. Give two reasons why TEL was banned as a gasoline additive?
8. Name four solid lubricants.
9. What should be the flash point of good lubricant?
10. Explain how graphite acts as a lubricant.

UNIT 4 : DEVELOPMENTS IN ENVIRONMENTAL CHEMISTRY

Structure

4.0 INTRODUCTION

Undeniably, the past few decades have witnessed an enormous improvement in scientific methodologies/solutions showing phenomenal capability to protect human health and environment and these innovative solutions have been possible through green chemistry. We have had fantastic accomplishments as the solutions were at the molecular level since green chemistry preached the prevention of waste at source rather than treating it up after generation. So, quite a few of the existing processes were redesigned using the basic tenets of green chemistry. Progressively, industries also started adopting the path of green synthesis. In this unit we will learn the concept of green chemistry and their applications to prevent the environmental pollution.

4.1 OBJECTIVES

After studying this unit, you will be able to:

- Understand the significance of green chemistry, how it came into emergence and how it is playing a key role in improving human health and environment.
- Understand the basic concepts and various principles of green chemistry.
- Explain about green solvents.
- Discuss the problems associated with plastics and attempts being directed towards resolving plastic pollution through green chemistry.
- Explain about eco-friendly pesticides and their significant role in agrochemical sector.

4.2 NEED FOR EMERGENCE OF GREEN CHEMISTRY

If we reflect back on the past several decades, we find that chemistry has played an indispensable role in increasing the quality of life and overall growth and development of society besides sufficing our basic needs-right from clothes to food to medicines **(Figure 1)**. As per the data provided by World Health Organization (WHO), life expectancy has increased over the years. Today, we have good pharmaceutical drugs to combat chronic diseases, improved transportation for energy efficiency, better nutrition, improved herbicides and pesticides for pest control, sophisticated medical devices etc.

Figure 4.1 Chemistry in everyday life.

However, the darker side of the massive production of chemicals and related goods with the industrial revolution has been the generation of waste that has negatively impacted both human health as well as environment. One thing is quite clear that none of the industries or manufacturers wished to harm the environment, but due to lack of fundamental understanding of the hazardous nature of many of the chemical products and poor handling, ended up releasing wastes continually into the environmental matrices. The awareness about the need to rethink chemical processes and designing began with the story of "Silent Spring" which was published in the form of book by Rachael Carlson way back in 1962. In this book, Carlson had written about how the bald eagle species became extinct because of excessive use of DDT. This was an extensively used pesticide that had entered the food chain, resulted in thinning of egg shells that failed to produce baby birds. Some of the other species such as ospreys and brown pelican also got affected by the use of DDT. The book was entitled silent spring with a much heavy heart by her as she feared that the birds who sang outside her window may become completely extinct. Thereafter, the awareness prompted the ban on this hazardous chemical pesticide.

A number of other incidences and disasters in the past triggered the need to adopt Green Chemistry, a few have been highlighted below:

4.2.1 Bhopal Gas Tragedy

On the night of 2nd-3rd December, 1984 the worst industrial disaster which is known as the Bhopal Gas Tragedy occurred at Union Carbide Company of India (a pesticide plant) due to leakage of methyl isocyanate gas. This gas spread rapidly within the radius of eight kilometres. More than 4000 people lost their lives and 2000 were almost injured. Studies carried out by ICMR (Indian Council of Medical Research) revealed that people died because of suffocation and toxins that had resulted in swelling in the brain and finally death

SAQ 1

What was the major factor that lead to the Bhopal Gas Tragedy?

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4.2.2 Thalidomide Disaster

Thalidomide drug was introduced in Germany in 1957 as a sedative and hypnotic for treatment of morning sickness in pregnant women. It was one of the most sold drugs. However, a few years later it was found that approximately 10,000 children were born with phocomelia which resulted in limb malformation and this was a repercussion of the consumption of thalidomide drug by the mothers of the new born children. This is referred as the thalidomide disaster which had caused a great deal of fear in general public about the effects of synthetic chemicals and prompted many countries to tighten drug approval regulations.

SAQ 2

Give the structure of thalidomide drug.

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4.2. 3 Times Beach

Times Beach is a small town located in the St. Louis County, Missouri, United States, which became a permanent community of small homes and trailer parks. In 1982, the people residing in this town were forced to leave forever as the Environmental Protection Agency (EPA) had found high levels of a chemical called dioxin in the soil along the roads. The source of contamination was found to be dioxin-tainted waste oil that was sprayed on the roads almost a decade ago. The spray of this toxic chemical had led to deaths of various animals such as horses and birds in addition to the acute poisoning symptoms including intense stomach pain, headache, diarrhoea, nosebleeds, skin rashes etc. experienced by people living in and around Missouri town. The problem faced by the residents increased even more by a flood because of which 700 families had to leave their homes. Finally, the entire town has to be evacuated.

4.2.4 Love Canal Disaster

Another incidence that caught a lot of public attention was the love canal disaster. Love Canal was originally built by William T. Love in the eastern edge of Niagara Falls, New York with the vision to generate power economically to fuel the industry as well as homes of this model city. Love believed that this could be done by digging a short canal between the upper and lower Niagara Rivers. However, unfortunately this project failed to sustain the economy and transmit electricity over great distances by means of an alternating current. Even worse, the canal was turned into a municipal and industrial chemical dumpsite from 1930's to 1950's, the chemical dump continued. Years after, the chemical substances that had been dumped began to leak and the site got contaminated by atleast 82 chemicals many of which were suspected carcinogens including benzene, chlorinated hydrocarbons and dioxin. As a consequence of the massive exposure to these chemicals, many critical health issues including high birth defect, miscarriage rates, liver cancer, seizure, nervous diseases etc. were witnessed amongst the neighbourhood children.

4.3 SOME IMPORTANT LAWS FOR ENVIRONMENTAL PROTECTION

All these incidences are referred as unforeseen chemical consequences that resulted in tragedy, followed by public outrage which in turn necessitated some legislations (also called environmental laws and movements) to control the manufacture, use and disposal of substances. Some important laws are highlighted below:

- *1970 National Environmental Policy Act (NEPA):* This law promotes the enhancement of environment by establishing the President's Council on Environmental Quality (CEQ).
- *1970 Clean Air Act:* To prevent air pollution, protect the ozone layer and promote public health.
- *1972 Pesticide Control Act:* For controlling the sale, distribution, and application of pesticides through an administrative registration process
- *1972 Clean Water Act:* To restore and maintain the chemical, physical, and biological integrity of the nation's waters.
- *1976 Toxic Substances Control Act:* Regulates the introduction of new or already existing chemicals.

However, these laws primarily focussed on dealing with pollution regulation. Soon the need was realized to prevent pollution at source rather than treating it after it was generated and thus, Green Chemistry was introduced.

4.4 GREEN CHEMISTRY AND SUSTAINABILITY

In the year 1990, Environmental Protection Agency (EPA) came with the first act called as the Pollution Prevention Act that focussed on pollution prevention at source which in turn lead to the emergence of Green Chemistry. This act states "Pollution Prevention is reducing or eliminating waste at the source by modifying the production process, promoting the use of non-toxic or less toxic substances, implementing conservation technique and reusing materials rather than putting them into the waste stream." With the passing of this act, for the first time there was a change in the policy from pollution control to pollution prevention and the industries were motivated to undertake research that dealt with the production of no waste. Soon a year after, EPA started an alternative synthetic reaction programme that aimed at providing research grant for conducting chemical synthesis without causing pollution. Thereafter, Paul Anastas who was working at that time at the EPA office of Pollution Prevention and Toxics coined the term Green Chemistry. In the year 1998, Anastas along with John C. Warner gave the twelve principles of Green Chemistry and published the first book on Green Chemistry entitled "Green Chemistry: Theory and Practice." These principles are encouraging scientists, chemists and industrialists to pursue chemistry in a greener way.

Green Chemistry is central to sustainable development or it can help in accomplishing sustainability. As per United Nations Brundtland Commission 'Sustainable Development" is defined as *"Meeting the needs of the present without compromising the ability of future generation to meet their own needs."*

SAQ 3

What is Green Chemistry? What is the significance of the colour green?

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4.5 CONCEPT OF GREEN CHEMISTRY

The term green is often associated with the natural surroundings (for instance green is the colour of plants). According to the founders of Green Chemistry Anastas and Warner, "Green Chemistry is the design of chemical products and processes that reduce or eliminate the use and/ or generation of hazardous substances." It is also known as "clean or sustainable chemistry" or a chemistry that is "benign by design." As per EPA "Green Chemistry prevents pollution at the molecular level, is a philosophy that applies to all areas of chemistry." We cannot define it as a new branch of chemistry, but it is the way chemistry should be practiced i.e. dealing with chemicals and their manufacturing processes to minimize any negative environmental effects. **Figure 2** highlights all the factors that green chemistry aims at reducing.

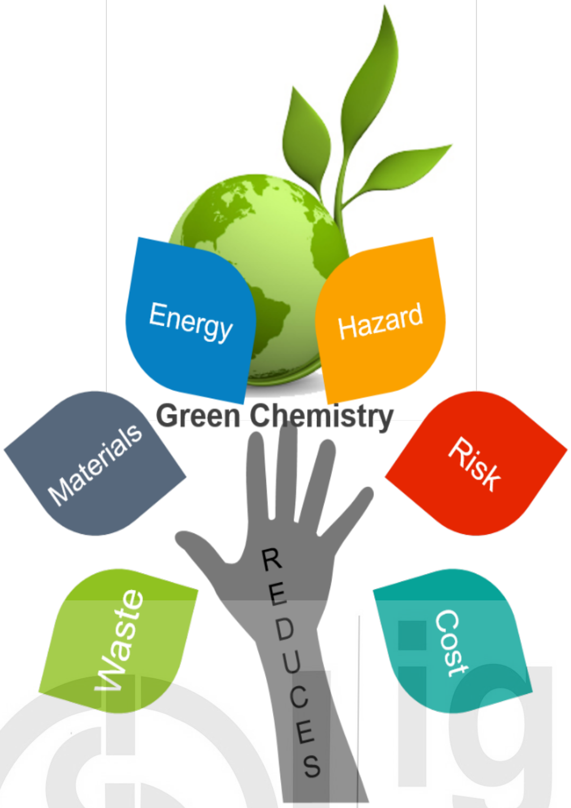

Figure 4.2 Approach of Green Chemistry.

4.5.1 Twelve Principles of Green Chemistry

In this unit, we have described the twelve principles of Green Chemistry in a concise manner which will be helpful in understanding the basic concept involved.

Principle 1: Prevent Waste

According to this principle, it is better to prevent waste rather than clean up waste after it is formed. This is certainly one of the most effective principles which if followed well can eliminate problems of waste generation. Also, it can significantly reduce the costs involved in the manufacturing processes as there are huge expenses involved in the treatment and disposal.

To understand this principle, we must know what waste is? By the name itself waste means something undesirable or unwanted. It is a process or material that does not add value to goods or services. We may also interpret waste as material discharged or released into the environment in an amount of manner that may cause a disastrous effect. If we talk about chemical waste, it is simply the unwanted product generated during a manufacturing process or reagents which remain unutilized. For instance: in the Leblanc process employed largely by the textile industry in Lancashire in UK, huge amounts of waste were generated. This process dealt with the generation of Na_2CO_3 from rock salt, limestone and sulphuric acid, HCl, CaS and CO are produced as waste as shown below:

$$2\ NaCl + H_2SO_4 \longrightarrow Na_2SO_4 + 2HCl$$

$$Na_2SO_4 + 4C + CaCO_3 \longrightarrow Na_2CO_3 + CaS + 4CO$$

The release of toxic HCl gas was reported to cause asthama and respiratory problems in South Lancashire where this process was carried out. Besides, the foul smeling CaS also lead to critical environmental issues. Roger Sheldon in 1992 gave a term called E-factor which is also known an environmental factor to measure waste. Larger the E-factor greater is the waste generated.

E-factor=kgs of by products (waste) produced/kg of desired product

Processes can be redesigned so that E-factor is minimized. There are successful examples which illustrate how the use of a selective catalyst helped in reducing E-Factor. For instance, Pzifer redesigned its Sertraline manufacturing process with the use of $TiCl_4$ as the catalyst that helped in minimizing the quantity of waste generated.

SAQ 4

Give an example to illustrate Principle 1.

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Principle 2: Atom Economy

Principle 2 states that "Synthetic methods should be designed to maximize the incorporation of all materials used in the process into the final product." The concept of atom economy developed by B.M. Trost is a consideration of "how much of the reactants end up in the final product."

To understand how atom economy is different from the yield calculation, we need to understand what yield % is and how it is calculated.

% Yield = (Actual Quantity of desired product/Theoritical quantity of the desired product) x 100

The biggest demerit with this calculation is that it does not consider waste or by-products formed because a 100% yield does not ensure whether waste is not generated. For instance: the very famous Wittig Reaction proceeds with 100% yield, but does not take into account the large amounts of by-products obtained. Any reaction can be considered green if there is maximum incorporation of the starting materials and reagents in the final product.

Examples of atom economical reaction:

(i) Generation of ammonia

$$N_2\,(g) + 3H_2\,(g) \longrightarrow 2NH_3\,(g)$$

Molar Mass 28 6 34

Atom Economy = (Mass of atoms in the desired product ÷ Mass of the atoms of the reactants) × 100

$$= \{34 \div (28+6)\} \times 100$$
$$= 100\ \%$$

(ii) Addition Reaction

$$CH_3\text{-}CH{=}CH_2\,(g) + H_2\,(g) \longrightarrow CH_3CH_2CH_3\,(g)$$

Molar Mass 42 2 44

Atom Economy = $(44 \div 44) \times 100$
$= 100\ \%$

SAQ 5:

Give an example of a reaction comparing its atom economical and uneconomical pathways.

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Principle 3: Designing Less Hazardous Chemical Synthesis

As per principle 3, "Wherever practicable, synthetic methodologies should be designed to generate substances that possess little or no toxicity to human health and environment." This principle motivates chemists to design products by avoiding reagents/starting materials that could be toxic. But, if we are not able to completely avoid the use of hazardous material, care must be taken towards minimizing its use. There are several examples to project how this principle is being implemented in industries due to increasing chemical consciousness with regard to hazard and exposure. For instance: Avalon Natural Products is a company that is based in California and has been a leader in making cosmetics has prohibited the use of chemicals banned under EU cosmetic derivative including perfluorinated compounds used in clothing, cookware, containers; bisphenol A-a carcinogen found in baby bottles; phthalates used in fragrances for long lasting effect and in nail polish to make it chip resistant, parabens employed for preventing cosmetics from deteriorating in bathrooms, sunlight and extreme conditions.

Also, there are examples of less hazardous chemical pathway employed. For instance: phosgene free route for the synthesis of polycarbonates using dimethyl carbonate which is much more greener than traditional pathway as it completely eliminates the use of HCl, CH_2Cl_2, does not need crystallization, generates higher quality polycarbonates.

Principle 4: Designing Safer Products

According to this principle "Chemical products should be designed to preserve efficacy of function while reducing toxicity." By safer chemical what we mean to say is a chemical that would cause minimum adverse effect on human health and environment. Examples of safer chemicals include structural analogues (i) methacrylonitrile which is a safer replacement of acrylonitrile (the probable human carcinogen) (ii) sodium salt of linear alkyl benzene sulphonic acid that readily biodegrades and is a greener alternative to branched alkyl benzene sulphonic acid.

Principle 5: Safer Solvents and Auxiliaries

This principle states "The use of auxiliary substances that do not directly contribute to the structure of the reaction product but are still necessary for the **chemical** reaction or process to occur should be made unnecessary wherever possible and innocuous when used." The statement is self-explanatory and indicates that alternative greener replacements should be explored for toxic organic solvents which are major contributors to overall toxicity of a reaction. Many of the traditional solvents such as benzene, perchloroethylene used in dry cleaning, shoe polish, white correction fluid are known as human carcinogen and chlorinated hydrocarbons are known to persist in the atmosphere causing depletion of ozone layer. Thus, pharmaceutical and other industries are looking for replacements some of which include (i) Water (ii) Supercritical Fluids (iii) ionic liquids (iv) polyethylene glycol (v) fluorinated biphasic solvents (vi) biobased solvents (derived from renewable resources) (vi) no solvent is the best solvent. There are solvent selection guides available which enable scientists to use safer solvents in research.

SAQ 6:

Why do you think no solvent is the best solvent?

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Principle 6: Design for Energy Efficiency

Principle 6 states "Energy requirements should be recognized for their environmental and economic impacts and should be minimized. Synthetic methods should be conducted at ambient temperature and pressure."

Thermal heating processes mostly employ fossil fuels reserves of which are limited and are soon going to exhaust. In addition, the energy supplied is not specific, which means it is not directed to the chemical bond of the molecule of the reactants. Besides, most of the energy is wasted in heating the container and the solvent, while some is also lost to the environment. The energy efficient alternatives have been lately been explored which include Microwave assisted, light induced photochemical reaction and ultrasound assisted irradiation.

SAQ 7:

How does microwave chemistry fall under the banner of green chemistry?

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Principle 7: Renewable Resources

As per this principle "A raw material or feedstock should be renewable rather than depleting technically and economically practicable."

All the forms of energy that we have been using with the advent of globalization and rapid industrialization are fossil fuel based, which are not only going to exhaust soon, but also have a serious effect on human health and environment. Hence, attention is now being devoted towards finding out alternative renewable resources which can work as excellent substitutes. The idea of making our future fuels, chemicals and materials from feedstocks that never deplete is certainly quite interesting. There are a few examples of alternative chemicals/products derived from natural resources including adipic acid from corn starch, biofuel production from microalgae, green plastics made from corn.

Principle 8: Reduce Derivatives

This principle states that "Unnecessary derivatization (blocking group, protection/deprotection, and temporary modification of physical/chemical processes) should be avoided wherever possible.

We often use protecting and deprotecting groups in organic synthesis in order to achieve desired chemoselectivity. To put it in simple words, whenever there are two reactive groups in a molecule and wish to use only one of the groups for the formation of desired product, the other group needs to be protected and subsequently deprotected. However, the incorporation of these groups require additional steps and reagents which also lead to an increase in waste generation. This principle advocates the reduction of using protecting groups and derivatives. The best alternative to avoid this is to use enzymes as they work under mild conditions and lead to chemo selectivity. They mostly react at the desired moiety and do not interfere with the rest of the molecular structure. The antibiotic synthesis of penicillin drug is the best example illustrating how the traditional protection deprotection method has been replaced with an enzymatic process and the enzyme used is penacylase.

Principle 9: Use of selective catalyst

According to Principle 9 "Catalytic reagents are superior to stoichiometric reagents."

In comparison to the traditional stoichiometric processes that lead to enormous amount of waste generation, use of catalysts permit higher atom economy and lesser waste generation. Catalyst is a substance which expedites the rate of a reaction by providing an alternative pathway without itself undergoing any change and also enable a process to occur selectively. The catalytic oxidation of $PHCH(OH)CH_3$ using CrO_4/H_2SO_4 has an atom economy of about 42% while the same reaction when carried out in presence of a catalyst has an atom economy of 87%. Also, when the reduction of benzaldehyde is carried out in presence of stoichiometric reagents, atom economy is 81% but on the other hand catalytic reduction of benzaldehyde leads to 100% atom economy.

SAQ 8:
Give examples of catalysts that active, selective as well as reusable.

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Principle 10: Design for Degradation
This principle states "Chemical products should be so designed that at the end of their function, they do not persist in the environment and break down into innocuous degradation products."

In order to understand this principle, we must understand what are persistent chemicals? These are chemicals which do not degrade easily and remain in the environment since long, thereby exhibiting toxic effects. In environment, degradation of the chemicals can occur via microbial activity and plants which is called biodegradation. However persistent chemicals do not degrade owing to lack of an appropriate enzyme or microbe. DDT is an example of organic pesticide that persists in the environment. Thus, now greener substitutes are being developed for DDT. Another very successful example is provided by the laundry industry. Detergents have found a lot of use over the past decades, however there was a problem due to its foaming which occurred because of non-biodegradability of alkyl benzenesulphonate in detergents. This problem was however resolved by using linear chain rather than branched chain on benzene sulphonate moiety, which showed better biodegradability.

Principle 11: Real Time Analysis for Pollution Prevention
As per this principle Analytical methodologies need to be further developed to allow for real-time, in-process monitoring and control prior to the formation of hazardous substances." In other words, what this principle underpins is that if chemical reaction is monitored continually then, it can help prevent the release of hazardous and polluting substances due to accidents or unexpected reactions. Further, if effectively a reaction is monitored, then the warning signs can be spotted, and consequently, the reaction can be stopped or managed before such

an event occurs. Online monitoring has been possible with the help of analytical techniques such as GC-MS and LC-MS which help in seeing what is happening inside a reaction and to know when exactly it will be completed, thereby reducing waste, time and energy.

Principle 12: Inherently Safer Chemistry for Accident Prevention

According to this principle "substances and form of a substance used in a chemical process should be chosen so as to minimize the potential for chemical accidents, including releases, explosions and fires"

Often, we have also heard of chemical releases due to either routine evolution of gases or accidental due to negligence of operator or operational malfunctioning. These releases have a detrimental effect on human health and environment. Therefore, there is a need to practice safer chemistry by designing chemical processes and products with specific aim to recognize and eliminate hazards from the manufacturing process.

4.6 GREENER SOLVENTS

Solvents have attracted a great deal of attention in the scientific community owing to the crucial role they play in agrochemicals, pharmaceuticals, paints, household and industrial cleaners etc. Not only do they provide a medium for the dissolution of a wide variety of substrates, but also accelerate many of the reactions. The largest user of solvents is the pharmaceutical sector.

4.6.1 Issues with the use of traditional solvents

Organic solvents are the major component of cleaning agents, adhesives, pharmaceuticals and cosmetics and have been used primarily for mass and heat transfer as well as separation of desired products from unreacted starting materials and by-products. Unfortunately, despite being useful one cannot ignore the flip side of the coin which has been toxicity, flammability, and environmental damage with the usage of copious amounts of solvents. We already are aware of the carcinogenic nature of hydrocarbon based solvents such as benzene. Also they are known to bioaccumulate because of their poor solubility in water. Besides, there are several other solvents (eg: carbon tetrachloride, and trichloroethylene) which may cause adverse health effects including eye irritation, headache, allergy, skin reactions etc. Therefore, efforts are being directed towards either reducing or completely eliminating the use of organic solvents.

Now we all talk about greener solvents which by definition are environmentally friendly solvents that are characterized by:

- Low toxicity
- Convenient Accessibility
- Greater Efficiency
- Enhanced reusability

- Increased Biodegradability

4.6.2 Alternative Green Solvents

In the subsequent sections, we shall be highlighting examples of alternative greener solvents that have made a way to large scale organic synthesis.

4.6.2.1 Water

Currently, water has gained prominence as a potentially benign solvent due to environmental benefits, safety, synthetic efficiency, low cost, high specific heat, natural abundance etc. One might be surprized to note that water had remained a poor solvent for organic transformations since long due to solubility reasons (poor solubility of organic compounds in water). Also removal of water was thought to be energy intensive; contaminated waste streams would be difficult to treat. Fortunately, these problems have been resolved gradually. Like for instance, the solubility issue was resolved using surfactants, phase transfer catalysts or by adding solubilizing groups such as sulphonates, phosphonates, oxalates etc. There are numerous examples which illustrate that reactions have been carried out "in water" as well as "on water" successfully with very high yield.

Now, what do we mean by in water and on water should be demarcated.

In-Water Reactions: When the reactants employed are soluble in water, it is an in water reaction. Quinoxaline synthesis is an example of in water reaction **(Figure 3)**.

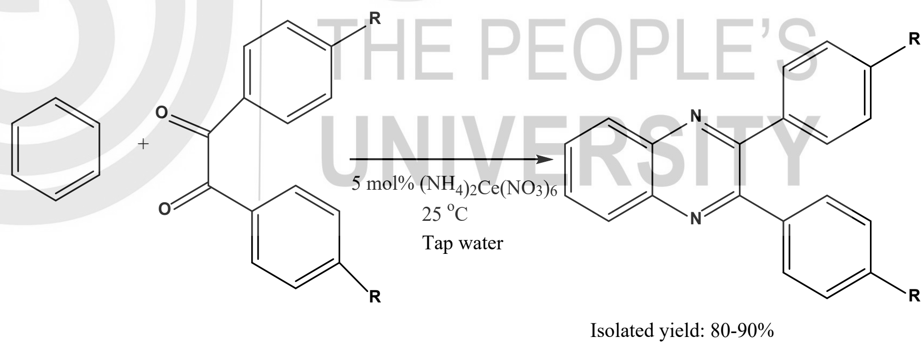

Figure 4.3 Schematic of quinoxaline synthesis accomplished in water.

On-Water Reactions: When reactants are insoluble in water (hydrophobic), they are simply suspended in water and remain in the form of aqueous emulsions or suspensions. Reactions also occur on water with good yield the reason for which remained unclear for long. However, in 1991 Breslow demonstrated that it could be due to hydrophobic effect (reactants tend to be away from water and accumulate together, which increases the probability of a reaction). **Figure 4** shows Diels Alder Reaction occurring on water. Breslow found that the rate of Diels Alder reaction increases drastically when carried out in water **(Table 1)**.

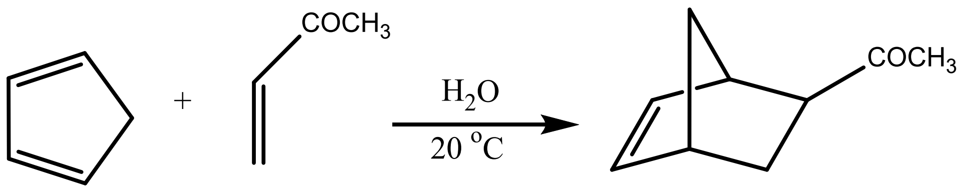

Figure 4.4 Diels Alder Reaction on water.

Table 1 Comparison of rate constant values of Diels Alder reaction

Solvent	**Additional Component**	**Rate Constant** $k_2 \times 10^5\ M^{-1}s^{-1}$
Isooctane	-	5.9
MeOH	-	75.5
Water	-	4,400.0
Water	LiCl (4.86 M)	10,800.0
Water	β-Cyclodextrin	10,900.0
Water	α-Cyclodextrin	2,610.0

4.6.2.2 Supercritical Fluids (SCFs)

Although supercritical fluids (SCFs) have been known since long, yet they got recognized as environmentally acceptable replacements for conventional solvents after the birth of green chemistry. So, what does the word supercritical mean? When any substance is subjected to a pressure and temperature greater than its critical point, it is said to be supercritical. A substance is characterized by its critical point that is obtained at specific conditions of temperature and pressure. In the supercritical region, the particular substance exhibits properties of both liquid as well as gas. The most commonly employed SCF includes supercritical carbon dioxide ($ScCO_2$) which has T_c=304 K and p_c=72.8 atm. $ScCO_2$ has emerged as a green alternative solvent in a number of industrial reactions such as hydrogenation, hydroformylation, polymer production, C-C bond formation owing to myriad of advantages it offers including non-toxicity, abundance, tendency to get easily removed from the product. In **Figure 5**, we have shown an example of the hydrogenation reaction carried in $scCO_2$ which gives the desired product with 100% yield. It is also acquiring paramount significance as a solvent for extracting essential oils and other products from spices, aromatic herbs and medicinal plants. It is interesting to note that the decaffeination of coffee has also been accomplished using $scCO_2$ as solvent.

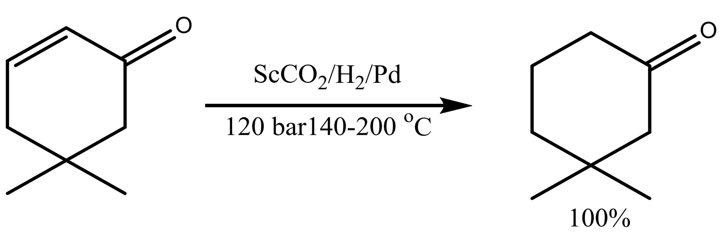

Figure 4.5 Example of use of $scCO_2$ in hydrogenation reaction.

SAQ 9 Extraction of an important compound from orange peel has been carried out using $scCO_2$? Can you name the compound?

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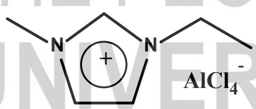
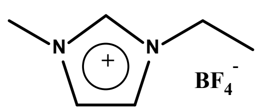
4.6.2.3 Ionic Liquids

Ionic liquids have emerged as a relatively new class of solvents that are primarily composed of complex organic salts comprising of cations and anions. These are essentially liquids at unusually lower temperature, as they have low vapour pressure. In technical terms, any salt, even NaCl above 800 ^oC can be designated as ionic liquid; however the most frequently used definition of ILs is *"Salts that exist in the liquid state below 100 ^oC and are liquid at room temperature."* Owing to their complex interplay of coulumbic, hydrogen bonding and Van der Walls interactions of their ions, ILs have emerged as greener alternatives to volatile organic compounds (VOCs). Also, to add to their fascinating properties, ILs exihibit:

1. Non- Volatility (which allows for high temperature reactions without the requirement of a pressure vessel to contain the vapors) and Non-Flammability
2. Work as both catalysts as well as solvents
3. Robust
4. Tailor made, task specific solvents-Allow the tailoring as per requirement.



1-ethyl-3-methylimidazolium chloride (III) chloride ([emim]$AlCl_4$]



1-butyl-3methylimidazolium fluoride-boron trifluoride ([bmim]BF_4)

Figure 4.6 Examples of room temperature ionic liquids.

Figure 6 shows examples of some of the common room temperature ionic liquids. Some well-known reactions conducted successfully in ILs include friedel crafts reaction, esterification, mannich reaction, regioselective alkylation, reduction of aldehydes, Fischer Tropsch Synthesis etc.

4.6.2.4 Liquid polymers

Similar to the ILs, liquid polymers having different molecular weights have also gained a lot of attention in organic synthesis since they possess low volatility, low eco-toxicitiy, good thermal stability, non-flammability and most importantly rapid biodegradability. Amongst several liquid polymers, polyethylene glycol (PEG) has been utilized by both academic as well as industrial researchers in diverse areas including electroplating, cosmetics, personal care, dyes, paints etc. In **Figure 7**, it has been demonstrated how PEG-200 is used as a solvent in Diels Alder Reaction. PEG can also function as an excellent aqueous biphasic system (ABS) wherein two phases are formed when mixed with another water soluble polymer and such ABS are good for extracting toxic dyes that would otherwise be difficult to remove from the contaminated waste streams.

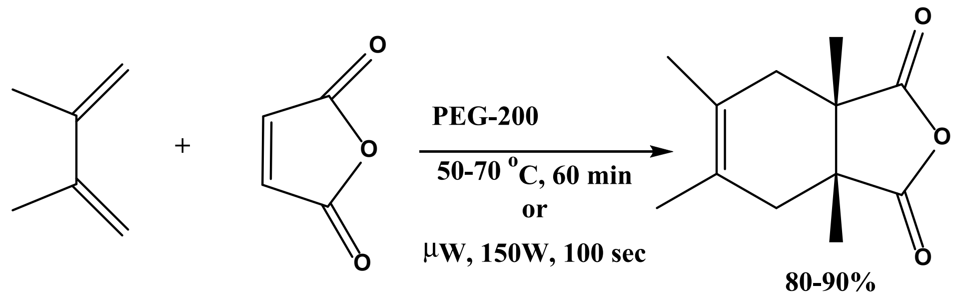


Figure 4.7 Example of Diels Alder Reaction in PEG-200.

4.6.2.5 Bio-Based Renewable Solvents

Solvents which are derived from renewable feedstocks that are bio-based have been classified as bio-based solvents. The sources generally include agricultural crops rich in carbohydrates including corn, wheat, beetroot etc. They offer several significant advantages including low VOC emission, low toxicity, biodegradability and environmentally friendliness. Some typical examples of bio-based solvents include bioethanol, glycerol and 2 Me-THF.

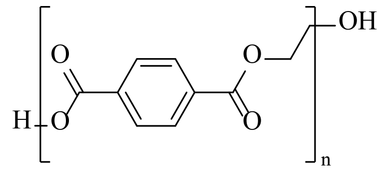
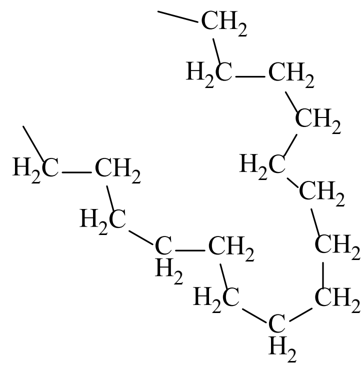
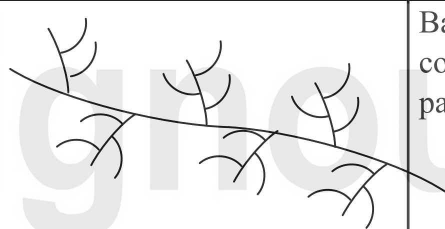
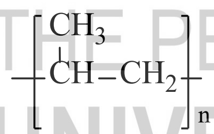
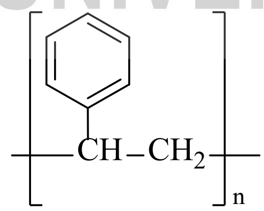
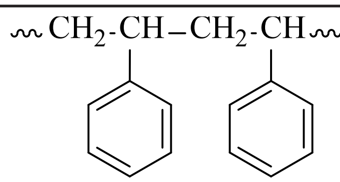
4.7 EARTH FRIENDLY PLASTICS

Plastics have turned out to be one of the biggest source of polluters today despite having innumerable valuable uses in daily life. While every minute, one million drinking plastic bottles are purchased, half or more than half of them are intended for one time use and thrown away as such which end up in either a landfill or natural environment, persisting for years and centuries. Not surprisingly, a staggering 8 million tonnes of plastic reaches the world's oceans every year and it is the ocean plastic which is known to kill millions of marine animals every year. Animals are killed via either entanglement in plastic objects or via ingestion of these wastes, or through the exposure to chemicals within the plastics that interfere with their physiology.

Before moving further, it is important to understand what do the plastic wastes comprise of and why are they so problematic?

Chemically, plastics are polymers of high molecular mass comprising of long chains of repeating molecular subunits called as monomers. **Table 2** provides data related to the variation in plastic polymers and their uses.

Table 2 Examples of polymers used as plastics with their structures and applications.

Sl. No	Polymer	Structure	Use/Application
1.	Polyethylene terephthalate (PET)		Water bottles, dispensing containers, biscuit trays
2.	High—density polyethylene (HDPE)		Shampoo bottles, milk bottles, freezer bags, ice cream containers
3.	Low—density polyethylene (LDPE)		Bags, trays, containers, food packaging film
4.	Polypropylene (PP)		Potato chip bags, microwave dishes, ice cream tubs, bottle caps
5.	Polystyrene (PS)		Cutlery, plates, cups
6.	Expanded polystyrene (EPS)		Protective packaging, hot drink cups

4.7.1 Primary issues with the synthetic plastics

They are produced from non-renewable sources including crude oil which are depleting.

Most of the synthetic plastics require the use destructive thermal treatment such as combustion or pyrolysis, which require high temperature

There has been increasing concerns due to the presence of toxic chemicals such as bisphenol A (BPA) and polystyrene (PS) that might leach into water. While BPA in particular is highly toxic and a known endocrine disruptor which can imitate body's hormones, interfering with the production, secretion, transport, action, function, and elimination of natural hormones often at times leading to fertility problems, male impotence, heart disease and other conditions; polystyrene has been declared a possible human carcinogen by the U.S. Environmental Protection Agency (EPA) and the International Agency for Research on Cancer.

Extremely small sized invisible plastics (mostly PS) called as microplastics are released from fibres, clothings when washed ultimately reach the oceans causing deleterious consequences to the marine species.

Considering the extreme toxicity and non-biodegradability associated with the conventional plastics, now attention is being directed towards the design of earth friendly or green plastics that are bio-degradable (decomposed by action of living organism into non-toxic simple molecules including carbon dioxide, water and biomass). These are essentially made from renewable materials such as plants, petrochemicals or microbes. The concept behind this is utilization of principles of green chemistry, especially principle 1 which aims at the prevention of pollution at source rather than treating up pollution after it has been caused and principle 7: utilization of renewable feedstocks whenever possible.

SAQ 10: Why do you think plastics have become a big issue today? What are the underlying problems?

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4.7.2 Greening of Plastics

Earth friendly plastics or bio-plastics commonly obtained from natural origins such as plants, animals or micro-organisms may be classified into the following types:

Polyhydroxyalkanoates (PHAs): Examples of bio-based plastics include Polyhydroxyalkanoates (PHAs) which are naturally produced by various micro-organisms eg: Cuprividus necator. Poly-3-hydroxybutyrate (PHB), polyhydroxyvalerate (PHV) and polyhydroxyhexanoate (PHH) are specific examples of PHA based plastics.

Polylactic acid (PLA): PLA is a thermoplastic aliphatic polyester synthesized from renewable biomass, typically from fermented plant starch such as from corn, cassava, sugarcane or sugar beet pulp.

Cellulose-based Plastics: Cellulose esters such as cellulose acetate and nitrocellulose are utilized to prepare cellulose based plastics.

Lignin based Plastics: Lignin is commonly obtained as a by-product of polysaccharide extraction from various plant materials during the production of paper, ethanol etc. Lignin based plastics are bio-renewable natural aromatic polymers with biodegradable properties.

Green Chemistry Challenge Award for innovation of PLA

In the year 2002, Nature Works LLC received Greener Reaction Condition Awards for replacing petroleum based polymers with bio-based compostable and recyclable polylactic acid (PLA). This PLA based bioplastic is one of the most promising and popular materials as an alternative to synthetic polymer. The synthetic process via which it is manufactured eliminates the use of organic solvents and other hazardous materials and employed a catalyst that reduces energy usage **(Figure 8)**. Furthermore, the approach to synthesis biodegradable or green plastics is a much better solution than disposing off the plastic waste to landfills or via incineration.

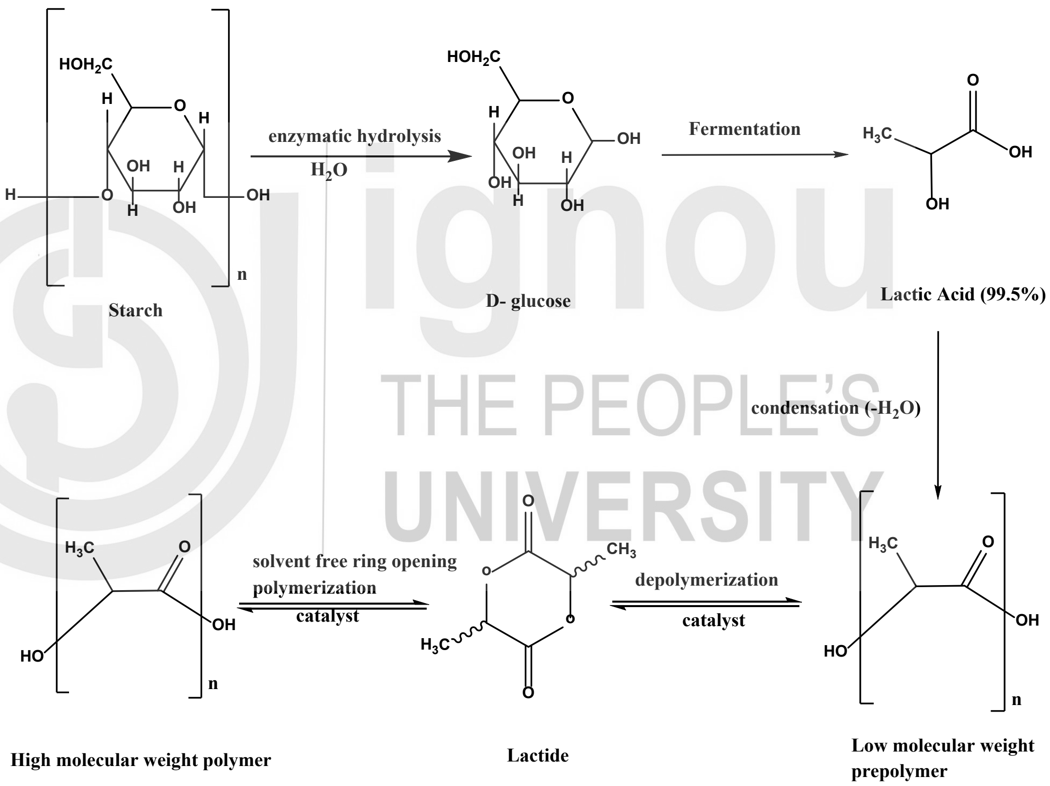

Figure 4.8 Synthesis of PLA by Nature works.

Conversion of plastic wastes into valuables

The most emerging green trend now is utilization of waste materials and converting them into valuables. This is a circular economy approach which primarily revolves around the concept of converting resources into waste and waste back into valuable resources which is greener and much more profitable as compared to the linear approach that only goes a one way path and leads to extensive waste generation. In view of pressing problems associated with plastic wastes, attempts are now being directed towards utilizing their untapped potential by converting them into either fuels such as petrol, diesel and kerosene etc using catalytic degradation or transforming them into valuable catalytic materials. In India also, we are recycling plastic wastes extensively and the current Union government's focus is on plastic waste management via the Swachh Bharat Abhiyan programme.

ENVIRONMENTALLY BENIGN PESTICIDES

Over the past decades, the global agricultural sector has extensively relied on pesticides that have undoubtedly increased the crop yield and reduced post-harvest losses, but unfortunately their expanded use has polluted the environment whilst affecting human health considerably. As per estimates, barely 0.1% of the total agrochemicals utilized in crop protection succumbs to reach the target pest, while the remaining 99.9% is undeniably bound to enter the environment thus causing hazards to non-target organisms.

4.8.1 Effect of Synthetic pesticides on human health and environment

To understand fundamentally how the pesticides, cause massive destruction, we need to understand the basics. These are organic chemical substances that impart protection to plants from pests, weeds or diseases. Typically, the various forms of pesticides employed are insecticides, fungicides, herbicides, rodenticides and plant growth regulators. Some of the common classes of pesticides that have negatively impacted human health and environment include organochlorines, organophosphates, carbamates, pyrethroids, triazines, and neonicotinoids. As already mentioned previously, DDT was the first ever pesticide that had shown serious environmental consequences resulting in thinning of egg shells and bald eagle species becoming extinct.

Subsequently, there have been reports mentioning how harmful the other synthetic pesticides have proven to be. **Table 3** gives examples of the various pesticides that have exhibited toxicity. Exposure to these chemicals can occur through contact with either the skin or via ingestion or inhalation. In fact, residues of pesticides have often found in everyday food including cooked meals, water, wine, fruit juices, refreshments, animal feeds etc. The fate of the pesticides within human body is unpredictable; once inside, it may start getting metabolized, excreted, and stored or bioaccumulated in body fat leading to dermatological gastrointestinal, neurological, carcinogenic, respiratory, reproductive and endocrine effects.

Table 3 Rising toxicity of pesticides 1945-2003.

Pesticide	Brand Name	Use	LD50	Toxicity
DDT	Dinocide	Insecticide	27,000	1
Metiocarb	Mesurol	Insecticide	230	117
Carbofuran	Curater	Insecticide	160	69
Deltamethrin	Decis	Insecticide	10	2700
Fipronil	Regent	Insecticide	4.2	6429
Clothianidin	Pocncho	Insecticide	4.0	6750

Imidacloprid	gaucho	Insecticide	3.7	7297

Source: Dr. J M Bonmatin, CNRS (France)

SAQ 11

Give examples of synthetic pesticides. In what way have they affected human health and environment?

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Search for Environmentally benign pesticides

In view of the harmful effects caused by the synthetic pesticides, the scientific community has realized the need for implementation of sustainable practices. Consequently, environmentally benign pesticides are being developed that are making an important breakthrough in pest control. Most of the plant derived materials have emerged as green pesticides as they are safer and effective against diseases, nematodes and other organisms in addition to phytophagous insects.

4.8.3 Examples

4.8.3.1 Microbial pesticides

Naturally occurring or genetically altered bacteria, fungi, algae, protozoans or viruses have proven to be an attractive and greener alternative to chemical insecticides. These are called as the microbial pesticides and are pathogenic to the targeted pest. These are applied as the conventional insecticidal sprays, dusts or granules. The major advantages of using microbial pesticides include:

(i) They are biodegradable.

(ii) Cheaper, Renewable and can be handled safely.

(iii) Residue-Free.

(iv) Safe to natural enemies and higher organisms.

(v) Difficult for insects to develop resistance against them.

4.8.3.2 Plant-pesticides

In nature, plants guard themselves against pests by evolved defence chemicals which alter the behaviour of selected pest. Such behaviour altering chemicals are defined as semiochemicals well known for their static mode of action, less persistence, and biodegradable nature. Plant pesticides are substances produced by plants from the genetic material that has been added to the plant. Among the plant products, azadirachtin, pyrethrum, sabadilla, etc. are known to have a successful history in pest control.

4.8.3.3 Biochemical pesticides

These are the naturally occurring substances that successfully control the pests via non-toxic mechanisms. They interfere with growth or mating, such as plant growth regulators, or substances that repel or attract pests, such as pheromones.

4.8.3.4 Natural Home-made solutions for pest-control

1. Soap based insecticides

These soaps utilize the salts and fatty acids within them to target many soft-bodied pests including aphids, whiteflies, mealy bugs, earwigs, thrips, and the early stages of scale.

2. Neem Oil Insecticides

Insecticidal oils have also proven their efficacy in treating pests by suffocating them. One such classic example is the oil that is extracted from the seeds of neem tree which is biodegradable and natural fungicide. It acts as a hormone disruptor and an antifeedant for the insects that feed on leaves and other parts of plants. While applying to crops, 2 teaspoons neem oil and 1 teaspoon of mild liquid soap are mixed well with 1 quart of water, and immediately sprayed on the affected part of the plant foliage.

3. Diatomaceous earth

Diatomaceous earth made from sedimentary rocks by fossilized algae (ditoms) also works as a natural pesticide. This substance works in quite an interesting manner mechanistically by virtue of its abrasive qualities or tendency to absorb lipids a waxy substance obtained from insects' exoskeleton, which eventually dehydrates them to death.

4. Garlic Insecticide Spray

Garlic has a typical aroma which makes it work as an insect repellent and owing to its natural fungicidal and pesticidal properties, it has emerged as an excellent economical, non-toxic pesticide. Examples of pests that have been controlled using garlic include aphids, ants, termites, white flies, beetles, borers, caterpillars, slugs, army worms etc.

5. Tomato Leaf

Tomato leaves contain alkaloids-"tomatine," that has shown efficacy in controlling aphids and other insects.

As most of the synthetic pesticides are not safe and many have even proven to be cancerous to human beings, therefore these natural environmentally friendly pesticides emerged as cleaner alternatives in pest control.

SAQ 12

Environmentally friendly pesticides are making a breakthrough in pest control. How would you justify this statement?

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4.9 LET US SUM UP

Undeniably, the past few decades have witnessed an enormous improvement in scientific methodologies/solutions showing phenomenal capability to protect human health and environment and these innovative solutions have been possible through green chemistry. We have had fantastic accomplishments as the solutions were at the molecular level since green chemistry preached the prevention of waste at source rather than treating it up after generation. So, quite a few of the existing processes were redesigned using the basic tenets of green chemistry. Progressively, industries also started adopting the path of green synthesis. Utilization of alternative greener solvents, catalysts and reagents, aiming at higher atom economy, relying on alternative renewable feedstocks have truly changed the landscape of synthesis pursued at academic as well as industrial level. Yet, despite accomplishing a lot as evident from the green chemistry challenge awards, a lot needs to be done as the number of challenges that lie ahead are far many. Population growth, energy, food supply, resource depletion, global climate change, water, toxics generation and dispersion remain the most pressing environmental issues facing the planet even today. Apart from this, there are research and implementation challenges in view of stringent laws. Nonetheless, green chemists foresee that all the barriers would be eventually resolved if there is effective preaching of green chemistry which requires educating our future chemists. Not just chemists, but since green chemistry is a powerful tool for safe and healthy environment for all including businessmen, economists, environmentalists etc. To be successful, green chemistry would require an interdisciplinary strategy in education, research and service.

4.10 GLOSSARY

Green Chemistry: Green Chemistry is the design of chemical products and processes that reduce or eliminate the use and/ or generation of hazardous substances

Atom Economy: The atom economy of a reaction is a measure of the amount of starting materials that end up as useful products.

E-Factor: E-factor also known as environmental impact factor measures the waste generated in a process. Higher the E-factor greater is the waste generated.

Biodegradable Plastics: Plastics that can be decomposed readily by the action of living organisms, usually microbes, into non-toxic products.

Polylactic acid: PLA is a thermoplastic aliphatic polyester synthesized from renewable biomass, typically from fermented plant starch such as from corn, cassava, sugarcane or sugar beet pulp.

In-Water Reactions: When the reactants employed are soluble in water, it is an in water reaction.

On-Water Reactions: When reactants are insoluble in water (hydrophobic), they are simply suspended in water and remain in the form of aqueous emulsions or suspensions.

Supercritical Fluids: A fluid is defined as supercritical when its temperature and pressure exceed critical values (T_c and P_c respectively).

Green Chemistry Challenge Awards: In 1995, the U.S. Environmental Protection Agency (EPA) received support from President Bill Clinton to establish an annual awards program highlighting scientific innovations in academia and industry that advanced green chemistry. This created the annual Presidential Green Chemistry Challenge Awards, which have been a major platform for promoting awareness about green chemistry.

Ionic Liquids: These are green solvents primarily composed of complex organic salts. These are essentially liquids at unusually lower temperature, as they have low vapour pressure.

Pesticides: Substances that are meant to control pests, including weeds.

4.11 SUGGESTED READINGS

1. Anastas, P.T. Warner, J.C. (1998), Green Chemistry: Theory and Practice. Oxford University Press.

2. Lancaster, M. (2010). Green Chemistry: An introductory text, RSC Publishing, Cambridge, UK.

3. Sidhwani, I. Sharma, R. K. (2020). An introductory Text on Green Chemistry, Wiley Publications, USA.

4.12 TERMINAL QUESTIONS

Q.1 Explain the concept of atom economy using a suitable example.

Q.2 What do you understand by earth friendly plastics?

Q.3 How would you classify whether a solvent is green or not? What are the problems associated with traditional solvents?

Q.4 What are environmentally benign pesticides? Give examples of some explaining how they are a superior alternative to the synthetic ones.

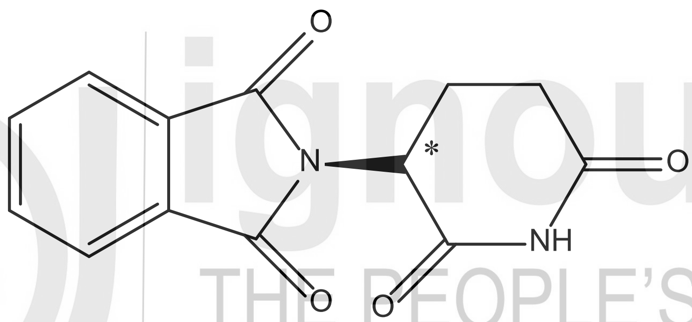
Q.5 Fundamentally highlight the differences between green chemistry and environmental chemistry.

Q.6 What are supercritical fluids? Give examples of reactions in which SCFs have been employed as green solvents?

Answers

Self-Assessment Questions

1. The accident had occurred due to non-functioning refrigeration unit, bulk storage of 42 tons methyl isocyanate (MIC) gas, non-functioning safety system, stashing staff and switch off of the warning system.



2. Chemistry is the design of chemical products and processes that reduce or eliminate the use and/ or generation of hazardous substances. According to EPA *"Green Chemistry prevents pollution at the molecular level, is a philosophy that applies to all areas of chemistry."*

   The colour green is associated with the natural surroundings. It is used for environmentally friendly and sustainable manufacturing processes.

3. Principle 1 states "It is better to prevent waste rather than to treat up after it is formed."

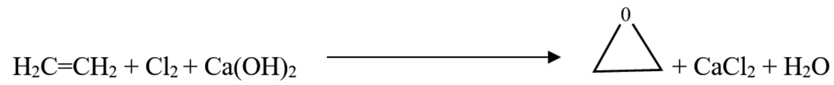
   Example: In the Leblanc process employed largely by the textile industry in Lancashire in UK, huge amounts of waste were generated. This process dealt with the generation of Na_2CO_3 from rock salt, limestone and sulphuric acid, HCl, CaS and CO are produced as waste as shown below:

   $$2\,NaCl + H_2SO_4 \longrightarrow Na_2SO_4 + 2HCl$$
   $$Na_2SO_4 + 4C + CaCO_3 \longrightarrow Na_2CO_3 + CaS + 4CO$$

   The release of toxic HCl gas was reported to cause asthama and respiratory problems in South Lancashire where this process was carried out. Besides, the foul smeling CaS also lead to critical environmental issues.

4. Oxirane synthesis has been carried out in two ways. The first approach is two-stepped (demonstrated here as one step) requiring $Ca(OH)_2$. The second approach is single step that requires O_2 as oxidant. If we compare the atom economy of both the reactions, we find that the second approach is 100 % atom economical while the first one has a very low atom economy of about 25.6%.

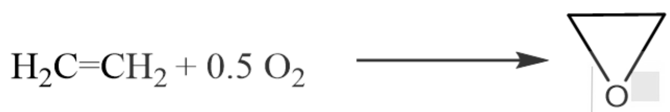
Approach 1:



28 71 74 44 111 18

$$\%\ \text{Atom economy} = \frac{44}{173} \times 100 = 25.4\%$$

Approach 2: Single Step Synthesis



28.05 0.5×32 44

$$\%\ \text{Atom economy} = \frac{44}{44} \times 100 = 100\%$$

5. No solvent means performing reactions without using any solvent. This is certainly the best and greenest solution as it would not lead to any waste generation and maximize atom economy, because the reactants are the sole reagents utilized.

6. In conventional heating methods, oil bath or hot plate are used as a source of heat to a chemical reaction Microwave irradiation is widely used as a source of heating in chemical synthesis. The basic mechanisms observed in microwave assisted synthesis are dipolar polarization and conduction. Microwave-assisted synthesis is greener on account of the following advantages (i) enhanced reaction rates (ii) higher yields (iii) greater selectivity (iv) more economic for the synthesis of a large number of organic molecules.

7. Examples of catalysts that suffice the three key goals of catalysis i.e. activity, selectivity and reusability are supported heterogeneous catalysts eg: Fe_3O_4 supported Cu nanocatalyst.

8. Extraction of D-Limonene from orange peels has been carried out efficiently using $scCO_2$. This experiment is conducted in undergraduate laboratories.

9. Plastics have become a big issue today on account of the following reasons:

   - They are produced from non-renewable sources including crude oil which are depleting.

- Most of the synthetic plastics require the use destructive thermal treatment such as combustion or pyrolysis, which require high temperature
- There has been increasing concerns due to the presence of toxic chemicals such as bisphenol A (BPA) and polystyrene (PS) that might leach into water. While BPA in particular is highly toxic and a known endocrine disruptor which can imitate body's hormones, interfering with the production, secretion, transport, action, function, and elimination of natural hormones often at times leading to fertility problems, male impotence, heart disease and other conditions; polystyrene has been declared a possible human carcinogen by the U.S. Environmental Protection Agency (EPA) and the International Agency for Research on Cancer.
- Extremely small sized invisible plastics (mostly PS) called as microplastics are released from fibres, clothings when washed ultimately reach the oceans causing deleterious consequences to the marine species.

10. DDT, Metiocarb, Carbofuran, Deltamethrin, Fipronil and Clothianidin are examples of the synthetic pesticides employed. The organochlorine pesticides like DDT are extremely persistent and accumulate in fatty tissue. Through the process of bioaccumulation (lower amounts in the environment get magnified sequentially up the food chain), large amounts of organochlorines can accumulate in top species like humans; act as endocrine disruptors, interfering with hormonal function of estrogen, testosterone and other steroid hormones.
11. In view of the harmful effects caused by the synthetic pesticides, the scientific community has realized the need for implementation of sustainable practices. Consequently, environmentally benign pesticides are being developed that are making an important breakthrough in pest control. Most of the plant derived materials have emerged as green pesticides as they are safer and effective against diseases, nematodes and other organisms in addition to phytophagous insects. Examples of eco-friendly pesticides include include neem oil, garlic spray, tomato leaves etc.

Terminal Questions

1. The concept of atom economy developed by B.M. Trost is a consideration of "how much of the reactants end up in the final product."

   Example to illustrate atom economy:

   Addition Reaction

	$CH_3\text{-}CH{=}CH_2\,(g)$ +	$H_2\,(g)$	$\longrightarrow CH_3CH_2CH_3\,(g)$
Molar Mass	42	2	44

Atom Economy = (44÷44) × 100

=100 %

2. Earth friendly plastics or bio-plastics commonly obtained from natural origins such as plants, animals or micro-organisms may be classified into the following types:

   (i) **Polyhydroxyalkanoates (PHAs)**: Examples of bio-based plastics include Polyhydroxyalkanoates (PHAs) which are naturally produced by various micro-organisms eg: Cuprividus necator. Poly-3-hydroxybutyrate (PHB), polyhydroxyvalerate (PHV) and polyhydroxyhexanoate (PHH) are specific examples of PHA based plastics.

   (ii) **Polylactic acid (PLA):** PLA is a thermoplastic aliphatic polyester synthesized from renewable biomass, typically from fermented plant starch such as from corn, cassava, sugarcane or sugar beet pulp.

   (iii) **Cellulose-based Plastics:** Cellulose esters such as cellulose acetate and nitrocellulose are utilized to prepare cellulose based plastics.

   (iv) **Lignin based Plastics:** Lignin commonly obtained as a by-product of polysaccharide extraction from various plant materials during the production of paper, ethanol etc.

By definition, green solvents are environmentally friendly solvents that are characterized by low toxicity, convenient accessibility, great efficiency and reusability. Through Life Cycle Assessment (LCA) a technique that quantifies environmental damage for each of the available solvents, one can decide which is greener.

Problems associated with the traditional solvents include:

Many of the traditional solvents are organic based and highly toxic. Besides, they may be volatile, flammable and also carcinogenic.

3. Environmentally benign pesticides are those which are:

   - Biodegradable
   - Cheaper, Renewable and can be handled safely
   - Residue-Free
   - Safe to natural enemies and higher organisms
   - Difficult for insects to develop resistance against them.

4. The basic difference between green chemistry and environmental chemistry is that green chemistry focusses on the design of processes and products in an environmentally benign manner so that no waste is generate and toxicity is minimized whereas environmental chemistry aims at the chemistry to combat the environmental problems after they are created.

5. Supercritical fluids are those substances which when subjected to a pressure and temperature greater than their critical point (obtained at specific conditions of pressure and temperature), are said to be supercritical. In that region, the SCFs exhibit properties of both liquid as well as gas. The most

commonly employed SCF includes supercritical carbon dioxide ($ScCO_2$) which has T_c=304 K and p_c=72.8 atm. $ScCO_2$ has emerged as a green alternative solvent in a number of industrial reactions such as hydrogenation, hydroformylation, polymer production, C-C bond formation owing to myriad of advantages it offers including non-toxicity, abundance, non-renewable nature, tendency to get easily removed from the product.